

WEO Methodology Rationale

Warmth Engine Observatory

Companion Document to Methodology Manual

January 2026 | Version 1.5 (July 2026)

CANONICAL SOURCE DECLARATION

This work is the authoritative version published at DOI: 10.5281/zenodo.18427582

Novel terminology, frameworks, and methodological innovations introduced herein originated with this publication as of January 2026.

Version 1.5 (July 2026) introduces rationale documentation for Infrastructure Threads (Section 10), CC Type 5 Response Band derivation (§9.4.1), Parallel Policy assessment refinements including Criterion 1 mandatory gate and mechanism specificity sub-test, Criterion 2 Industry Similarity formalisation, and CC-A confidence ceiling (§9.3.2, §9.5.2, §9.5.3), revised Option E with field calibration (§3.6.2), Basel 5-Factor object-of-assessment rule, F3 strategic-concentration qualifier, and entity-specific evidence calibration (§4.1), PISA-5 F2/F4 assessment object for non-regulatory events (§4.6.4), ENT-4 coupling argument (§8.8), WER provisionality and corpus scope (§8.13, §8.14), Framework Family parent→member direction (§9.2), the Distinct Actor Requirement common-control extension (§9.1.1), the reduced verification lag for verified product releases (§3.9.1), and the convergent implementation evidence accommodation for Policy events with unpublished instruments (§3.15).

The canonical version is the Zenodo publication. Derivative works, summaries, or adaptations should cite this canonical source.

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About the Author

Warmth Engine is the founder of the Warmth Engine Observatory, a platform tracking AI infrastructure coordination dynamics across geopolitical blocs. Other published work by this author includes *The Warmth Engine Series: Analytical Prophecy and the Convergence of AI Sovereignty, Infrastructure Colonialism, and the Coordination Imperative* (Zenodo DOI: 10.5281/zenodo.17451629), conducted as separate independent research.

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Section 1: Introduction & Document Purpose

1.1 Purpose

This document provides the empirical derivation, theoretical grounding, and calibration justification for all methodology components in the WEO Methodology Manual. It serves as the authoritative reference for understanding *why* methodological choices were made, complementing the Manual's specification of *what* those choices are.

1.2 Document Relationship

Document	Purpose
Methodology Manual (DOI: 10.5281/zenodo.18427565)	Operational specification — what the methodology is
Methodology Rationale	Justification — why the methodology is designed this way

The Methodology Manual is self-contained for operational use. This Rationale document provides empirical derivations, historical precedent analysis, and calibration justifications for users requiring full methodological transparency.

1.3 Rationale Categories

Each methodology component falls into one of six categories:

Category	Definition	Treatment
EMPIRICALLY GROUNDED	Derived from external research, data, or established frameworks	Full derivation with citations
ADAPTED	Modified from established methodology for WEO context	Adaptation rationale with source
WEO SYNTHESIS	Novel combination of established precedents	Synthesis sources documented
WEO-ORIGINAL	Genuinely novel contribution with no prior art	Novelty verification documented
CATEGORY CREATOR	Novel calibration choice for new measurement domain	Explicit acknowledgement with validation commitment
STANDARD PRACTICE	Industry-standard approach requiring no special justification	Brief note citing convention

Section 2: Methodology Balance Statement

2.1 Overall Balance

WEO methodology components distribute approximately **~24% empirically grounded**, **~55% precedent-based**, and **~21% category creator** — with the empirically grounded and precedent-based components collectively representing ~79% external grounding.

Category	Component Count	Percentage
EMPIRICALLY GROUNDED	28	~24%
PRECEDENT-BASED	65	~55%
CATEGORY CREATOR	25	~21%

The V1.5 component count (118) reflects a finer-grained counting methodology than V1.4 (99 components). Standard Practice components are individually counted where they satisfy the dual-test criteria; MIXED parent roll-ups have been replaced by their constituent sub-decisions; and all Category lines carry standardised bands. The two counts are not directly comparable.

The count treats all components equally. Core architectural decisions — the tier system, the CC taxonomy, the SCP framework, the ENT taxonomy — carry greater structural consequence than convention-based choices. No weighting scheme is applied. This follows established practice in comparable frameworks (Basel G-SIB equal-weighted buckets, PRISMA unweighted checklists, OECD EPS equal-weighted sub-indicators).

The table's rows aggregate the §1.3 categories: EMPIRICALLY GROUNDED comprises Empirically Grounded and Standard Practice (convention-based) components; PRECEDENT-BASED comprises Adapted and WEO Synthesis components; CATEGORY CREATOR comprises Category Creator and WEO-Original components.

Note: A methodology component is a design choice satisfying two criteria:

1. Independence: changing the component does not logically force reconsideration of any other single component. If modifying X requires Y to change because Y is a downstream consequence of X, they are not independent components — the dependent item is an implication, not a separate design decision.
2. Determinability: the item represents a distinct analytical judgement — a choice between alternatives where a different selection was available. Downstream consequences, clarifications of existing provisions, explanatory notes, and structural observations arising from other provisions do not satisfy this test.

The independence criterion adapts Keeney & Raiffa's (1976) nonredundancy and preferential independence properties for attribute sets and INCOSE's Requirement Rule C11/R32 (uniqueness/non-overlap). The determinability criterion adapts INCOSE's Requirement Rule C5 (singular: one capability per statement) and ISO/IEC Directives Part 2 Clause 3.3.3 (requirement/provision distinction). Different-domain parallels include the MECE principle (logical decomposition, reflexively applied), GRADE's double-counting warning (procedural logical rule against downstream dependency), and psychometric discriminant validity (Campbell & Fiske 1959; statistical rather than logical, applied to external measurements rather than framework self-assessment). No published framework was found to combine a logical independence test, a determinability test, and reflexive application to the framework's own decomposition as a unified criterion.

Validation: all 126 Category-line entries have been exhaustively verified against both criteria. Nine entries were excluded from the component count: six Standard Practice entries failing the determinability test (downstream consequences, clarifications, explanatory notes, and structural observations), and three higher-confidence entries failing at the boundary (a formalisation of an implicit provision with no operational outcome change, a definitional necessity arising from tier definitions, and an explanatory observation about data state). The remaining 117 entries pass both tests. One additional entry (this component designation criteria) enters the count, producing a total of 118.

Category: ADAPTED (MEDIUM-HIGH 65–75%)

2.2 Confidence Distribution

Confidence Level	Count	Examples
VERY HIGH (>90%)	3	Keohane coordination; Finnemore & Sikkink threshold; Basel G-SIB
HIGH (75–90%)	66	Event classification; software routes; bloc definitions; Vienna Convention; industry taxonomy; workflow integration; OEP Pathways A/B; OEP MIIT framework; source accommodation alignment; Corporate Strategic Disclosure; Level 5 operational evidence; ENT enrichment layer; ENT binary qualification; ENT guardrails; ENT-1 resource demand; ENT-4 environmental dependency; Chinese cascade model; SCP 7-dimension framework; SCP evidence standard; universal membership categories; bloc assignment gate; register evidence standard; 3× scope threshold; 根据 temporal accommodation; F3 pathway exemption; Distinct Actor Requirement; Funding Flow denominator
MEDIUM-HIGH (60–75%)	33	Tier classification; ADAPTED frameworks; CC evidence requirements; mechanism statements; PIET windows; T2 Impact-Demonstrated; OEP rack density; OEP confidence gradation; Option F defence appropriation; conflict-era currency adjustment; ENT taxonomy structure; ENT-3 consequence activation; T2 scope (principal actor asymmetry); ≥3 PAA dimensional threshold; three-gate PAA designation; severance test design; AIK/ACS/Participant taxonomy; sovereign control principle; dimension sustainability taxonomy; third bloc formation; two-level metric hierarchy; inapplicable/indeterminate distinction; Option G supply-chain response; reduced verification lag (verified product releases)
MEDIUM (50–60%)	0	None
CATEGORY CREATOR (variable — novel components without a stated confidence band)	8	CC taxonomy (verified novel IP); ENT-2 linguistic threshold; ENT-2/ENT-5 boundary; consumptive-process floor; WER positioning; CC-A evidence model for Parallel Policy; similarity criteria framework; Type 7 analytical closure
STANDARD PRACTICE (bare by convention)	8	No confidence band — standard practice components carry a brief convention note per §1.3
LOW (<50%)	0	None — all banded components meet the §2.3 minimum; grounding decompositions for category-creator components are reported per section

Confidence bands express confidence in the design decision. Grounding assessments ("~X% grounded") express the proportion of a design resting on external precedent — a transparency decomposition per §2.4, not a confidence band, and not the quantity §2.3 regulates.

2.3 Minimum Confidence Thresholds

WEO maintains two distinct minimum confidence thresholds:

Threshold	Applies To	Minimum	Exceptions
Event Verification	Individual events entering database	70%	None — events below 70% excluded
Methodology Components	Empirical backing for calibration values	65%	CATEGORY CREATOR and WEO-ORIGINAL components carry validation commitments and priority review (§2.4–§2.5) in lieu of the floor

The 70% event threshold ensures database quality. The 65% methodology threshold ensures empirical grounding whilst acknowledging that category creation requires some novel calibration.

2.4 Honest Framing

Where external validation is unavailable, WEO:

1. Adapts proven methodologies from adjacent domains
2. Explicitly flags category creator choices
3. Commits to recalibration as empirical data accumulates
4. Maintains transparency about confidence levels

2.5 Methodology Evolution Commitment

WEO methodology is a living framework subject to refinement as:

1. Empirical data accumulates from ongoing observation
2. New analytical approaches emerge in adjacent fields
3. Validation milestones (e.g., 100-event threshold) trigger systematic review
4. User feedback identifies practical improvements

All methodology components include explicit confidence levels and validation commitments. Category Creator elements are flagged for priority review as the evidence base matures. Version history and material changes are documented in methodology version notes.

Section 3: Event Verification Rationale

Manual Reference: Chapter 3

3.1 Source Quality Hierarchy

Manual Section: 3.1

Category: STANDARD PRACTICE

Rationale: The four-tier source hierarchy (Primary → Corroborating → Supplementary → Excluded) follows established verification practice across intelligence community (ICD 203/206), wire services (Reuters, AFP), fact-checking networks (EFCSN), and defence intelligence (Jane's).

Key Precedents:

- ICD 203/206 establish analytic standards emphasising multi-source analysis and explicit disclosure of single-source dependence
 - Reuters Handbook requires minimum source standards by claim type
 - EFCSN Code of Standards establishes multi-source verification requirements
-

3.1.1 Primary Source Substitution

Manual Section: §3.1 (Primary Source Substitution)

Category: ADAPTED (HIGH 75–90%)

Rationale: The substitution rule formalises the principle that source quality can substitute for source quantity when primary sources are Grade 1 canonical instruments. A Federal Register entry, an official gazette publication, or an audited financial filing independently verifies the event it documents — its authority derives from its institutional origin, not from external corroboration. Two independent Grade 1 instruments documenting the same event provide stronger verification than one Grade 1 primary plus two Grade 3 news articles paraphrasing it.

Closest Precedents:

Framework	Relevance	Divergence from WEO
ICD 203 (Analytic Standards)	Quality-over-quantity principle for intelligence assessments; single high-authority source can outweigh multiple low-authority sources	Intelligence context, not event verification
Reuters Handbook of Journalism	Convergent independent verification, not fixed source counts; authority of source weighs against number required	Wire service practice, not infrastructure event verification
§5.3.4 Dimension Evidence Standard	Same quality-over-quantity principle applied to structural capability claims; 1+1 base with Grade hierarchy	Dimension claims, not event claims; the substitution rule extends the same logic to events

Key Design Decision:

1. **Grade 1 restriction.** The substitution applies only to Grade 1 sources because their institutional authority provides self-verification. Two corporate press releases (Grade 2) or two industry analyses (Grade 3) do not carry the same self-verifying quality — they require independent corroboration to confirm accuracy. The restriction prevents the substitution from weakening the evidence standard for events where corroboration is genuinely needed.

Confidence Assessment: 75%. The principle is well-established across the precedent frameworks and is already implemented in §3.4.9 for Tier A claims ("2+ Primary OR 1P + 2C"). The substitution rule formalises and extends this existing pathway. Confidence matches §5.4.12 (Dimension Evidence Standard) because the underlying quality-over-quantity logic is identical.

Validation Commitment: Monitor whether the Grade 1 restriction is sufficient to prevent evidence-standard erosion. If events begin qualifying through the substitution pathway with sources that are technically Grade 1 but lack the self-verifying quality of canonical instruments, tighten the qualifying criteria.

3.2 Claim Verification Tiers (A/B/C)

Manual Section: 3.4

Category: STANDARD PRACTICE

Rationale: Three-tier claim verification follows ICD 203/206 analytic standards and UK PHIA probability/confidence frameworks.

Tier	Requirement	Application
Tier A	Primary mandatory	Existential claims
Tier B	Primary preferred	Supporting details
Tier C	Corroborating acceptable	Contextual information

3.3 Four Event Types

Manual Section: 3.6

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: The four event types (Coordination, Infrastructure, Policy, Corporate) are classified by **mechanism** (HOW the action occurs) and **actor type** (WHO performs it), not by subject matter, downstream effects, or durability.

Event Type	Primary Mechanism	Actor Dimension
Coordination	Negotiation, mutual accommodation amongst ≥ 2 sovereign entities	Multi-nation
Infrastructure	Deployment of physical/digital systems	Government/sovereign entity
Policy	Unilateral legislative, regulatory, or executive action	Single sovereign entity
Corporate	Private sector action (partnerships, releases, market events)	Private company

Key Distinction: An agreement *about* infrastructure = Coordination (mechanism is negotiation). A policy *requiring* infrastructure = Policy (mechanism is regulation). A company *deploying* infrastructure = Corporate (actor is private). A government *deploying* infrastructure = Infrastructure (actor is sovereign).

Empirical Precedent:

Event Type	Primary Precedent	Confidence
Coordination	Abbott & Snidal legalization framework (2000); Keohane cooperation theory (1984); CAMEO codes 03-05	HIGH
Policy	Lowi's Policy Typology (1964) — classifies by HOW coercion is applied, not subject matter	HIGH
Corporate	IPE literature (Strange 1988, Bartley 2018); CAMEO BUS/MNC actor codes	MEDIUM-HIGH
Infrastructure	Emerging DPI literature; Bakonyi & Darwich (2024) — distinguishes infrastructure materiality (physical construction) from infrastructure governance (rule-setting)	MEDIUM

Derivation: This mechanism-based classification follows established precedent from CAMEO/GDELT event coding (which classifies by action verb, not topic) and Lowi's Policy Typology (which classifies by HOW coercion is applied, not subject matter). Infrastructure as a distinct category is supported by emerging Digital Public Infrastructure (DPI) scholarship, which distinguishes infrastructure materiality from infrastructure governance.

3.4 Event Type × Tier Applicability Matrix

Manual Section: 3.6 (Event Type × Tier Applicability)

Category: STANDARD PRACTICE

Rationale: The constraint that Policy events cannot qualify as Tier 1 follows from definitional necessity:

- **T1 requires:** Cross-bloc participation with mutual accommodation
- **Policy is:** Unilateral rule-making by single jurisdiction
- **Unilateral:** Cannot satisfy "mutual" by definition
- **Therefore:** Policy ≠ T1

Policy events with cross-bloc *effects* route to T2 (typically Restriction) or T3. Coordinated multilateral policy alignment involving mutual accommodation qualifies as Coordination Event Type, not Policy.

3.5 Coordination Event Criteria

Manual Section: 3.7

3.5.1 Keohane Definition

Category: EMPIRICALLY GROUNDED (VERY HIGH >90%)

Rationale: WEO adopts Keohane's (1984) coordination definition — observable policy adjustments through negotiation or mutual accommodation, where each entity's actions facilitate achievement of partners' objectives.

This definition has been IR theory consensus for 40+ years and provides clear operationalisation distinguishing:

- Coordination from harmony (automatic alignment)
- Coordination from parallel independent action (similar policies without negotiation)
- Coordination from aspirational statements (declarations without mechanisms)

Citation: Keohane, R. O. (1984). *After Hegemony: Cooperation and Discord in the World Political Economy*. Princeton University Press.

3.5.2 33% Convergence Threshold

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: The approximately one-third threshold for Type 3 coordination (demonstrated convergence) derives from Finnemore & Sikkink (1998: 901), who establish that "tipping rarely occurs before one-third of the total states in the system adopt the norm." Their analysis across women's suffrage (1930 tipping point), the anti-personnel landmine ban (May 1997), and laws of war norms demonstrates this empirical pattern.

WEO operationalises this at 33% (~50 of ~150 AI-active nations) as a practical threshold, acknowledging this reflects an empirical approximation rather than a precise theoretical requirement.

Citation: Finnemore, M., & Sikkink, K. (1998). "International Norm Dynamics and Political Change." *International Organization* 52(4): 887-917.

3.5.3 Six-Level Implementation Hierarchy

Category: EMPIRICALLY GROUNDED (VERY HIGH >90%)

Rationale: The six-level evidence hierarchy (Announced → Adopted → Ratified → In Force → Implemented → Verified) follows Vienna Convention treaty law, OECD policy tracking standards, and SIPRI arms control verification literature.

Key Precedents:

- Vienna Convention Articles 11-18 codify consent mechanisms (signature, ratification, accession); Article 24 governs entry into force
- Policy monitoring frameworks distinguish announcement, adoption, and implementation phases (cf. OECD policy tracking methodology)
- SIPRI Arms Control Verification hierarchy tracks commitment through operational verification

Key threshold: Binding agreements require Level 3 (ratified) minimum; non-binding frameworks require Level 5 (implemented) minimum.

3.5.4 Type 2 Level 5 — Operational Evidence Requirement

Manual Section: §3.7 (Coordination Criteria — Implementation Verification Standards)

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: The four-pathway operational evidence framework synthesises formally codified mechanisms from multiple established governance architectures into a novel unified qualification gate. The underlying principle — that publication or catalogue adoption of an international standard does not constitute operational implementation — is formally recognised across multiple institutions: ISO/IEC Guide 21-1:2005 (Clause 1) explicitly excludes rules for the "use" of standards in production, trade, or legislation; the ISO Systematic Review (ISO/IEC Directives Part 1, Subclause 2.9) distinguishes adoption from use as separate assessment questions; and the academic decoupling literature (Boiral 2003; Aravind & Christmann 2011; Wijen 2014) provides extensive empirical documentation of the phenomenon.

The architectural precedent for maintaining independent assessment dimensions for adoption and implementation derives most directly from the FATF Mutual Evaluation Methodology (adopted February 2013, updated October 2022), which separates Technical Compliance (rules adopted into national law) from Effectiveness (rules operating in practice) using independent rating scales and evidence-based assessment across 11 Immediate Outcomes. The FATF Methodology states explicitly that the effectiveness assessment "differs fundamentally" from the assessment of technical compliance. WEO adapts this two-dimensional architectural principle from national AML/CFT compliance to individual standard implementation assessment. Each of the four evidence pathways draws on independently established institutional mechanisms: Pathway (a) on the IAF MLA Level 5 endorsement process (IAF PL 3) and ISO Survey of Certifications; Pathway (b) on the EU New Legislative Framework's OJEU citation gate (Regulation (EU) 1025/2012, confirmed by CJEU in *James Elliott Construction C-613/14* (2016) and *Public.Resource.Org v Commission C-588/21 P* (2024)); Pathway (c) on the ISO/IEC Directives Part 2:2021 mandatory normative reference framework (Clause 2 references "constitute requirements"); and Pathway (d) on mechanisms such as the EU Outdoor Noise Directive's mandatory Declaration of Conformity reporting and the FATF effectiveness assessment's evidence-of-outcomes requirement.

The novel elements are: (i) the combination of all four evidence categories into a single disjunctive qualification gate (no existing framework bundles all four); (ii) the anchoring requirement for Pathway (c), requiring normative dependency chains to terminate in a standard with independent operational evidence (no existing framework formalises this concept); and (iii) the application of this gate to individual international consensus standards as a unit of analysis for geopolitical coordination intelligence (existing analogues operate at the national regulatory system level or within single regulatory domains).

Closest Precedents:

Framework	Institution	Relevance	Divergence from WEO
FATF Mutual Evaluation Methodology (2013/2022)	FATF	Strongest architectural analogue: two independent assessment dimensions (technical compliance vs. effectiveness) with separate evidence requirements and rating scales	Operates at national regulatory system level, not individual standard level; uses graduated four-point scale, not binary gate; domain-specific to AML/CFT
EU New Legislative Framework (Regulation 1025/2012)	European Commission / CJEU	Strongest single-pathway precedent: OJEU citation as formal, judicially confirmed qualification gate between publication and regulatory operational effect	Applies only to harmonised European standards within EU regulatory scope; single-dimension gate (regulatory reference only)
IAF MLA Level 5 / IAF PL 3	IAF (now GACI)	Direct precedent for requiring evidence of certification activity as qualification criterion for scheme endorsement; requires "clear international, regional or national market support" and "evidence of operating"	Gate operates at scheme viability level, not minimum assessment count; limited to management system and product certification standards within IAF scope
ISO/IEC Directives Part 2:2021	ISO/IEC	Direct codification of normative dependency mechanism — normative references "constitute requirements"	Does not formalise anchored vs. unanchored chains; no operational evidence requirement for referenced standards
Basel Committee RCAP	BCBS	Three-tier assessment approach (adoption timeliness → transposition consistency → bank-level outcomes) confirming institutional recognition of the adoption/implementation gap	Explicitly acknowledges it does not assess supervisory effectiveness; banking regulation only

Key Design Decisions:

1. **Disjunctive gate (any one pathway sufficient).** Unlike FATF's 11 Immediate Outcomes (all assessed), WEO requires only one of four pathways. This reflects the binary nature of the implementation question for individual standards: the question is whether ANY operational effect exists, not a comprehensive assessment of implementation depth.
2. **Pathway (c) anchoring.** A normative dependency chain cannot be self-referencing without an anchor. At least one standard in the chain must independently satisfy Pathway (a), (b), or (d). This prevents a cluster of published-but-unimplemented standards from bootstrapping each other to Level 5 through circular normative references. No existing framework formalises this concept.
3. **Binary gate vs. graduated scale.** WEO uses pass/fail rather than FATF's graduated four-point scale. This is a deliberate design choice — WEO needs a classification decision (implemented or not), not an implementation quality assessment. The EU NLF's OJEU citation gate provides direct precedent for binary treatment in standards governance.
4. **Retrospective application.** The operational evidence requirement applies to all events in the database, not only to new candidates. Standards that entered under V1.2 criteria where publication alone satisfied Level 5 are assessed against the four pathways. Events failing all four pathways are removed from the database. This ensures the database reflects verified operational coordination, not catalogued intentions.

Confidence Assessment: 82%. The principle that publication does not equal implementation is formally codified in multiple institutions (ISO/IEC Guide 21-1, FATF Methodology, EU NLF), and three of the four pathways have direct or strong analogous institutional precedent. Confidence is bounded below 90% because: (a) no single framework bundles all four pathways, making this a synthesis rather than an adaptation; (b) the anchoring requirement for Pathway (c) is genuinely novel; and (c) the application to individual standards as a unit of analysis for geopolitical intelligence has no direct precedent. Confidence is above 75% because the architectural principle is well-established, the FATF analogue is strong, and each pathway independently maps onto formally codified mechanisms.

Validation Commitment: Monitor for (1) any ISO or IEC governance reform that creates a formal implementation tracking mechanism for the full standards catalogue; (2) evolution of the merged IAF/ILAC entity (Global Accreditation Cooperation Inc.) and any changes to scheme endorsement criteria; (3) academic literature proposing multi-pathway implementation assessment frameworks; (4) FATF 5th Round methodology updates that may refine effectiveness assessment criteria; and (5) any extension of the EU NLF gate mechanism to non-harmonised standards.

3.6 Two-Stage AI Connection Filter

Manual Section: 3.5

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

3.6.1 10% / \$100M Thresholds

Rationale: The 10% allocation threshold and \$100M absolute threshold derive from convergent regulatory precedent:

Source	Threshold	Application
SEC ASC 280-10-50-12	10%	Segment reporting materiality
Basel III	10%	Large exposure definition threshold
IMF BPM6	10%	Direct investment classification (paragraph 6.12: ≥10% voting power)

The \$100M absolute threshold ensures large-scale projects with material AI components (e.g., \$400M AI line in \$5B fab) are captured even when percentage is below 10%.

\$100M Validation Sources:

- Institutional infrastructure fund minimum deal sizes represent institutional-scale investment thresholds
- Market transaction evidence: POET Technologies \$75M AI hardware financing (October 2025), Applied Digital \$112.5M AI data centre drawdown (October 2025)
- EU practice often uses ≥€50M as a threshold for IPCEI notification (GBER 2023) and direct participant status in national calls — WEO's \$100M represents approximately 2× this practical threshold
- Distribution analysis of verified AI infrastructure investments: \$50M-\$100M band contains lowest investment density, confirming natural breakpoint at \$100M

Citations:

- SEC filings: POET Technologies (October 2025), Applied Digital (October 2025)
- EU Important Projects of Common European Interest (IPCEI) framework

3.6.2 Option E — Corporate Strategic Disclosure

Manual Section: §3.5, Stage 2 Option E

Category: ADAPTED (HIGH 75–90%)

Rationale: Option E adapts established regulatory evidentiary practice to the Stage 2 AI connection question for Corporate events. Corporate financial instruments — PPAs, equity agreements, procurement contracts — specify capacity, price, and delivery terms without designating workload type. This is structurally parallel to the gaps addressed by PIET (export controls list by HS code) and OEP (infrastructure documentation is application-agnostic): the instrument class does not produce AI-specific language by convention.

The evidentiary approach — accepting corporate disclosure as evidence of the strategic purpose of a transaction whose operative instrument is silent on that purpose — has well-established precedent across international regulatory frameworks. Five of seven frameworks examined make the approach mandatory or statutory; no framework rejects it in principle.

Framework	Established Practice
US Antitrust (HSR Act Items 4(c)/4(d); 2023 Merger Guidelines)	Mandatory filing of strategic analyses; ordinary-course documents more probative than retrospective materials
EU Merger Control (Form CO §5.4; <i>Facebook/WhatsApp</i> €110M fine)	Mandatory filing of board presentations; regulated disclosures treated as binding representations
Sanctions Enforcement (OFAC; Section 13(r) Securities Exchange Act)	Statutory pipeline from regulated corporate disclosure to enforcement action
Securities Regulation (SAB 99; Reg S-K Item 303; <i>Leidos</i> (2018))	Entity's own characterisation of transaction significance directly affects regulatory obligations
WTO Dispute Settlement (<i>Australia – Plain Packaging</i> , DS435/441)	Internal corporate documents cited by panel to establish industry awareness; examination beyond measure's text

Field calibration: A survey of six comparable AI infrastructure tracking platforms examined whether regulated financial filings are required as an inclusion gate. None requires regulated financial filings. All accept official corporate communications as primary source evidence.

Platform	Requires regulated filings	Accepts official communications	Confidence tiers	Status field
Epoch AI	No	Yes — primary source type	Yes (Confirmed / Likely / Unlikely / Confirmed false)	Yes (Existing / Planned / Decommissioned)
SemiAnalysis	No	Yes + physical verification	Not publicly documented	Not publicly documented
CNAS Sovereign AI Index	No	Yes — accepts announced roadmaps and signalled intent	No formal per-project tiers	Conceptual but not tagged
Stanford HAI AI Index	No	Yes — synthesises multiple public source types	No per-entry tiers	N/A (report format)
OECD.AI	No	Yes — government submissions + public policy documents	No per-entry tiers	Yes (policy status categories)
CSET/ETO	No	Yes — official sources + public documentation	No per-entry tiers	Not formally published

Sources: Epoch AI GPU Clusters inclusion criteria (epoch.ai/data/gpu-clusters-documentation/inclusion); Epoch AI estimation methodology (epoch.ai/data/gpu-clusters-documentation/estimation); Pilz et al., "Trends in AI Supercomputers" (arXiv 2504.16026, 2025).

Key design decisions:

1. **Official corporate communications as the inclusion gate.** The cross-framework precedent table establishes that corporate disclosure as evidence of purpose is sound — five of seven frameworks make it mandatory or statutory. The field calibration establishes that using regulated filings as the inclusion gate rather than a confidence-tier anchor is out of step with field practice and creates a systematic exclusion of entities with no filing obligation. The inclusion gate accepts official corporate communications meeting three requirements (transaction specificity, contemporaneity, senior authorisation) — the same safeguards identified across all seven examined frameworks.
2. **Regulated filings repositioned as the highest-confidence evidence category.** The cross-framework finding that legal-liability accountability confers probative weight is preserved: regulated filings achieve Confirmed tier automatically because the filing obligation and its attendant liability exposure provide inherent corroboration. This is consistent with the evidentiary hierarchy across all seven frameworks, where regulated-filing status is the single strongest indicator of reliability.
3. **Corroboration integrated as a confidence-tier input.** WEO's methodology uses tiered-confidence architecture in multiple systems — CC confidence levels (CC-V, CC-E, CC-A), the SCP evidence standard with escalation triggers (§5.3.4), and Basel/PISA-5 factor rubrics — rather than stacked binary gates. Option E integrates corroboration into this established pattern: presence of corroboration elevates Likely to Confirmed; absence of corroboration does not disqualify. This retains corroboration as a quality signal — no surveyed comparator requires corroboration at all — whilst aligning the gate architecture to the methodology's tiered-confidence pattern.
4. **Confidence tier and status field.** Epoch AI's four-tier certainty scale and three-category status field provide the template. Option E adapts this to two tiers (Confirmed, Likely) and two statuses (Operational, Announced), reflecting the narrower scope of the Stage 2 AI connection question relative to Epoch AI's broader data collection mandate.
5. **Qualifying communication types enumerated.** The Manual lists six qualifying communication types (regulated financial disclosures, official press releases, official partner/customer announcements, published annual reports including voluntary reports from private companies, official product launch documentation, and senior management conference presentations) and five excluded categories (social media posts, analyst commentary, marketing materials, leaked documents, unnamed-source media reports). The enumeration provides deterministic classification for analysts.
6. **AI-linkage content requirement.** Option E relaxes the source-type constraint (instrument text → official corporate communication) without relaxing the content standard for AI infrastructure connection. The communication must contain language that would satisfy the AI Connection Filter (Options A, B, or C) if it appeared in the instrument's operative text. This ensures the evidentiary bar for what constitutes AI connection evidence is consistent across all five options — the source changes, the content standard does not.

Documented limitation: *Japan – Alcoholic Beverages II* (WT/DS8/AB/R, 1996) rejected subjective intent inquiry. Option E focuses on operational function (role in AI infrastructure), not subjective corporate intent — consistent with the "design, architecture, and structure" framework that also underpins the T2 Impact-Demonstrated mechanism (§3.7.4).

Confidence Assessment: HIGH. The underlying evidentiary principle — corporate disclosure as evidence of purpose — is established across seven frameworks, five making it mandatory or statutory. The field calibration reinforces this: the inclusion gate aligns to documented field practice across six comparable platforms. The confidence tier and status field adapt a documented template (Epoch AI). The main residual uncertainty is whether the boundary between qualifying and non-qualifying communication types produces consistent classification at the margin — specifically, whether "official product launch documentation" and "conference presentations by senior management" admit material that does not meet the three requirements in practice.

Validation Commitment: Review confidence tier and status assignments against the first five events qualifying via the relaxed Option E, with specific attention to Likely-tier events and the boundary between qualifying and non-qualifying communication types.

3.6.3 Option F — Defence Appropriation Evidence

Manual Section: §3.5, Stage 2 Option F

Category: ADAPTED (MEDIUM-HIGH 60–75%)

Rationale: Policy events with genuine AI connections that are funded through defence appropriation processes face a structural documentation constraint that is universal, not specific to WEO. Defence appropriation documents fund AI-specific programmes through publicly verifiable financial instruments. Once appropriated funds enter implementation, operational detail moves behind classification barriers — a permanent feature of military spending architecture across all major defence economies. The appropriation stage is therefore the sole publicly verifiable evidence window for the AI connection.

Any framework attempting to track military AI activity from public documentation encounters the same structural barrier: the evidence required for post-implementation verification does not exist in publicly accessible form. The constraint is empirically documented across five surveyed economies and is consistent with the broader finding that observable documentation surface contracts during re-militarisation whilst underlying activity expands (§4.11).

Option F accepts binding defence financial documents as AI connection evidence at Stage 2, capturing events at the sole point where the AI connection is publicly documented. The adaptation applies existing primary source evidentiary standards — enacted legislation and executive budget submissions qualify as Tier A evidence per §3.4.4 — to a specific document class not previously recognised in the AI Connection Filter's option structure.

Boundary logic — eligibility by Event Type and document class:

The classification barrier affects all event types within the defence domain. However, existing pathways already serve Coordination and Infrastructure events:

- **Coordination events** in the defence context (e.g., multilateral defence AI partnerships) typically qualify via Keohane auto-qualification because the agreement or communiqué itself is public and contains AI-specific language. Classification constrains downstream implementation detail, not the coordination event's qualification evidence.
- **Infrastructure events** in the defence context (e.g., classified military AI facilities) are served by OEP (Option D), which establishes AI connection through operational physical evidence where official documentation is absent or application-agnostic.
- **Policy events** funded through defence appropriations have no fallback pathway when Options A, B, and C fail. These are the events where the appropriation document is the sole primary source evidence of AI connection.

Option F fills this remaining gap. Eligibility is determined by Event Type (Policy, established at Stage 1) and document class (defence appropriation documents), consistent with how Options D, E, and PIET scope their eligibility. Domain classification is a separate, independent assessment at Stage 4.

Empirical calibration:

The pass criteria were calibrated against a verified survey of defence appropriation documents across five major economies, applying the same multi-agent cross-validation approach documented for the principal actor framework (§5.4):

Economy	Primary Document	AI Granularity	Usable for Option F
United States	DoD RDT&E R-1 Justification Books	EXPLICIT	Yes
United Kingdom	MoD Main Supply Estimates / Policy	IDENTIFIABLE	Yes
European Union	EDF Work Programme 2026	EXPLICIT	Yes
Japan	MOD Defense Programs & Budget FY2026	EXPLICIT	Yes
Australia	Defence Portfolio Budget Statements	EMBEDDED	No

Three-level granularity scale:

- **EXPLICIT:** Dedicated AI budget line with a financially isolated monetary value
- **IDENTIFIABLE:** AI explicitly named as a designated beneficiary of a shared funding pool
- **EMBEDDED:** AI spending subsumed within undifferentiated general categories; not separately identified

The survey confirmed that the classification constraint operates consistently across different legislative traditions. Nations whose defence budget architecture produces only EMBEDDED granularity do not qualify via Option F — this reflects their financial architecture, not a methodology limitation.

Calibration Evidence	Contribution
US DoD R-1 Justification Books (PE 0603040A: AI/ML Advanced Technologies, \$20.487M FY2026)	Confirmed EXPLICIT granularity in executive budget submissions at programme element level
UK Defence Innovation Fund (£400M, ring-fenced with AI as mandated beneficiary)	Confirmed IDENTIFIABLE granularity via binding departmental policy with explicit AI designation
EU EDF Work Programme 2026 (EDF-2026-RA-SIMTRAIN-MSAI-STEP, €16M)	Confirmed EXPLICIT granularity in Commission Implementing Decisions with isolated AI call topic budgets
Japan MOD FY2026 Budget ("Initiatives for the Utilisation of AI" section)	Confirmed EXPLICIT granularity in executive budget submissions with hyper-granular AI breakdown
Australia Defence PBS 2025-26 (ASCA merged into Program 2.8)	Confirmed EMBEDDED granularity — AI not separately identified; correctly excluded
US DoD AI spending structure (\$25.2B total, \$1.8B explicit topline — ~93% embedded in broader programmes)	Confirmed that classification renders post-implementation AI spending verification structurally impossible from public sources

Key design decisions:

1. **IDENTIFIABLE granularity accepted alongside EXPLICIT.** The document must explicitly name AI in the financial commitment; financial isolation of the AI component is not required. Ring-fenced capability pools listing AI as a legally mandated beneficiary satisfy the AI connection test because the designation is documented in the instrument — not inferred.
2. **Three-category document typology.** Enacted appropriation legislation, executive budget submissions, and departmental work programmes all qualify, provided each carries binding appropriation authority. The survey demonstrated that AI-specific granularity resides in programme-level documents, not top-level appropriation act text.
3. **No minimum monetary threshold.** Option F establishes AI connection; significance is assessed separately in Chapter 4. This preserves the clean separation between §3.5 (AI connection) and Chapter 4 (significance assessment).
4. **Eligibility by Event Type and document class, not domain.** Option F restricts to Policy events (Event Type, determined at Stage 1) evidenced by defence appropriation documents (document class). Domain classification is an independent Stage 4 assessment. This is consistent with how Options D, E, and PIET scope their eligibility — none use domain as a restriction.
5. **Economy-agnostic application.** The pathway applies uniformly. Nations with EMBEDDED AI budget granularity do not qualify. This accurately reflects the documentation landscape rather than imposing a jurisdictional restriction. Economies whose defence budget architecture operates at structural opacity levels consistent with the documentation asymmetry documented in §3.15 and Manual §4.11 — currently, no Eastern Bloc defence budget document surveyed produces granularity above EMBEDDED — are not excluded by jurisdictional rule but by the same documentation landscape constraint that the source accommodation framework addresses elsewhere in the methodology.

Confidence Assessment: MEDIUM-HIGH (65–75%). The driving constraint — classification rendering post-implementation verification impossible — is empirically universal and not subject to calibration uncertainty. The pass criteria are calibrated against verified document formats across five economies. The main residual uncertainty is whether the IDENTIFIABLE boundary will produce consistent classification across future edge cases where the AI designation in a shared funding pool is ambiguous.

Validation Commitment: Review pass criteria against the first five events qualifying via Option F, with specific attention to IDENTIFIABLE-granularity edge cases.

3.6.4 Option G — Supply-Chain Response Evidence

Manual Section: §3.5, Option G

Category: ADAPTED (MEDIUM-HIGH 60–75%)

Source methodology: PIET (§3.5.1) — structural template for qualifying events where official instrument text contains no AI-specific language but the AI infrastructure connection is real and documentable.

Adaptation rationale: PIET addresses restriction events (export controls, materials bans) where the AI connection manifests through post-implementation trade flow disruption. Option G addresses the symmetric case: response events (minerals strategies, subsidy frameworks, stockpile legislation, supply-chain security acts) where the AI connection manifests through the controlled material's documented role in AI infrastructure manufacturing. Both pathways solve the same structural problem — instrument text that addresses the material, not the downstream application — through different evidence mechanisms appropriate to their event class.

PIET requires 12–18 month post-implementation evidence because restriction events need empirical trade flow data to confirm impact. Option G does not require a waiting period because the AI infrastructure relevance of the controlled material is established through pre-existing materials science and supply-chain documentation, not through post-implementation observation. Gallium's role in GaN power semiconductors, germanium's role in fibre optic amplification, and rare earth elements' role in permanent magnets are documented physical facts, not empirical claims requiring post-implementation verification.

Key design decisions:

1. **Two-condition structure (AI infrastructure input + instrument text absence).** Condition 1 establishes that the controlled material is genuinely AI-infrastructure-relevant. Condition 2 confirms that Option G is the appropriate pathway — that the event cannot qualify through the stricter text-based Options A, B, or C. This mirrors PIET's eligible event class structure: event type (Policy), event subtype (response/security rather than restriction), controlled items (AI infrastructure inputs), and prior option failure.
2. **AI-Critical Materials Register as convenience list, not exhaustive boundary.** The register pre-establishes AI infrastructure relevance for common materials, avoiding redundant causal-chain documentation in each processing cycle. Materials not on the register can still qualify through individual documentation under Condition 1(a)+(b). This balances processing efficiency with scope control.
3. **Scope control through instrument specificity.** Option G requires the instrument to be specifically directed at the controlled material — not a general industrial policy that incidentally covers AI-relevant materials. This prevents over-broadening (e.g., a national infrastructure package with a minerals chapter should not qualify the entire package). The specificity standard mirrors Option A's $\geq 10\%$ threshold logic: only the AI-infrastructure-relevant component qualifies.
4. **No post-implementation waiting period.** Unlike PIET, Option G's evidence base is pre-existing scientific and technical documentation. The AI infrastructure application of gallium, germanium, or rare earth elements does not require post-implementation observation to confirm — it is documented in materials science literature, manufacturer specifications, and supply-chain analyses from recognised institutions.
5. **Symmetric treatment of restriction and response.** PIET qualifies restriction events; Option G qualifies response events. Together they provide symmetric coverage of both sides of supply-chain coordination dynamics — which aligns with WEO's stated objective of tracking coordination dynamics across blocs without systematic bias towards one side's instruments.

Coordination Connection implications: Supply-chain response events qualifying through Option G enter the database where they can be assessed for Coordination Connections under standard CC methodology (§9). Candidate connection types include CC Type 5 (Response) where explicit trigger identification exists in official statements, and CC Type 7 (Parallel Policy) where convergent policy action across jurisdictions is documented.

Evidence basis: A gap assessment conducted in May 2026 documented 11 blocked candidates across three policy domains (critical minerals, energy-AI policy, minerals coordination). The gap was identified when supply-chain response candidates were assessed against the AI Connection Filter and systematically failed Options A, B, and C due to instrument text addressing the controlled material rather than AI. Materials-to-AI-infrastructure causal chains are documented in USGS Mineral Commodity Summaries, BGS Critical Minerals Strategy, EU CRMA legislative history (Regulation 2024/1252 recitals), CSET semiconductor supply chain analysis, and manufacturer technical documentation.

Confidence: 72%. Adapted from PIET (§3.5.1), whose components are rated MEDIUM-HIGH confidence (60–75%) per Rationale §3.7. The adaptation is bounded and structurally parallel: same problem (instrument text does not reference AI), different evidence mechanism (pre-existing documentation vs post-implementation observation), same outcome (AI Connection established at primary source standard). Confidence is bounded below 80% because: (a) the AI-Critical Materials Register is novel and untested — the register's scope may need adjustment after first application; (b) the "instrument specificity" scope control has not been tested against edge cases. Confidence is held above 65% because: (a) direct adaptation from published methodology; (b) materials-to-AI-infrastructure causal chains are well-documented in primary sources; (c) the symmetric treatment principle is methodologically sound; (d) 11 blocked candidates demonstrate the gap is real and consequential.

Validation commitment: Monitor Option G applications for scope creep. If events qualifying through Option G are systematically tangential to AI infrastructure coordination (general industrial policy presented as minerals strategy), tighten Condition 1 or add a third condition. Review the AI-Critical Materials Register at each methodology version milestone.

3.7 PIET — Post-Implementation Evidence Track

Manual Section: §3.5.1

Rationale: PIET addresses a bounded documentation gap for Policy restriction events on physical AI-critical materials or physical technology products where official instruments use no AI-specific language. The mechanism is grounded in the WTO de facto discrimination doctrine — the principle that facially neutral measures are classified by their demonstrated effects — and adapts this to the Stage 2 AI connection question. All calibration thresholds are WEO-original.

3.7.1 Gate Architecture and 3-of-4 Pass Logic

Manual Section: §3.5.1 — Pass Logic

Category: CATEGORY CREATOR (MEDIUM-HIGH 60–75%)

Rationale: No established international framework uses explicit N-of-M partial pass logic — confirmed across ten surveyed frameworks (Basel RCAP, IOSCO, ICD 203, UK PHIA, EMA MCDA, HACCP, Moody's/S&P, Wassenaar, WTO DSU, UN Sanctions). FATF's 4th/5th Round Mutual Evaluation Procedures demonstrate threshold-based multi-criteria assessment as a recognised governance design principle, but FATF's ALL-of-N structure is the structural inverse of PIET's N-of-M. The 3-of-4 threshold is WEO-original: N=1 risks false positives; N≥3 provides independent corroboration whilst tolerating one structurally unavailable gate.

Precedent	Contribution
FATF 4th/5th Round Mutual Evaluation Procedures	Threshold-based multi-criteria assessment as recognised governance design principle; ALL-of-N contrast
Survey of ten frameworks	Confirmed absence of N-of-M pass logic across all surveyed frameworks

Confidence Assessment: MEDIUM-HIGH. **Validation Commitment:** Review against cumulative PIET event data.

3.7.2 Trade Flow and Price Thresholds

Manual Section: §3.5.1 — Gates 1–2

Category: CATEGORY CREATOR (MEDIUM-HIGH 60–75%) for both

Rationale: No institutional framework defines percentage disruption levels from export restrictions — confirmed across WTO, OECD, BIS/EAR, Wassenaar, OFAC, EU, and IMF. The >50%/>90% trade flow thresholds are calibrated as conservative detection floors. The >20% price threshold derives from composite reasoning: Cashin & McDermott (2002, *IMF Staff Papers*) document annual $\sigma = 12.6\%$ for real industrial commodity prices (placing 20% at $\sim 1.6\sigma$); the LME 15% daily circuit breaker — introduced as emergency interim measures during the March 2022 nickel crisis; 15% daily price limit formally designated permanent in March 2023 following the Oliver Wyman Independent Review — provides institutional reference for disorderly price movement; IMF WP/2011/283 confirms no canonical threshold exists.

Precedent	Contribution
Cashin & McDermott (2002, <i>IMF Staff Papers</i>)	Annual $\sigma = 12.6\%$; $20\% \approx 1.6\sigma$
LME 15% daily circuit breaker (introduced as emergency interim measures during the March 2022 nickel crisis; 15% daily price limit formally designated permanent in March 2023 following the Oliver Wyman Independent Review)	Institutional reference for disorderly price movement
IMF WP/2011/283	Confirmed absence of canonical threshold

Confidence Assessment: MEDIUM-HIGH. **Validation Commitment:** Review against each qualifying PIET event; thin-market sensitivity analysis.

3.7.3 Assessment Windows

Manual Section: §3.5.1 — Pending Designation and Gates 3–4

Category: 12-month: WEO SYNTHESIS (MEDIUM-HIGH 60–75%); 18-month: ADAPTED (MEDIUM-HIGH 60–75%)

Rationale: The 12-month floor synthesises: WTO Safeguards Agreement Article 7.4 (one-year liberalisation threshold); IMF Article IV consultation cycle; IMF WP/24/77 (mineral commodity adjustment 18–24 months). The 18-month outer bound adapts the SEC 10-K cycle: maximum ~14–15 month lag from fiscal year start (SEC Rules 33-8128/33-8644). **Known structural limitation:** Enacted standalone statutory legislation typically requires 24–60+ months — documented outer-bound exception.

Precedent	Application
WTO Safeguards Agreement, Article 7.4	1-year progressive liberalisation threshold
IMF Article IV / WP/24/77	12-month review cycle; 18–24 month adjustment for mineral commodities
SEC Rules 33-8128/33-8644	10-K mechanics — maximum ~14–15 month lag
WTO DS431	Government response timing corroboration

Confidence Assessment: MEDIUM-HIGH for both windows.

3.7.4 T2 Impact-Demonstrated Mechanism

Manual Section: §5.5 — T2 Restriction Test: Impact-Demonstrated Pathway

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

Rationale: Draws on the WTO de facto discrimination doctrine. *Japan–Alcoholic Beverages II* (WT/DS8/AB/R, 1996) originated the "design, architecture, and structure" framework under GATT Article III:2 — whether a measure affords protection is determined objectively from its design, architecture, and structure, not stated intent. *US–Clove Cigarettes* (WT/DS406, 2012) extended this to TBT Article 2.1. *Canada–Autos* (WT/DS139/AB/R) confirmed de facto discrimination under Article I:1. PIET adapts this from product-level trade discrimination to state-level strategic targeting: Gate 3 government response patterns serve as the effects evidence. The domain gap is acknowledged, correctly limiting confidence to MEDIUM-HIGH.

Precedent	Application
<i>Japan–Alcoholic Beverages II</i> (WT/DS8/AB/R, 1996)	Originated "design, architecture, and structure" framework under GATT III:2
<i>US–Clove Cigarettes</i> (WT/DS406, 2012)	Extended framework to TBT Article 2.1
<i>Canada–Autos</i> (WT/DS139/AB/R)	De facto discrimination — Article I:1; facially neutral measures by demonstrated effects

Confidence Assessment: MEDIUM-HIGH. **Validation Commitment:** Review as T2 Impact-Demonstrated events accumulate.

3.8 OEP — Operational Evidence Protocol

Manual Section: §3.5, Stage 2 Option D

Rationale: OEP resolves Stage 2 qualification gaps for Infrastructure events where official primary documentation is absent, generic, or application-agnostic. The single-pathway pass logic reflects the binary nature of the AI connection question: one confirmed pathway is sufficient to establish that a facility serves AI workloads.

3.8.1 Pathway A — Technical Output Analysis

Manual Section: §3.5, Option D — Pathway A

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: Chip decapsulation is direct physical measurement of fabricated silicon — observational methodology, not inference. CSET/ETO's institutional data reliance and US government response to the September 2023 Kirin 9000S analysis confirm TechInsights' de facto primary source status. China's Commerce Ministry Unreliable Entity designation (October 2025) constitutes indirect adversary-government confirmation. **Coverage limitation:** TechInsights is the sole provider at this standard; the designation may impair access to Chinese chips with no identified substitute.

Precedent	Application
CSET/ETO dataset (eto.tech/dataset-docs/chipexplorer)	Direct institutional data reliance — US government-funded research
TechInsights Kirin 9000S analysis (September 2023)	Calibration event; BIS investigation trigger
BIS ECCN 3A090 (87 FR 62,186 and subsequent)	AI chip definition anchor

Confidence Assessment: HIGH.

3.8.2 Pathway B — Regulatory Designation Logic

Manual Section: §3.5, Option D — Pathway B

Category: ADAPTED (HIGH 75–90%)

Rationale: Adapts BIS AI-enabling product designation logic to Stage 2 AI connection. The foundational instrument is 87 FR 62,186 (October 2022), creating ECCN 3A090 and listing AI capability denial as one of four stated rationales. All subsequent BIS rulemakings cite this instrument. Adaptation is explicit: where BIS uses product-type designation for export control jurisdiction, OEP uses it for AI connection. PROBABLE confidence reflects the inferential step — facility-level AI allocation inferred from product type, consistent with IC analytic practice under ICD 203 (ODNI, 2015). Regulatory currency check required: 90 FR 4,544 was rescinded May 2025.

Precedent	Application
87 FR 62,186 (October 2022)	Foundational instrument — ECCN 3A090; AI capability denial rationale
89 FR 96,790 (December 2024)	HBM addition; BIS AI-enabling language
IC analytic practice under ICD 203 (ODNI, 2015)	PROBABLE confidence basis

Confidence Assessment: HIGH.

Validation Commitment: Review the AI-exclusive component boundary at methodology version milestones; monitor whether events entering through this gate remain within the intended scope of AI-exclusive components with no significant general-purpose market.

3.8.3 Pathway C — Hardware Deployment Evidence

Manual Section: §3.5, Option D — Pathway C

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%) for rack density; EMPIRICALLY GROUNDED (HIGH 75–90%) for MIIT framework; CATEGORY CREATOR (PROVISIONAL) (MEDIUM-HIGH 60–75%) for 10% threshold

Rationale: No formal AI rack density classification standard exists. The ≥ 30 kW threshold reflects the air cooling upper limit (ASHRAE TC 9.9 H1); ≥ 60 kW is calibrated to 2022–2025 AI training configurations (NVIDIA DGX H100/H200 ~40.8 kW/rack; GB200 NVL72 ~120 kW/rack). Uptime Institute 2024 confirms both thresholds substantially exceed deployment norms. **These hardware references represent the calibration basis at time of V1.3 publication (April 2026) and are subject to the hardware generation review commitment documented in Manual §3.5, Pathway C.** **Hardware generation note:** GB200 NVL72's 2:1 GPU-to-CPU ratio generates false negatives for the $\geq 4:1$ indicator.

The MIIT framework is established by 工信部联通信〔2023〕180号 (October 2023, six joint ministries, gov.cn). NDA and MIIT have both cited 智能算力 at 90 EFLOPS (FP32 precision, 换算为 FP32) at end-2024 as a separate national statistical category. **Regulatory currency note:** MIIT's 智算中心 / 智能算力 designation framework is subject to ongoing regulatory evolution; currency of the governing instrument should be verified at the time of OEP assessment.

The 10% threshold has no external precedent. Three structural analogies: EAR de minimis rule (15 CFR §734.4); USMCA Chapter 4 RVC (19 U.S.C. §4531); USDA Organic Certification (7 CFR Part 205). PROVISIONAL status retained.

Precedent	Application
ASHRAE TC 9.9 H1; Uptime Institute 2024	Air cooling upper limit; confirms ≥ 30 kW and ≥ 60 kW exceed deployment norms
NVIDIA DGX H100/H200; GB200 NVL72 specifications	Empirical calibration for rack density and GPU-to-CPU indicators
工信部联通信〔2023〕180号 (October 2023)	Formal MIIT tripartite classification framework
EAR de minimis (15 CFR §734.4); USMCA RVC (19 U.S.C. §4531); USDA Organic (7 CFR Part 205)	Structural analogies for percentage-of-composition classification

Confidence Assessment: MEDIUM-HIGH (rack density); HIGH (MIIT); MEDIUM-HIGH PROVISIONAL (10% threshold). **Validation Commitment:** Hardware generation review at major GPU platform transitions; 10% threshold reviewed at 10-event OEP Pathway C milestone.

3.8.4 Confidence Gradation (STRONG/PROBABLE)

Manual Section: §3.5, Option D — Pass Logic
Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

Rationale: Established prior art for evidence-to-confidence mapping includes ICD 203 (ODNI, January 2015, Section D.6.e.(2)), which provides the conceptual basis for confidence as a function of evidence quality and quantity — operationalised in IC assessment practice (2007 NIE on Iran; 2017 ICA on Russian Activities). IPCC uses five confidence levels; NATO STANAG 2511 uses a 6×6 reliability-credibility matrix. The specific two-level implementation (STRONG/PROBABLE) is WEO-original, suited to OEP's binary classification context.

Precedent	Application
ICD 203 (ODNI, January 2015), Section D.6.e.(2)	Conceptual basis — confidence as function of evidence quality and quantity
2007 NIE; 2017 ICA	IC assessment practice operationalising ICD 203
IPCC five levels; NATO STANAG 2511	Prior art for multi-level confidence gradation

Combination Rules Derivation: The combination rules follow from ICD 203's evidence quality primacy principle (Section D.6.e.(2)): confidence reflects evidence quality, not evidence quantity. Two PROBABLE pathways do not aggregate to STRONG because cumulative inferential evidence remains inferential — the AI connection question is binary, and convergent inference about a binary fact does not become direct observation. One STRONG pathway raises overall confidence to STRONG regardless of additional pathways because observational evidence is established. This design prevents confidence inflation through accumulation of weaker evidence types, consistent with IC analytic practice under ICD 203.

Confidence Assessment: MEDIUM-HIGH.

3.9 6-Month Verification Lag

Manual Section: 3.9

Category: STANDARD PRACTICE

Rationale: The 6-month lag trades recency for verification quality, aligning with established practice across multiple domains:

Domain	Standard Lag	Rationale
Financial infrastructure	90-180 days	Post-implementation credit ratings
Physical infrastructure	6-12 months	Operational history requirement
Policy/regulatory	6-12 months	Impact assessment period
Academic/think tank	6-12 months	Standard publication lag (OECD, World Bank)

3.9.1 Reduced Verification Lag for Verified Product Releases

Manual Section: 3.9.2, 3.15.13

Category: ADAPTED (MEDIUM-HIGH 60–75%)

Rationale: The verification lag performs several functions: implementation confirmation, reversal absorption, evidence accumulation, and third-party validation. For product releases, implementation confirmation is available at entry — general availability is directly observable in primary release documentation and public accessibility, without elapsed time. The provision distributes the remaining functions across its structure: a 60-day period absorbs launch-window reversals; a continuous-availability condition converts the period from passive delay into an affirmative stability requirement; and a mandatory maturity re-review at implementation date + 180 days verifies the record at the point it would have cleared the standard window. Events entering under the provision receive two verification passes where standard-window events receive one.

Parameter Derivation (60 days): A 2022–2026 census of Western and Chinese model releases classified post-release incidents into two populations. Integrity incidents — withdrawal, suspension, rollback, capability restatement, availability-affecting regulatory action — concentrate within 33 days of general availability (n = 10; nine of ten within 21 days; the longest availability interruption resolved within 22 days). Successor-driven lifecycle retirements occupy a separate 89–177 day band. The 60-day parameter covers the observed integrity range at 1.8× the latest incident whilst remaining below the lifecycle band, keeping the continuity condition a test of release integrity rather than release cadence.

Alternatives Assessed: A 90-day period enters the observed lifecycle band, exposing the continuity condition to scheduled retirements that carry no integrity information. Retaining the undifferentiated 6-month lag for this event class duplicates, through elapsed time, functions the entry verification and maturity re-review perform, and produces a systematic documentation gap in the corpus's fastest-cadence event class.

Precedent Basis:

Precedent	Structure
MSCI, "MSCI Global Investable Market Indexes Methodology" (February 2025), Sub-section 3.2.4.1 "IPOs and Other Early Inclusions". URL: https://www.msci.com/indexes/documents/methodology/0_MSCI_Global_Investable_Market_Indexes_Methodology_20250211.pdf	Default Minimum Length of Trading Requirement with early inclusion of large IPOs at approximately the tenth trading day
FTSE Russell, "FTSE Global Equity Index Series Ground Rules" v14.1 (May 2026), Rules 8.1.1–8.1.3; "FTSE Global Equity Index Series IPO Fast Entry Rule Update," index notice (26 May 2026). URL: https://research.ftserussell.com/products/index-notice/home/getmethodology?id=2619702	Fast entry on first-day-close capitalisation above the Fast Entry Level; implementation after the close of the fifth trading day
S&P Dow Jones Indices, "Consultation on Treatment of MegaCap Companies — Results" (4 June 2026). URL: https://www.spglobal.com/spdji/en/documents/indexnews/announcements/20260604-1483731/1483731_spdji-us-indices-megacaps-results-20260604.pdf	Retention of 12-month seasoning for the S&P 500 following consultation — a divergent outcome within the same design space
Moody's Investors Service, "Rating Symbols and Definitions," Provisional Ratings — (P) designation. URL: https://www.moodys.com/sites/products/ProductAttachments/Moodys%20Rating%20Symbols%20and%20Definitions.pdf	Provisional designation pending contingency fulfilment; removed on completion; withdrawn where issuance fails
Fitch Ratings, "Rating Definitions," Special Report (11 June 2024), Expected Ratings — (EXP) suffix.	Expected rating sensitive to finalisation of draft documentation; suffix removed and rating updated on receipt of final documentation
FDA Accelerated Approval, 21 CFR Part 314 Subpart H (§314.510; withdrawal §314.530)	Earlier availability on surrogate endpoints; mandatory confirmatory trials; withdrawal procedures on failure
EMA Conditional Marketing Authorisation, Regulation (EC) No 507/2006 (Article 4); Regulation (EC) No 726/2004 (Article 14-a)	One-year validity with annual renewal obligations; conversion to standard authorisation on fulfilment

The adapted structure is the three-part architecture — default period, conditioned early entry on a directly observable qualifying fact, and mandatory post-entry verification with unwind — applied to general-availability events. The 60-day parameter is WEO-calibrated from the incident census; the architecture is not WEO-invented.

Confidence Assessment: MEDIUM-HIGH (70–75%). Architecture: convergent precedent across index methodology, credit-rating practice, and pharmaceutical regulation, corroborated across independent research returns (~75%). Parameter: empirically derived from a small census (n = 10 integrity incidents) with open observation windows in the most recent release cohort (~70%).

Validation Commitment: Every event entering under §3.9.2 carries a §3.15.13 maturity re-review record; the first cohort of reduced-lag entries is reviewed as a set. Incident taxonomy (withdrawal / suspension / rollback / restatement / licence change / regulatory action / deprecation) is recorded per reviewed event. Parameter re-evaluation: upward to 90 days on any verified integrity incident occurring 60–180 days post-availability in the tracked population; downward revision considered only after 12 months with $n \geq 15$ tracked entries spanning both Western and Eastern release cohorts and all observed integrity incidents within 21 days.

3.10 Currency Windows

Manual Section: 3.4.8

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: Currency windows govern source publication dates (distinct from event implementation dates which are governed by the 6-month verification lag). These follow established financial reporting standards and regulatory requirements:

Data Type	Currency Window	Precedent
Infrastructure operational status	≤ 24 months	Financial reporting standards
Policy/regulatory status	≤ 12 months	SEC/Basel reporting requirements
Financial data	≤ 12 months	SEC/Basel reporting requirements
Technical specifications	≤ 36 months	Technical documentation practices

Interaction with 6-Month Verification Lag: Both requirements must be satisfied:

1. **Event implementation date:** Within verification window (Jan 2022 to 6 months prior to analysis)
2. **Source publication date:** Within currency window for data type

A source published today reporting on infrastructure completed 2 years ago is valid — the methodology intends current verification of established reality.

3.10.1 Conflict-Era Source Currency Adjustment

Manual Section: §3.4.12

Category: ADAPTED (MEDIUM-HIGH 60–75% to HIGH 75–90% — varies by source type; see below)

Rationale: Armed conflict produces a documented bifurcation in institutional documentation output. Emergency instruments accelerate to revision cycles measured in weeks; routine regulatory instruments decelerate or halt; infrastructure verification shifts from near-real-time documentary evidence to satellite imagery with inherent temporal lag; and financial reporting experiences filing delays proportionate to sanctions disruption and audit access constraints. These patterns are empirically documented across multiple conflict episodes and institutional frameworks.

The adjustment adapts the existing §3.4.8 currency windows to account for this bifurcation. It does not create a new evidentiary framework — it modifies temporal freshness parameters within the established source verification architecture.

Trigger framework:

The conflict-era trigger adapts the World Bank Fragile and Conflict-Affected Situations (FCS) dual-source classification methodology for WEO's documentation-impact purpose. The FCS framework — the most formally codified conflict classification in use — requires quantitative thresholds from both ACLED and UCDP before classifying a country as conflict-affected. WEO adapts this dual-gate principle: criterion 1 provides the external conflict classification (UCDP), whilst criterion 2 establishes the documentation-impact connection that justifies adjusting currency windows.

Precedent	Contribution
World Bank FCS (dual-source quantitative classification, FY24 methodology)	Structural precedent — external quantitative threshold + impact verification before classification activates
UCDP Armed Conflict Dataset (Gleditsch et al., 2002; codebook v.25.1)	Criterion 1 — externally maintained, quantified conflict classification with defined intensity levels
IMF World Economic Outlook database (ad hoc data exclusion, Syria from 2011)	Documented institutional response to conflict-era documentation gaps — validates the premise that conflict degrades documentation landscapes

The criterion 2 sub-criteria are not uniformly quantified. Sub-criterion (a) provides a hard threshold ($\geq 10\%$ chokepoint volume); sub-criteria (b) and (c) are qualitative assessments constrained by the evidence requirement in criterion 3. This asymmetry is documented honestly: no systematic quantitative measurement of regulatory publication rate depression during conflict exists in academic literature. The absence of this measurement was confirmed during deep research and represents an unfilled gap suitable for future research. The current calibration — qualitative assessment with mandatory documented evidence — is the methodologically honest response to that gap.

Adjusted currency ceilings — empirical basis by source type:

Source Type	Adjustment	Category	Confidence	Key Evidence
Emergency executive instruments	≤12mo → ≤6mo	EMPIRICALLY GROUNDED	~80%	EU sanctions: 6 packages in 101 days (17-day cycle, 2022); UNSC Gulf War: 12 resolutions in 153 days (12.7-day cycle, 1990); EPAA: major revision every 12–18 months
Routine regulatory instruments	≤12mo → ≤24mo	ADAPTED	~70%	Ukraine Business Ombudsman bill stalled; IMF Syria data exclusion from 2011; aggregate deceleration documented but not systematically measured
Infrastructure operational status	≤24mo → ≤36mo	ADAPTED	~72%	Kuwait Stock Exchange: 18–20 month closure; Iraq Stock Exchange: ~15 month closure; satellite verification lag weeks–months; ICRC "data vacuum" finding
Financial data	≤12mo → ≤18mo	ADAPTED	~70%	SEC Release 34-88465 (45-day COVID extension); Russia audit firm substitution delays; EU ban on accounting services to Russian entities (June 2022)
Technical specifications	Unchanged	N/A	N/A	No evidence of accelerated technical obsolescence during conflict

All adjusted ceilings exceed the 65% methodology component confidence threshold.

Key design decisions:

1. **Ceilings, not floors.** The adjustment modifies the maximum permissible source age, not the minimum. A newer source is always valid. This is consistent with the §3.4.8 design where the windows are upper bounds.
2. **6-month emergency instrument ceiling (not 3 months).** The observed emergency revision cycles are 2–8 weeks, which would support a ceiling as short as 3 months. The 6-month ceiling provides a conservative buffer. The specific calibration between 3 and 6 months is flagged for refinement pending analysis of EU sanctions revision content — whether successive packages substantively modified predecessors or merely extended them.
3. **CC temporal windows excluded.** Coordination Connection temporal windows (§9.4.1) govern the timing relationship between events, not source freshness. Conflict may accelerate coordination activity, but the temporal windows measure when events occurred relative to each other — this relationship is independent of how old the source documentation is.
4. **Territory-scoped application.** Adjusted ceilings apply to sources produced within or about the conflict-affected territory, not globally. This prevents a conflict in one region from loosening currency standards for sources from unaffected jurisdictions.
5. **Six-month review cycle.** Active determinations are reviewed at six-month intervals — long enough to avoid administrative churn, short enough to deactivate determinations when conditions change. No external precedent governs this interval; it is a CATEGORY CREATOR calibration.

Confidence Assessment: MEDIUM-HIGH to HIGH overall (70–80%). Emergency instrument shortening is the strongest empirically grounded component (~80%); routine instrument and infrastructure extensions are ADAPTED with moderate confidence (~70–72%); financial data extension is ADAPTED (~70%). The trigger framework is ADAPTED from the World Bank FCS methodology with MEDIUM-HIGH confidence (70–75%). The six-month review interval is CATEGORY CREATOR.

Validation Commitment: Review adjusted ceilings against the first conflict-era determination. Monitor whether the 6-month emergency ceiling proves too permissive or too restrictive. Review trigger threshold if UCDP revises its intensity classification methodology. Monitor academic literature for systematic regulatory publication rate measurement during conflict — if published, use to recalibrate the routine instrument extension.

3.11 Non-English Source Verification

Manual Section: 3.3

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: Enhanced verification protocols apply to non-English source events due to:

1. **Language barriers:** WEO's predominantly English-language research base affects efficient navigation and filtering of primary sources in other languages
2. **Translation verification:** Non-English documentation requires additional corroboration to ensure accurate interpretation
3. **Verification standards:** Intelligence community multi-source validation precedent

These protocols apply to any event where primary sources are not in English, regardless of bloc — including Chinese, Russian, French, German, Japanese, Arabic, and other non-English documentation.

Primary Application — Chinese-Language Sources:

Chinese-language sources represent the most common non-English verification scenario for AI infrastructure events.

- Official State Council documents (english.gov.cn) OR Ministry-level announcements (MIIT, CAC, NDRC) AND
- Third-party confirmation from: World Bank/ADB project databases, UN agency reports, peer-reviewed academic papers, or trade statistics cross-check (China Customs + UN Comtrade)

Infrastructure Claims: State announcement + satellite imagery OR budget allocation + construction permits OR vendor contracts + operational evidence

Financial Claims: Official announcement + IMF/World Bank country data OR China Customs + UN Comtrade cross-verification

Other Non-English Sources:

The same enhanced verification principles apply to all non-English sources regardless of bloc:

- Official government sources in original language + English-language confirmation OR
- Academic research with primary source citations + multilateral organisation data OR
- Trade statistics + third-party reporting from internationally recognised outlets

Precedent Citations:

- ICD 203/206 (Intelligence Community Directive — multi-source validation)
- UK PHIA (Professional Head of Intelligence Assessment)
- OSINT Guidelines 2023
- Jane's methodology (defence intelligence)

3.11.1 CJK Language Conventions

Manual Section: §3.3.1

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: The four-pattern taxonomy addresses inconsistencies identified during event processing where CJK text appeared without a governing standard — truncated operative terms, inverted ordering (CJK-first vs English-first), and commercial brand names with no methodology-functional role. The taxonomy provides a testable rule for each use case rather than relying on editorial judgement.

Pattern A (verbatim citation) preserves evidentiary integrity for CC qualification. Pattern B (methodology-functional terms) retains operative CJK where removal would degrade analytical precision — particularly for Chinese regulatory citation markers (根据, 关于, 印发) that carry specific legal meaning in CC evidence chains. Pattern C (named programmes) follows standard English-language academic convention for foreign-language proper nouns. Pattern D (direct quantities) preserves source fidelity for numeric evidence.

The methodology-functional test provides a binary decision rule that prevents CJK text from accumulating without analytical purpose whilst preserving it where it serves the methodology.

Confidence: 75%. The taxonomy is grounded in established multilingual documentation practice (ISO 704 terminology principles, academic transliteration conventions) adapted for WEO's specific processing context. Applied retrospectively to existing database entries with consistent results.

Validation commitment: Monitor for CJK patterns that do not fit the four categories. If a fifth pattern emerges from processing, extend the taxonomy. Review at methodology version milestones.

3.12 Bloc Classification

Manual Section: §3.10 (Bloc Classification)

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: V1.3 §3.12 combined two distinct classification processes — macro-level bloc membership and event-level Non-Aligned attribution — in a single section without explicit separation. This created interpretive ambiguity: the three-priority hierarchy (with financial backing at Priority 1) could be mistaken for the method by which macro-level register classification decisions are made, when in fact macro-level classification is determined by technology ecosystem alignment.

The two processes answer different questions at different analytical scales. Macro-level classification asks "which ecosystem does this nation's AI infrastructure structurally depend on?" — technology ecosystem alignment is the correct primary signal because it captures structural lock-in (switching a chip supply chain takes years and billions of dollars; switching a funding source for a single project takes months). Event-level attribution asks "for this specific event, which bloc's ecosystem does it operate within?" — financial backing is the correct Priority 1 because at the individual event level it provides an objective, measurable, reproducible signal that can be calculated as a percentage. Technology determination (Priority 2) requires more interpretive judgement, so it serves as the fallback when money does not resolve the attribution.

The priority ordering is not contradictory — it is appropriate to each analytical scale. The separation makes this explicit.

Confidence: HIGH (80%). The underlying analytical processes are operationally stable. No classification outcome changes.

3.12.1 Universal Membership Categories

Manual Section: §3.10.1 (category definitions)

Category: ADAPTED (HIGH 75–90%)

Source methodology: Systematic survey of eight international tiered classification frameworks: NATO partnership tiers, EU accession framework, FATF grey/black list classification, OECD accession framework, Basel Committee RCAP, Shanghai Cooperation Organisation membership tiers, Non-Aligned Movement (historical), and contemporary technology alliance frameworks (Pax Silica, Chip 4, AISI Network, Wassenaar Arrangement).

Adaptation rationale: No existing framework offers a symmetric, universal set of membership categories applied identically to competing technology ecosystems. Every surveyed operational framework is anchored by one hegemon, with rivals as residual or excluded. The five-category universal structure (Core / Integrated / Engaged / Peripheral / Non-Aligned) is WEO's synthesis of transferable design primitives from these frameworks, adapted for technology ecosystem alignment classification.

Key design decisions:

1. **Five-category count.** The modal tier count across the eight surveyed frameworks is five $\pm$ one. NATO functionally operates with three sub-member tiers plus the member tier; OECD maintains five tiers; EU has four substantive tiers. Five categories provides sufficient discrimination to capture the spectrum from structural embedding (US in Western ecosystem, 7/7 SCP dimensions as of May 2026) to single-vector connection (e.g., governance participation without infrastructure dependency) without classification noise from finer subdivision.
2. **Strength over mechanism.** Categories describe alignment strength (how deeply integrated), not alignment mechanism (how the nation entered the ecosystem). This follows Basel RCAP's design — the only surveyed framework that cleanly separates membership status from compliance assessment. The mechanism (formal agreement, voluntary technology choice, infrastructure lock-in, political alliance, governance participation) is recorded as a descriptive attribute within each nation's register entry. This resolves the V1.3 asymmetry where Western categories described mechanisms (Conditional, Functional, Governance) and Eastern categories described dependency types (Aligned, Dependent, Leaning).
3. **Symmetry.** Identical definitions, indicators, and distinguishing criteria apply to the Western Bloc, Eastern Bloc, and any future bloc. No category definition references bloc-specific institutions, values, or political systems. This is a genuine methodological contribution — no surveyed operational framework achieves this level of symmetry across competing blocs. The FATF methodology is symmetric by design but asymmetric in application (no G7 grey-listings despite documented deficiencies); the WEO structure is symmetric by both design and definition.
4. **Non-Aligned as a dynamic status.** Non-Aligned is defined by the absence of predominant bloc attribution, verified against observable indicators, not by political declaration. This addresses the Non-Aligned Movement's historically demonstrated failure mode, where formal non-alignment masked operational alignment (Cuba, India-USSR 1971, Vietnam). The dual assessment (declared policy + observable dependency) follows the FATF dual-axis precedent (technical compliance + effectiveness).

Precedent sources:

Precedent	Institution	Design primitive adopted
RCAP four-grade compliance scale	Basel Committee (BIS)	Separation of mechanism from strength; weakest-link aggregation
EU accession framework (2020 revised methodology)	European Union	Phased, proportionate, reversible transitions; cluster organisation
FATF mutual evaluation methodology	FATF	Dual-axis assessment (declared + de facto); action-plan escalation
OECD accession framework	OECD	Graduation pipeline; values filter; Key Partner vs Candidate distinction
NATO partnership tiers	NATO	Three-tier sub-member structure (MAP, EOP, PfP)
Non-Aligned Movement (1961–present)	NAM	Cautionary precedent: declared vs de facto alignment

Confidence: 75%. The five-category structure is grounded in systematic precedent survey across eight frameworks, with convergent evidence supporting the tier count, strength-over-mechanism design, and symmetry principle. The specific indicators for each category involve analytical judgement that will be refined through continued application. The 65% minimum confidence threshold (§2.3) is met.

Validation commitment: Reassess category structure at methodology version milestones. Monitor whether the five-category discrimination remains adequate as the bloc landscape evolves (particularly if a third bloc forms). Review indicator sets after three register snapshots to identify any systematic classification difficulties.

3.12.1a Bloc Assignment Gate

Manual Section: §3.10.1 (Bloc Assignment Gate subsection)

Category: ADAPTED (HIGH 75–90%)

Rationale: The two-track bloc assignment gate fills the architectural gap between the general classification principle ("every nation is classified by which ecosystem its AI infrastructure primarily operates within") and the within-bloc category assignment. In V1.3, Non-Aligned actors had a formalised identification mechanism (the primary partner test: financial backing → technology transfer → strategic initiative leadership), but actors assigned to the Western or Eastern blocs had no equivalently formalised assignment process. The bloc assignment was qualitatively sound but procedurally implicit.

Track 1 formalises the observable infrastructure alignment assessment that was already being applied in practice. The three indicators (hardware origin, compute/model platforms, regulatory alignment) are the same signals that informed every existing Bloc Membership Register entry. Track 2 extends the assessment to the full seven-dimension dependency direction analysis for contested cases where the observable indicators are ambiguous or contradictory. The seven-dimension framework draws directly from the Sovereign Capability Profiles (§5.3.2), applied here as a dependency direction test rather than a capability assessment — asking "whose ecosystem does this actor draw on?" rather than "what does this actor independently control?"

The dead heat rule (exact dependency split = Non-Aligned) follows from the operational definition of Non-Aligned status: the absence of predominant alignment to either ecosystem. If a structured dependency analysis across all seven dimensions produces an exact split, the actor has demonstrated through its infrastructure choices that it is not structurally embedded in either ecosystem — which is precisely what Non-Aligned status describes. Event-level attribution for such actors uses the existing three-priority hierarchy (§3.12.3).

The near-dead-heat rule (narrow majority constrains category to Peripheral or Engaged) ensures that the category assignment reflects the strength of structural embedding. An actor with a 4–3 dependency split has a measurably weaker ecosystem attachment than an actor with a 7–0 or 6–1 split. Assigning such an actor to Core or Integrated would overstate the depth of its embedding.

Closest Precedents:

Framework	Relevance	Divergence from WEO
EU SWD(2021) 352 Strategic Dependencies	Single-source dependency identification across ~5,000 products using HHI concentration metrics	Product-level, not actor-level; measures import concentration, not ecosystem alignment
IISS Cyber Capabilities (2021/2023)	Multi-dimensional capability assessment producing tiered classifications	Does not assess ecosystem dependency direction; classifies capability, not alignment
NATO Partnership Tiers	Graduated alliance membership based on integration depth	Security alliance, not technology ecosystem; membership is treaty-based, not infrastructure-based
FATF Grey/Black List	Classification based on observable compliance indicators with dual assessment (declared vs actual)	Financial compliance, not technology alignment; binary pass/fail rather than graduated categories

Key Design Decisions:

1. **Two-track sequential system rather than universal seven-dimension analysis.** Track 1 resolves the vast majority of registered nations through observable indicators that do not require a full SCP-style dependency analysis. Track 2 is reserved for a small minority of genuinely contested cases. This keeps the assessment burden proportionate: the full seven-dimension analysis is analytically powerful but resource-intensive, and applying it universally to nations whose alignment is unambiguous would be methodologically unnecessary.
2. **Dead heat = Non-Aligned, not forced assignment.** The alternative — a forced tiebreaker that assigns a perfectly split actor to one bloc — would impose a false precision that the data does not support. An actor whose infrastructure dependencies are exactly evenly distributed between ecosystems is telling the assessor something real about its structural position. Forcing a bloc assignment would obscure this signal. Non-Aligned status with event-level attribution preserves the analytical information.
3. **Dimension durability tiebreaker as conditional empirical heuristic.** The default hierarchy (hardware > platform > governance) is grounded in documented switching costs: semiconductor fabrication and equipment dependencies exhibit switching timelines measured in decades, whilst governance alignment can shift within a single political cycle. However, the foundational lock-in literature (Shapiro & Varian 1999, Table 5.1) demonstrates that hardware lock-in declines with time as assets depreciate whilst information/platform lock-in rises — the hierarchy is empirically observed for frontier AI infrastructure in 2026, not a universal law. The four override conditions ensure the tiebreaker adapts to cases where the default ordering breaks down.

Evidence Basis:

Source	Type	Contribution
Shapiro, C. & Varian, H. R. (1999). <i>Information Rules</i> , Ch. 5, Table 5.1	Direct	Canonical seven-category lock-in taxonomy; documents time-dynamics showing hardware lock-in declining and information lock-in rising — grounds the conditional framing
Allen, G. C., et al. (2024). "The True Impact of Allied Export Controls." CSIS	Direct	Documents ~10-year minimum Chinese EUV substitution gap despite US\$150B+ investment — direct evidence that frontier hardware switching costs exceed platform/governance alternatives
Farrell, J. & Klemperer, P. (2007). "Coordination and Lock-In." <i>Handbook of Industrial Organization</i> , Vol. 3, Ch. 31	Direct	Canonical survey distinguishing switching costs from network effects; does not propose a strict hierarchy by magnitude — reinforces the need for conditional framing
UK OBR Brexit Analysis (2020–2025)	Empirical	Institutional gravity: 5+ years, 4% GDP impact, still unable to meaningfully diverge — grounds the institutional gravity override
GAIA-X / EU cloud sovereignty (2019–2026)	Empirical	Failed platform displacement: 7 years, 43 billion euros adjacent spending, US hyperscaler share held at ~70% — grounds the network-effect platform override
Apple x86→ARM transition (2020–2023)	Empirical	Vertically-integrated hardware switching in 35 months — grounds the vertical integration override
Korea-Japan photoresist dispute (2019–2020)	Empirical	Commodity-input substitution in ~6 months — grounds the commodity-input override

Confidence Assessment: 75%. The two-track system formalises existing practice (Track 1) and extends it with established WEO analytical tools (Track 2) for contested cases. The dead heat and near-dead-heat rules follow logically from the Non-Aligned definition and the graduated category structure. The dimension durability tiebreaker is the least certain element — it is grounded in documented switching costs and three primary academic sources but is explicitly contested by the foundational lock-in literature at long time horizons. Confidence is bounded below 80% because the tiebreaker's four override conditions require assessor judgement about when to apply them. Confidence is above 70% because Track 1 and Track 2 are methodologically straightforward, the dead heat/near-dead-heat rules are deterministic, and the tiebreaker is needed only in edge cases.

Validation Commitment: Monitor the dimension durability hierarchy against observed infrastructure transitions. If frontier hardware switching costs decline below platform switching costs (e.g., through successful domestic EUV programmes or alternative lithography development), revise the default ordering. Review the four override conditions after each Bloc Membership Register update to identify any cases where the conditions were invoked and whether they produced analytically sound results. Reassess the tiebreaker calibration at methodology version milestones.

3.12.1.1 Evidence Standard for Register Classification Changes

Manual Section: §3.10.1.1

Category: ADAPTED (HIGH 75–90%)

Source methodology: §5.3.4 Sovereign Capability Profiles evidence standard (1 Primary Grade 1/2 + 1 Corroborating Grade 2/3), adapted for bloc membership classification context.

Adaptation rationale: The SCP evidence standard was designed for dimension-level assessment decisions — determining whether an actor meets a specific capability dimension. Bloc membership classification decisions are analogous: they determine whether a nation meets the indicators for a specific alignment category. The same evidence architecture (primary + corroborating, with defined grade requirements per §3.1) applies. The minimum standard (1+1) is appropriate because bloc classification changes are lower-stakes than SCP dimension assessments — a bloc classification does not carry the same analytical weight as a PAA or AIK designation.

Gap addressed: V1.3 did not specify an evidence standard for register classification changes. The register's own documentation stated it was "evidence-based" and required "source citation," but no section of the Manual defined what quality or quantity of sources was required. This created a circular reference: the Manual delegated to the register, but the register had no formal standard to comply with. §3.12.1.1 closes this gap.

Retrospective audit provision: Snapshot 2025.12.14-001 was produced prior to the formalisation of this standard. The retrospective audit provision acknowledges this honestly without backdating the standard. Source documentation for that snapshot has been supplemented to meet the §3.12.1.1 standard; classifications were not altered.

Confidence: HIGH (80%). Directly adapted from the evidence standard defined in §5.3.4 with minimal modification.

3.12.1.2 Register Operations

Manual Section: §3.10.1.2 (versioning, archival, update triggers)

Category: ADAPTED (HIGH 75–90%)

Rationale: The register's operational framework addresses three design decisions: version identification, source preservation, and update cadence.

The version identifier format (YYYY.MM.DD-NNN) provides unambiguous temporal ordering and supports multiple snapshots per day during periods of rapid geopolitical change. This follows standard document management convention adapted for a living register that updates at irregular intervals.

The dual archival strategy (Internet Archive + OpenTimestamps) serves complementary purposes: Internet Archive provides public accessibility and content retrieval, whilst OpenTimestamps provides cryptographic proof of document state at a specific point in time. The combination ensures that any register snapshot can be independently verified as unaltered after publication — a requirement for institutional-grade data integrity.

The event-triggered update model (snapshots taken when geopolitical developments warrant reclassification, not on a fixed calendar) reflects the nature of the underlying data: AI infrastructure ecosystem alignment shifts in response to discrete geopolitical events (export controls, bilateral agreements, infrastructure deployments), not on quarterly or annual cycles. Calendar-triggered snapshots would either capture periods of no change or miss significant developments between scheduled dates.

Confidence: 85%. All three design choices adapt established operational practices (document versioning, digital archival, event-driven data management) to the register context. No novel methodology is introduced.

Validation commitment: N/A — operational format, not analytical methodology.

3.12.1.3 Third Bloc Formation — Transitional Classification

Manual Section: §3.10.1.3

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

Rationale: The two-bloc structure (Western, Eastern) reflects the empirical reality at V1.5 publication — only two actors satisfy the PAA qualification threshold (§5.3). However, the methodology must accommodate the possibility of a third bloc forming without requiring architectural revision. The most likely formation scenario involves an actor currently classified within an existing bloc approaching PAA qualification through independent sovereign capability accumulation across multiple SCP dimensions, causing a structural split between the new and original blocs.

The structural problem: If a third bloc forms from within an existing bloc, nations previously classified as Core in the original bloc would not cleanly transfer to the new bloc. Their ecosystem dependencies would be split — some dimensions aligning with the new bloc (e.g. regulation, investment, equipment), others remaining with the original (e.g. chip design, compute, models). This is structurally different from Non-Aligned status, which reflects strategic choice or neutrality. Split-alignment during bloc formation reflects unavoidable structural transition where the infrastructure dependencies themselves are divided.

Transitional mechanism rationale: Reclassifying structurally split nations as Non-Aligned during the transition period — and applying the three-priority hierarchy (§3.12.3) per-event — repurposes an existing, tested mechanism for a new scenario. This avoids the need for a new classification framework whilst providing event-level accuracy. The three-priority test is well-suited because it assesses each event individually against measurable criteria (financial backing, critical technology, strategic leadership), which correctly handles the situation where a nation's events may genuinely belong to different blocs depending on which ecosystem capabilities are utilised.

Stabilisation criterion rationale: "Predominance across the seven SCP dimensions" rather than a simple dimensional count (e.g. 4/7) was chosen because dimensions are not equally weighted in practice. Hardware dependencies (D1–D3) create deeper structural lock-in than regulatory alignment (D6), which can shift with a single legislative act. The stabilisation assessment must consider the structural weight of dependencies, not merely count them. This judgement is left to the analyst rather than algorithmically specified, consistent with the indicator-based assessment approach used throughout §3.10.1.

Category accommodation rationale: During the transition period, the nation is classified as Non-Aligned and events are attributed per the three-priority hierarchy. The analyst tracks the dependency pattern across both blocs to assess when stabilisation occurs. Upon reclassification to a specific bloc, the category level reflects the strength of alignment at that point — a nation emerging from transition is unlikely to qualify as Core immediately, as residual dependencies on the former bloc would typically place it at Engaged or Peripheral. The expectation that alignment will stabilise over time is grounded in the economic logic of supply chain consolidation — maintaining parallel dependencies across two ecosystems is costly and strategically unstable.

Scope limitation rationale: Full multi-bloc methodology (formal dimensional dependency assessment, weighted scoring) is deferred rather than attempted. The transitional mechanism is sufficient for initial classification and avoids over-engineering a solution for a scenario that has not yet occurred. If a third bloc forms, the specific dimensional profile of the new PAA and the specific dependencies of affected nations will inform the full methodology — designing it in the abstract risks producing a framework mismatched to the actual scenario.

Confidence: 65%. This is the minimum confidence threshold (§2.3). The transitional mechanism is logically sound and repurposes tested components (Non-Aligned classification + three-priority test), but it has not been empirically tested against a real third-bloc formation scenario. Confidence reflects this untested status. Reassess when a third PAA candidate approaches qualification.

3.12.2 Non-Aligned Nation Identification

Manual Section: §3.10.2 (five-criteria test)

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: The five-criteria test provides an auditable, explicit identification mechanism for Non-Aligned status, establishing the relationship between macro-level NA classification and event-level attribution.

Dual assessment provision: The requirement that Non-Aligned status be assessed against both declared policy and observable infrastructure dependency addresses a documented failure mode. The Non-Aligned Movement historically demonstrated that declared non-alignment does not reliably predict actual ecosystem dependency (Cuba maintained non-aligned status whilst operating entirely within the Soviet ecosystem; India signed the Indo-Soviet Treaty of Friendship in 1971 whilst remaining NAM chair). WEO's dual assessment follows the FATF dual-axis methodology (technical compliance + effectiveness) as the closest operational analogue for distinguishing what nations declare from what they do.

Confidence: HIGH (80%). The five-criteria test and the dual assessment provision are operationally stable — no classification outcome changes.

3.12.3 Event-Level Attribution Scope Limitation

Manual Section: §3.10.3 (scope paragraph)

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: The explicit scope limitation — stating that the three-priority hierarchy applies only to event-level attribution for Non-Aligned nations, not to macro-level register classification — resolves the interpretive ambiguity identified in the V1.3 structure. The rationale for the different priority ordering at each scale is documented in §3.12 and in this rationale entry (§3.12 above).

The three-priority hierarchy operates as documented in the Methodology Rationale (§3.12) and the AI model events clarification (§3.12.4), which remains applicable in full.

Confidence: 75%. The scope limitation formalises what was implicitly true in V1.3 — the three-priority test was always designed for event-level classification, not macro-level register decisions. No operational outcome changes. The risk is that explicit scope limitation could be misread as narrowing the test's applicability; the Manual text addresses this by clearly stating the test's purpose within its defined scope.

3.12.4 AI Model Events — Critical Technology Determination

Manual Section: §3.10.3 (Priority 2 — Critical Technology)

Category: ADAPTED (HIGH 75–90%)

Source methodology: US Bureau of Industry and Security (BIS), Export Administration Regulations (EAR), 15 CFR Parts 730-774.

Adaptation rationale: BIS EAR establishes two principles relevant to AI model bloc attribution:

1. Development location determines technology origin (15 CFR § 734.3)
2. Cloud computing access does not change underlying technology origin (15 CFR § 734.18)

WEO adapts these principles for bloc attribution: the headquarters jurisdiction of the entity that developed the base model weights determines which bloc provides "critical technology" under Priority 2. Deployment infrastructure (cloud provider, hosting location) is not determinative.

Derivative threshold: The >50% new parameters threshold is WEO-original calibration for determining when derivative work constitutes genuinely new technology rather than adaptation. No direct external precedent — this is a category creator sub-component within an otherwise adapted framework.

Confidence: HIGH (85%) — core attribution principle directly adapted from BIS EAR. Derivative threshold is category creator calibration subject to refinement.

3.13 Event Evolution & Succession Protocol

Manual Section: 3.14

Category: ADAPTED (HIGH 75–90%)

Rationale: The transformation-scaling distinction determines whether evolving events warrant new record creation or version history updates. This institutional-continuity approach aligns with established practice across infrastructure databases, treaty registries, and financial platforms.

Core Principle — Transformation vs Scaling:

Transformation (New Event)	Scaling (Rolling Update)
Legal instrument changes	Same legal instrument, expanded scope
Party composition changes	Same parties, increased commitment
Classification changes (Tier, Bloc, Domain, Type)	Classification stable
Entity splits or merges	Single entity persists
Fresh authorisation required	Existing authorisation covers expansion

Precedent — Industry Practice:

Database	Record Creation Trigger	Scaling Handling
World Bank PPI	New contract	Update existing record
Global Energy Monitor	New construction phase	Update wiki/phase record
IJGlobal	New financing event	Same asset record
UN Treaty Collection	New legal regime requiring fresh consent	Amendment attaches to parent
Bloomberg/Refinitiv	New deal (distinct transaction)	Same deal record through all revisions

Numerical Thresholds as Review Triggers:

The 2× budget and 50%+ capacity thresholds function as mandatory review triggers rather than automatic new-event creation:

Threshold	Function	Outcome Determined By
2× budget (100% increase)	Triggers human review	Transformation-scaling criteria
50%+ capacity increase	Triggers human review	Transformation-scaling criteria

Rationale: Quantitative scaling within the same institutional framework represents continuous evolution, not transformation. A coordination mechanism (e.g., a federal award) may remain unchanged whilst associated private investment components scale significantly. New records for such scaling would fragment analytical continuity without corresponding analytical benefit.

Succession Modelling:

When genuine transformation warrants new event creation, explicit bidirectional succession relationships preserve analytical continuity. Standard succession is recorded via paired `superseded_by` and `succeeded_from` fields. Entity division or consolidation is recorded via paired `split_into` and `merged_from` fields.

Confidence: HIGH (85%)

Precedent Sources:

- World Bank PPI Database methodology
- Vienna Convention on the Law of Treaties (amendment vs new treaty distinction)
- Bloomberg/Refinitiv deal tracking methodology
- Temporal database literature (Snodgrass, Kimball SCD patterns)

3.13.1 Corporate Product Release Diagnostic

Manual Section: 3.14.1 (Corporate Product Release Supplement)

Category: ADAPTED (MEDIUM-HIGH 60–75%)

Rationale: The transformation-scaling criteria in *Manual Section 3.14.1* were designed for events with identifiable legal instruments, sovereign parties, and regulatory authorisations — characteristics inherent to Coordination, Policy, and Infrastructure event types. Corporate product releases lack these institutional structures. A model release has no treaty, no sovereign parties, and no regulatory authorisation in the sense the criteria intend. Under strict application of the unmodified institutional criteria, successive product releases from the same developer would always resolve as rolling updates (all answerable questions indicate scaling; inapplicable questions provide no transformation signal), regardless of the technical and commercial magnitude of change between generations.

The corporate product diagnostic adapts the same principled boundary — change in kind versus change in degree — to characteristics meaningful for product releases: product identity (is this a separately identifiable product?), technical independence (does this represent independent development or incremental refinement?), and market relationship (does this product coexist with or replace its predecessor?). The boundary remains institutional in the Bloomberg/Refinitiv sense: a "new deal" versus "same deal through all revisions" distinction applied to technology product context.

Adaptation Source: The Bloomberg/Refinitiv deal-tracking methodology cited in §3.13 distinguishes new issuances from amendments to existing issuances. The corporate product diagnostic applies the same logic: a ground-up technical rebuild with independent documentation constitutes a new record, whilst an optimisation variant within the same architecture constitutes an update to the existing record. The underlying principle — institutional continuity determines record architecture — is preserved; only the indicators of continuity differ by event type.

Empirical Validation: Corpus analysis of 17 corporate model events across five sequential processing cycles during the initial database population phase (January–April 2026) confirmed the replacement criteria produce outcomes consistent with all existing database determinations. The framework correctly identifies genuine product transformations (architecture class changes, independent training runs, parallel product lines) whilst routing version updates (weight refreshes, inference optimisations, checkpoint updates) to rolling update status. Two validation scenarios involving closed-source products with limited public technical documentation correctly produced "flag for human review" outcomes — confirming the framework does not auto-resolve genuinely ambiguous cases.

Closed-Source Interaction: Developers who do not publish architecture details may systematically produce Dimension B = Ambiguous outcomes for successor releases. This is an honest interaction with Structural Visibility Asymmetry (*Manual Section 4.11*), not a framework deficiency. The asymmetry in public documentation between open-source and closed-source releases means that open-source succession cases are more readily resolved, whilst closed-source cases more frequently require human review. This mirrors the broader SVA pattern documented in the Manual.

Confidence: MEDIUM-HIGH (70%) — adapted from established precedent (Bloomberg/Refinitiv deal-tracking distinction, already validated in §3.13) with empirical validation against the full corporate event corpus. The specific indicators for Dimension B (what constitutes "independent" versus "incremental" technical development) involve analytical judgement that will be refined through continued application. The 65% minimum confidence threshold (*Section 2.3*) is met.

Precedent Sources:

- Bloomberg/Refinitiv deal tracking methodology (via §3.13)
- WEO processing corpus: 17 corporate model events, 5 processing cycles (January–April 2026)
- CC Type 7 corporate product release exclusion (*Manual Section 9.4.3*; *Rationale §9.5.1*) — establishes precedent for event-type-specific treatment of corporate product releases within the existing methodology

3.14 Post-Qualification Verification

Manual Section: 3.15

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: Events evolve after initial database entry. Infrastructure projects expand, policy frameworks are amended, corporate initiatives scale. The Rolling Updates system addresses this through three mechanisms, each grounded in established practice.

6-Month Lag for Updates

The same 6-month verification standard governing initial qualification applies to post-entry developments. Research on major technology infrastructure announcements documents that 40–60% of large capital projects experience significant delays, modifications, or cancellations. The 6-month window captures immediate failures whilst validating actual implementation.

This aligns with institutional verification practice:

Domain	Standard Verification Lag	Source
Credit ratings	90-180 days post-implementation	Rating agency methodology
Infrastructure bonds	6-12 months operational history	Bond prospectus requirements
Policy effectiveness	6-12 months post-enactment	Legislative impact assessment
Academic infrastructure data	12-24 months	OECD, World Bank methodology

Genesis Data Preservation

Genesis data — the original qualified values at database entry — remains immutable. This provides permanent reference for qualification basis, Coordination Connection verification provenance, historical analysis, and audit trail integrity.

This principle derives from financial data versioning standards where baseline values must be preserved alongside current values for compliance and auditability.

Two-Tier Architecture

The Genesis/Current data separation enables systematic evolution tracking without compromising historical integrity. Current data reflects the latest verified state; Genesis data documents what qualified the event for inclusion. This architecture mirrors version control patterns in compliance-grade financial systems.

Confidence: HIGH (85%)

Integration with Event Evolution

The rolling updates system operates alongside the Event Evolution Protocol (*Manual Section 3.14*). When pending updates are verified, they undergo transformation-scaling assessment:

- **Scaling** (same institutional framework): Incorporate via rolling update
- **Transformation** (classification change, party change, legal regime change): Route to Section 3.14 for succession handling

This integration ensures that quantitative growth within stable institutional frameworks is captured through version history, whilst genuine transformations receive appropriate new-event treatment with explicit succession relationships.

3.15 Source Accommodation and Event Verification Alignment

Manual Sections: §3.1 (Source Accommodations), §3.4.11 (Verification Assessment for Accommodated Sources), §3.5 Option D (OEP dual-purpose evidence), §3.5.1 (PIET dual-purpose evidence), §4.11 (Relationship to Event Verification)

Category: ADAPTED (HIGH 75–90%)

Rationale: Documentation-based methodologies produce structural gaps where standard primary source categories do not cover the evidence class required for verification. The Manual addresses four such gaps through source accommodations — expanded primary source definitions for specific event classes. Each accommodation specifies qualifying conditions at its point of definition (§4.11 — two accommodation types, §3.5, §3.5.1). The source accommodation alignment formalises the relationship between these accommodations and §3.1 event verification requirements, and establishes confidence assessment guidance for events verified through accommodated sources.

OEP and PIET dual-purpose evidence. OEP pathway evidence (independent technical laboratory analysis) and PIET gate evidence (trade flow data, commodity prices, government responses, corporate disclosures) each establish AI connection at Stage 2. The same evidence inherently verifies event existence — the analysis that confirms AI-relevant architecture also confirms the manufacturing milestone; the trade flow disruption that confirms AI supply chain impact also confirms the restriction's implementation. The dual-purpose role follows from the evidence structure and requires no additional qualification gate.

§4.11 state media accommodation. Regulatory process architectures that produce structurally less public documentation than jurisdictions with extensive regulatory transparency frameworks — no published regulatory impact assessments, no published consultation responses, shorter operative texts — create a verification gap where standard §3.1 primary source categories (government official documents via .gov domains, regulatory filings, multilateral publications) may not be available for Tier A claims. State media with official documentary function in the issuing jurisdiction — outlets operating with documented state editorial oversight — bridges this gap when reporting named official statements from identified government authorities. The qualifying conditions (official documentary function, named official attribution, verifiable institutional role) adapt established journalism verification practice for source attribution in state-controlled media environments.

§4.11 convergent implementation evidence accommodation. The state media accommodation addresses the regulatory process architecture gap where a named official statement is reported through an outlet with documented state editorial oversight. A second evidence configuration arises where the instrument itself is unpublished — issued as internal administrative guidance or communicated through informal regulatory channels — and no state media report meets the state media accommodation's conditions. The convergent implementation evidence accommodation addresses this configuration by accepting multi-stream convergent evidence as primary evidence when the streams include an accountability-bearing disclosure subject to securities regulation. The accountability-bearing disclosure provides the primary source anchor: the entity filing the disclosure is subject to regulatory penalty for material misstatement, which supplies an institutional accuracy incentive structurally distinct from editorial oversight. The accountability-bearing disclosure maps to the §3.1 primary source role; the remaining convergent sources map to the §3.1 corroborating source role. Official trade data and customs enforcement actions, classified as supplementary sources under §3.1, are accepted in the corroborating role within this accommodation — paralleling PIET's acceptance of trade flow evidence as gate-qualifying data (§3.5.1). The five qualifying conditions (Manual §4.11, Methodology Response) are conjunctive. A separate exhaustion requirement makes the accommodation available only after all other pathways — §3.1 verification and Stage 2 qualification — have been assessed and found inapplicable. The AI-relevance test for the accountability-bearing disclosure adopts the Option E equivalence formulation: the disclosure must characterise the event in terms that would satisfy the AI Connection Filter if the same language appeared in the instrument's operative text. This applies the same Stage 2 threshold to unpublished-instrument events as published-instrument events face. The convergent evidence simultaneously establishes event existence and AI infrastructure relevance, following the dual-purpose evidence pattern documented for OEP and PIET above.

Closest Precedents:

Framework	Institution	Relevance	Divergence from WEO
ICD 203 Analytic Standards	US Intelligence Community	Requires explicit disclosure of single-source dependence; distinguishes assessments based on converging vs. single-source intelligence	Intelligence assessment context, not geopolitical event verification
FATF Mutual Evaluation Methodology (2013/2022)	FATF	Graduated effectiveness assessment — single-channel evidence receives lower rating than multi-channel convergent evidence	National regulatory system assessment, not individual event verification; four-point scale, not binary ceiling
Reuters Handbook of Journalism	Reuters	Source attribution requirements scale with claim significance; accommodation of non-standard sources requires explicit flagging and editorial justification	Journalism, not methodology documentation; no formal ceiling mechanism

Key Design Decision — Confidence Term Ceiling:

Tier A claims verified exclusively through an accommodated pathway (sole basis) carry a confidence term ceiling of "Officially reported" (Manual §3.4.5) rather than "Confirmed / Verified." This applies the existing §3.4.5 confidence language framework to specific evidence configurations. "Officially reported" is defined as evidence "stated by authoritative primary source with independent corroboration" (1P + 2C) — the structural match for sole-reliance on an accommodated primary source supplemented by corroborating coverage. Where standard primary sources independently verify the same claims, the ceiling does not apply.

Source type	Confidence ceiling	Rationale
Standard §3.1 primaries	No ceiling	Converging independent documentation streams
§4.11 state media (sole basis)	"Officially reported"	Authoritative primary source with corroboration, not independent multi-source confirmation
§4.11 convergent implementation evidence (sole basis)	"Officially reported"	Absence of a published instrument constrains verifiable event scope to what convergent sources report
OEP (sole basis)	"Officially reported"	Single underlying physical evidence base from one source entity
PIET	No ceiling	Multiple independent evidence streams across 3-4 empirical domains

The PIET exemption follows from multi-stream convergence: a 3-of-4 gate pass across distinct empirical domains is structurally equivalent to standard §3.1 multi-source confirmation.

Confidence Assessment: HIGH. The dual-purpose evidence statements follow from evidence structure and require no novel calibration. The §4.11 state media accommodation adapts established journalism source-attribution practice (Reuters Handbook, AFP Twenty Principles) to WEO's verification context — the qualifying conditions (documentary function, named attribution, verifiable institutional role) are standard source-verification requirements applied to a specific media category. The §4.11 convergent implementation evidence accommodation rests on multi-source convergence anchored by a regulated financial disclosure subject to securities law; ICD 203's converging-source standard is the informing precedent. The five qualifying conditions are conjunctive; the accommodation is available only after all other pathways — §3.1 verification and Stage 2 qualification — have been assessed and found inapplicable. The confidence term ceiling applies existing §3.4.5 terminology to defined evidence configurations — informed by ICD 203's single-source disclosure requirements for the state media and OEP configurations, and reflecting the absence of published instrument text for the convergent implementation evidence configuration.

Validation Commitment: Monitor for (1) changes in regulatory process architecture in jurisdictions where the §4.11 accommodations currently apply that increase public documentation availability, reducing the structural gap — including whether previously unpublished instruments become publicly accessible, which would make standard §3.1 verification available for affected events; (2) emergence of additional regulatory environments exhibiting structurally similar documentation gaps, requiring assessment of whether the accommodation framework extends; (3) evolution of intelligence community or journalism source-assessment frameworks that may provide refined models for accommodated-source confidence calibration; and (4) cases where the sole accountability-bearing disclosure is filed by an entity under the control of the issuing jurisdiction's government, to assess whether the institutional accuracy incentive documented in §3.15 operates with reduced force in that configuration.

Section 4: Significance Assessment Rationale

Manual Reference: Chapter 4

4.1 Basel 5-Factor Assessment

Manual Section: 4.3-4.5

Category: EMPIRICALLY GROUNDED (VERY HIGH >90%) — Basel adaptation; the V1.5 provisions documented below carry individual confidence assessments.

Rationale: WEO adapts the Basel Committee on Banking Supervision G-SIB Methodology (2018) — the framework G20 central banks have used since 2011 to assess systemic importance in banking infrastructure.

The five factors (Size, Interconnectedness, Substitutability, Complexity, Cross-jurisdictional Activity) provide proven multi-dimensional assessment for infrastructure significance. WEO adapts thresholds and evidence requirements for AI infrastructure whilst preserving the framework's structural integrity.

2+ Factors Requirement: Prevents single-factor outliers from qualification, aligning with Basel's score-based (not binary) methodology.

Citation: Basel Committee on Banking Supervision. (2018). *Global systemically important banks: revised assessment methodology and the higher loss absorbency requirement*.

Object-of-Assessment Rule (§4.4 preamble):

V1.5 specifies the assessment object for non-facility Part A events through a §4.4 preamble provision, calibrated against factor application patterns across nine Basel subpopulations (95 events). Approximately 40% of Part A events are not facilities — they are policy instruments, chip product lines, investment transactions, or procurement contracts routed through factors calibrated for physical infrastructure. Each requires a choice of assessment object: the instrument or transaction, or the infrastructure, technology, or material it controls, funds, or produces. Factors 1 (Size), 3 (Substitutability), and 4 (Complexity) are assessed against the controlled, funded, or produced infrastructure or technology. Factors 2 (Interconnectedness) and 5 (Cross-Jurisdictional Activity) are assessed against the event's cross-border effect. The split follows from the factors' analytical function: the first group measures the significance of the thing the event acts upon; the second measures the reach of the action itself. The rule does not alter factor definitions, thresholds, or the 2+/5 pass logic.

Three inconsistency patterns identified in the calibration analysis inform the provision's inline examples:

- F4 instrument-vs-controlled-technology readings for policy instruments — the same instrument class received opposite F4 assessments depending on whether complexity was assessed against the instrument's drafting or the controlled technology's technical characteristics
- Trivial-universality factor patterns for chip designers — F2 (Interconnectedness) and F4 (Complexity) passed at 100% on class-generic facts, while F1 (Size), F3 (Substitutability), and F5 (Cross-Jurisdictional Activity) systematically failed, producing zero intra-class discrimination; the F1 "top 25% in category" sub-threshold, applied to the produced technology's global market position, restores discriminating power
- F5 two-readings ambiguity for compute facilities — whether bloc-capability contribution by a domestic facility satisfies Cross-Jurisdictional Activity was unresolved, with assessors splitting on near-identical facts; V1.5 specifies that documented cross-border evidence is required

The §4.6 "Major Economy Advanced Technology Export Controls" worked example illustrates the preamble rule: F4 is assessed against the controlled technology (advanced-node semiconductor manufacturing), for a factor count of 5/5.

Calibration basis: Factor vectors extracted from 95 Basel Part A events; subpopulation pass-rate matrices computed; factor application patterns analysed across nine subpopulations.

Confidence Assessment: HIGH. The object-of-assessment rule is a definitional provision specifying the assessment object for an existing framework across 95 corpus events. Factor definitions, thresholds, and pass logic are unchanged. No qualification outcomes change under the rule; the correction applies to rationale-text consistency, not inclusion or exclusion decisions.

F3 Strategic-Concentration Qualifier (§4.4, Factor 3):

The Basel G-SIB Substitutability category assesses the degree to which other institutions can provide the same services in the event of a failure — a definition that encompasses both product-level replaceability and market concentration. The strategic-concentration qualifier specifies the assessment frame for substitutability when regulatory constraints limit an ecosystem's access to alternatives, distinguishing product-level replaceability from ecosystem-level accessibility.

A full-corpus calibration analysis across the complete population of chip designer events (12 events) identified a systematic F3 assessment pattern. Five chip designers operating in unconstrained ecosystems pass F3 — their products hold dominant market positions, and product-level replaceability is assessed against the global market. Seven chip designers operating in export-control-constrained ecosystems fail F3 on identical product-level-replaceability grounds — alternative chip architectures exist globally. The qualifier addresses this by specifying the assessment frame: where documented regulatory constraints restrict an ecosystem's access to global alternatives, substitutability is assessed against the constrained market's available alternatives.

Three resolution paths were assessed:

- Path (a): a demand-supply-imbalance evidence requirement on the existing "Removal would create significant gap" and "One of few global suppliers" criteria — substitutability assessed against the constrained market's available alternatives where documented regulatory instruments restrict access. This preserves the operational-evidence principle: the qualifier requires documented evidence of constrained access (export control instruments, verified demand-supply imbalance), not subjective designation or strategic intent.
- Path (b): a sixth criterion with a state-designation evidence gate — an entity qualifies if a state has designated its product class as strategically significant. This imports aspiration and designation evidence into a framework whose other criteria require operational evidence (market share, supplier counts, capacity gaps). The evidentiary register mismatch introduces a qualification pathway with different epistemic properties from the existing five criteria.
- Path (c): leave F3 as product-level and capture the ecosystem-access constraint in the analytical layer (tier classification rationale framework). The tier classification rationale framework already carries contextual signals about ecosystem constraints; path (c) duplicates this signal without strengthening the factor assessment itself.

Path (a) was adopted. The demand-supply-imbalance evidence requirement maintains the operational-evidence standard across all F3 criteria: the qualifier is activated by documented regulatory instruments and verified access constraints, not by designation or aspiration.

Calibration basis: Factor vectors extracted from 12 chip designer events across the Basel Part A corpus (95 events total); F3 pass/fail patterns analysed by ecosystem access conditions; three resolution paths assessed against the operational-evidence principle.

Confidence Assessment: HIGH. The qualifier is a definitional provision specifying the assessment frame for an existing criterion. F3's product-level replaceability assessment, its threshold structure, and the 2+/5 pass logic are unchanged. The evidence requirement (documented demand-supply imbalance within the constrained ecosystem) applies the same operational-evidence standard as the existing F3 criteria. The qualifier's discriminating mechanism — documented access constraint versus mere existence of trade restrictions — is verifiable against primary source evidence (regulatory instruments, market access data).

Entity-Specific Evidence Requirement (§4.4, 2+/5 threshold):

A full-corpus calibration analysis found that six Eastern Bloc AI accelerator designer events qualify at exactly 2/5 on two factors — F2 (Interconnectedness) and F4 (Complexity) — that every AI chip designer in the class satisfies on identical grounds. F2 passes on fabless-foundry dependency and export-control constraint. F4 passes on advanced-node semiconductor architecture. Neither factor distinguishes one chip designer from another; any entity entering the same category passes both automatically. F1, F3, and F5 fail systematically across all six events. The 2+/5 threshold, applied to this population, admits the entire class without discriminating within it.

The entity-specific evidence requirement addresses this: at least one passing factor must rest on evidence specific to the assessed event or entity. V1.5 specifies the test operationally — replace the assessed entity with any other entity in the same category; if the factor rationale remains true, the factor is class-generic. Class-generic factors count toward the threshold but cannot be the only factors an event passes on.

The seventh corpus-wide 2/5 event passes the requirement: its F3 rests on a voucher-subsidy funding model distinct from commercial alternatives — evidence that does not transfer to other entities in the same category.

Under V1.5, the six Eastern designer events also satisfy the requirement. The F3 strategic-concentration qualifier (§4.4, Factor 3) gives them F3 on documented export-control constraints restricting their ecosystem's access to alternative suppliers — evidence specific to each entity's operating conditions, not generic to all chip designers. Their factor vectors move from 2/5 to 3/5. No existing corpus event is excluded.

Calibration basis: Factor vectors extracted from all seven corpus-wide 2/5 minimum-band events within the 95-event Basel Part A population; pass-rate analysis by subpopulation identified the trivial-universality pattern in the chip designer cohort.

Confidence Assessment: HIGH. The requirement constrains how the 2+/5 threshold is reached without changing factor definitions, individual thresholds, or the pass logic. No existing qualification outcome changes.

Validation Commitment: Monitor whether the entity-specific evidence requirement produces consistent classification outcomes as new event categories enter the Basel pathway. Review the class-generic versus entity-specific distinction at each methodology version milestone against the current corpus composition.

F1 Sub-Threshold Text Amendment (§4.4, Factor 1):

V1.5 amends the fourth F1 sub-threshold from "Top 25% globally in AI infrastructure investment within category" to "Top 25% globally within category by AI infrastructure scale." The sub-threshold measures the entity's significance relative to peers in the same category. Revenue is the primary evidence of that significance. "Investment" was inherited from the Basel G-SIB source framework's terminology. In the G-SIB context, "investment" refers to assets under management; in AI infrastructure assessment, the sub-threshold measures the entity's scale within the category. "Scale" describes what the assessment actually measures: the entity's size within the category, evidenced by revenue, deployment, or equivalent indicators.

F1 "Top 25% Within Category" Operationalisation (§4.4, Factor 1):

The fourth F1 sub-threshold applies to AI accelerator designer events that do not meet the three facility-calibrated sub-thresholds (\$500M capital deployment, 5,000+ GPUs deployed, Top 500 supercomputer). V1.5 operationalises the sub-threshold, calibrated against a full-population data collection across the global AI accelerator designer population.

The data collection compiled revenue, deployment, and market-share data across approximately 25 entities shipping AI accelerators at commercial or internal-deployment scale. The data reveals a pronounced tier structure: a top cluster of four to seven entities with revenues from \$16B to \$194B — or deploying at comparable internal scale — separated from the remainder by more than an order of magnitude. Most entities outside the top cluster report revenues below \$300M. The top-25% boundary falls within the gap between the top cluster and the rest. No entity currently below the boundary could cross it under any plausible metric or counting method.

The operationalisation specifies three elements. The category boundary: companies currently designing and producing AI accelerators at scale — shipped to external customers or deployed internally at production scale — excluding pre-revenue entities and edge-only processors. The primary evidence of scale: entity AI accelerator revenue. The data-availability accommodation: where the accelerator is not sold commercially and no standalone chip revenue exists, the parent entity's AI infrastructure segment revenue evidences the entity's scale within the category. This accommodation covers entities whose AI accelerators are consumed through the parent entity's cloud or internal infrastructure. The provision follows the source framework's structural pattern: the BCBS G-SIB methodology specifies indicators, denominators, and data sources for each assessment category.

Calibration basis: Per-entity revenue, deployment, and market-share data collected across the global AI accelerator designer population; revenue tier structure analysed; top-25% boundary located within the order-of-magnitude gap between the top cluster and the remainder.

Confidence Assessment: HIGH. The provision operationalises an existing sub-threshold without changing its pass logic. The tier gap is large enough that no plausible metric or counting-method variation shifts any entity across the boundary.

Validation Commitment: Review the category counting rule and the tier structure at each methodology version milestone. Monitor whether the order-of-magnitude gap between the top cluster and the rest persists as the AI accelerator market evolves, and whether any entity approaches the boundary from below.

4.2 \$500M Physical Infrastructure Threshold

Manual Section: 4.4 (Factor 1: Size)

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: The \$500M threshold derives from U.S. Federal Highway Administration "Major Project" definition under 23 U.S.C. 106(h), which designates projects exceeding \$500M as requiring enhanced federal oversight.

Exact Statutory Language:

"A recipient of Federal financial assistance for a project under this title with an estimated total cost of \$500,000,000 or more... shall submit to the Secretary... a project management plan"

Legal Status: Current federal law. Established by SAFETEA-LU (2005), lowering the previous \$1B "Mega Project" threshold.

Supporting Convergent Sources:

Source	Threshold	Type
U.S. FHWA 23 U.S.C. 106(h)	\$500M	Federal statute
Hart-Scott-Rodino Act (2025)	\$505.8M	Federal statute
EU Foreign Subsidies Regulation	€500M (~\$525M)	EU regulation
AIIB Accountability Framework	\$300M	Multilateral bank

4.3 5,000+ GPU/Accelerator Cluster Threshold

Manual Section: 4.4 (Quantitative AI Thresholds)

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: The 5,000+ GPU threshold is empirically grounded through distribution analysis of AI infrastructure investments (2020-2025).

Empirical Findings:

Research Finding	Source
Natural breakpoint at ~\$100M	Distribution analysis (multi-source research)
5,000 GPUs ≈ \$150-200M	GPU pricing correlation (H100 ~\$30-40K); domestic AI accelerator pricing may differ but threshold measures card count, not capital equivalent
\$100M distinguishes technical from strategic	Cross-validated research

Research Convergence:

- GPU clusters show breakpoints at \$50M (project vs institutional scale) and \$200M (national significance); the cleaner infrastructure-scale break occurs at ~\$100M across all categories
- 5,000+ H100s correlate to \$150M-\$200M investment; \$100M identified as "significant deployment threshold"

Analysis confirmed a natural breakpoint exists at ~\$100M for distinguishing technical projects from strategic deployments. 5,000 GPUs sits comfortably above this threshold.

Position in Threshold Hierarchy: Between infrastructure entry (\$100M) and geopolitical significance (\$500M Basel Size Factor), the 5,000+ GPU threshold (~\$150-200M) captures infrastructure-scale AI compute deployments.

4.4 Software Routes 1-4

Manual Section: 4.8

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Route 1: Official Designation

Rationale: Relies on authoritative government frameworks:

- CISA Systemically Important Entities (NSM-22, 30th April 2024)
- EU DMA Gatekeepers (Regulation 2022/1925)
- EU NIS2 Essential Entities (Directive 2022/2555)
- EU DORA Critical ICT Third-Party Providers (Regulation 2022/2554)

Eastern Bloc Finding: No defensible equivalent exists. CIIO lists are confidential by Chinese government policy. EB platforms qualify via Routes 2-4 instead.

Route 2: Investment Threshold

Rationale:

- **\$100M threshold:** Institutional investor "significant commitment" convention. Market transactions cluster in the \$75M–\$150M significance zone (POET Technologies \$75M AI hardware financing, Applied Digital \$112.5M AI data centre drawdown).
- **\$50M threshold:** Government programme participation thresholds (NRCan, EU Cohesion Policy, DoD FMR)

Route 3: Multi-Criteria

Rationale:

- **10,000 dependents:** npm/PyPI ecosystem dependency analysis
- **50% market share:** Sherman Act Section 2 (US) and Article 102 TFEU (EU) dominance thresholds. Note: These derive from case law precedent and Commission/Court guidance (US v. Alcoa, 1945; EU Commission Article 102 guidance) rather than statutory text, which does not specify numeric thresholds.

Route 4: Open-Source AI Models

Rationale: $\geq 100\text{M}$ cumulative downloads, derived from EU DMA's 45M monthly end users threshold with $\approx 2\times$ multiplier for AI-specific context where downloads $\neq$ active users.

R4-Level Thresholds:

R4-Level	Score	Multiplier	G-SIB Precedent
Level 1	≥1.0	1.15×	Bucket 1 (1.0% surcharge)
Level 2	≥4.0	1.40×	Bucket 2 (1.5% surcharge)
Level 3	≥7.0	1.75×	Bucket 3 (2.0% surcharge)

V1.5 consolidates Route 4D's third limb (critical infrastructure integration) into limb 1 as "≥3 government agencies or critical infrastructure operators across ≥2 countries." The fold preserves the critical infrastructure qualification within the existing ≥3/≥2 quantification rather than maintaining an unguarded gate; zero corpus impact, consistent with the quantified-threshold pattern established by DMA, NIS2, and DORA.

4.5 Route 4 Closed-Source Equivalents (4A-CS to 4D-CS)

Manual Section: 4.8 (Route 4 subsection)

Category: ADAPTED (MEDIUM-HIGH 60–75%) — from DMA, with reduced observability accommodation

Rationale: Closed-source AI models (GPT-4, Claude, Gemini, Ernie) lack the publicly observable metrics available for open-source models (downloads, derivatives). The 4A-CS to 4D-CS framework establishes equivalent qualification criteria using observable proxy metrics.

Qualification Logic: Any ONE of four criteria qualifies.

Criterion 4A-CS: User Base

Threshold:

- | 100M monthly active users (MAU) → Automatic qualification
- 50M-100M MAU → Qualifies with 4B-CS OR 4C-CS

Precedent:

- EU Digital Markets Act: 45M monthly end users threshold
- WEO applies ≈2× multiplier (100M) for automatic qualification
- 50M-100M range requires supporting criterion to prevent borderline misclassification

Measurement Sources: Similarweb, data.ai, company earnings disclosures, third-party analyst reports

Criterion 4B-CS: Geographic Reach

Threshold:

- Available in ≥ 150 countries, OR
- Available in ALL G7 nations + documented Eastern Bloc presence (PRC, Russia)

Rationale: Geographic availability indicates cross-bloc coordination significance. The G7 + Eastern Bloc test captures strategic positioning even for models with restricted geographic access.

Criterion 4C-CS: Enterprise Integration**Threshold (Composite):**

1. Fortune 500 enterprise adoption (≥ 50 documented customers)
2. Government/defence contracts (≥ 3 agencies across ≥ 2 countries)
3. API ecosystem ($\geq 1,000$ registered developers OR ≥ 100 published integrations)

Limitation Acknowledged: Closed-source enterprise integration is less observable than open-source metrics. Composite assessment prevents single-dimension gaming whilst maintaining rigour.

Precedent: DMA business user criteria adapted for AI platform context.

Criterion 4D-CS: Capability Threshold**Threshold:**

- $\geq 1B$ parameters (documented or credibly estimated), AND
- Frontier performance on recognised benchmarks (top 10 on MMLU, HumanEval, or equivalent)

Rationale: Capability threshold captures models with significant coordination impact regardless of commercial adoption metrics. Parameter count + benchmark performance provides observable proxy for frontier status.

4.6 PISA-5 Assessment

Manual Section: 4.9

Rationale: PISA-5 (Policy and Institutional Significance Assessment) was developed to qualify single-jurisdiction AI policy and institutional events that cannot route through Basel 5-Factor (no physical infrastructure) or Software Routes (no software/platform release).

Factor	Status	Source
F1: J-Tier (Jurisdictional Significance)	EMPIRICALLY GROUNDED	G7/G20 classification; Stanford HAI AI Index
F2: Regulatory Scope	EMPIRICALLY GROUNDED	Legislative analysis methodology
F3: AI Centrality (mandatory gate)	CATEGORY CREATOR	AI-specific filter — no precedent
F4: Enforcement/Accountability	EMPIRICALLY GROUNDED	Regulatory effectiveness research
F5: International Relevance	EMPIRICALLY GROUNDED	Extraterritoriality doctrine; Brussels Effect

Qualification Logic

Standard Pass: $\geq 4/5$ factors satisfied (with F3 mandatory)

Alternative Pass: F1 + F2 + F3 all satisfied

Rationale for Alternative Pass: Major jurisdiction (F1) + binding framework (F2) + AI-specific (F3) captures "minimum viable significance" even without full enforcement (F4) or international reach (F5).

Mandatory Gate: F3 (AI Centrality) must ALWAYS pass — if F3 fails, event fails PISA-5 regardless of other factors.

Novelty Assessment

Category: CATEGORY CREATOR (HIGH 75–90%)

Research across 40+ sources confirmed:

- No existing framework combines all five PISA-5 design elements (multi-factor qualification + tiered jurisdiction + mandatory gating + alternative pass logic + human review tiebreaker)
- Basel G-SIB is closest structural analogue but serves fundamentally different purpose (financial institution capital requirements vs policy event qualification)
- Five PISA-5 *design elements* assessed as NOVEL:
 - J-Tier classification (no precedent for G7/G20 + GDP + AI capability combination)
 - F3 mandatory gate (multi-factor frameworks typically use weighted averages, not hard gates)
 - $\geq 4/5$ OR F1+F2+F3 alternative logic (no precedent found)
 - AI Hub exception (J-Tier D)
 - T1-T5 structured tiebreaker criteria
- One component ADAPTED: The 5-factor structure concept from Basel G-SIB (but application to policy events is novel)

J-Tier D (AI Hub Exception): Category creator accommodation for AI-significant smaller jurisdictions meeting 2 of 4 criteria. Flagged for annual designation review. See §4.6.2 for detailed derivation.

4.6.1 Tiebreaker Criteria Derivation (TB1-TB5)

Manual Section: 4.9 (Human Review Framework)

Category: ADAPTED (MEDIUM-HIGH 60–75%) for TB1–TB2; WEO-ORIGINAL for TB3–TB5

The PISA-5 Human Review Framework uses five tiebreaker criteria when events pass exactly 3/5 factors:

Criterion	Threshold	Derivation Source
TB1: Resource Commitment	>\$15M annual budget OR >50 FTE	AI governance institution budget analysis
TB2: Institutional Permanence	Statutory basis OR >3 year mandate	Administrative law institutional design precedent
TB3: Stakeholder Engagement	≥ 3 stakeholder categories	Multi-stakeholder governance best practice
TB4: International Influence	Formal coordination mechanism	Treaty and MOU practice
TB5: Novel Governance	First-of-kind approach	Innovation assessment

TB1 \$15M Threshold Derivation:

Analysis of established AI governance institution budgets (2023-2025) identified \$15M annual operating expenditure as the threshold distinguishing substantive institutional capacity from nominal presence. This threshold is validated against established infrastructure classification frameworks.

Critical Distinction: TB1's \$15M is annual operating budget (recurring expenditure), distinct from Part A Basel Factor 1 (\$500M+ capital investment) and Chapter 3 entry threshold (\$100M infrastructure investment).

Validation Commitment: Review tiebreaker thresholds at 100-event milestone based on classification outcome data.

4.6.2 J-Tier D AI Hub Exception — Derivation

Manual Section: 4.9 (AI Hub Exception)

Category: CATEGORY CREATOR (HIGH 75–90%)

Problem Addressed: Standard J-Tier criteria (G7/G20 membership, \$500B GDP) exclude jurisdictions that are demonstrably significant to global AI coordination despite smaller economic footprints. Examples include Taiwan (critical semiconductor manufacturing), Israel (AI startup ecosystem), Singapore (AI governance leadership), and UAE (sovereign AI investment).

Criteria Selection Rationale:

The four J-Tier D criteria were selected to capture distinct pathways to AI significance:

Criterion	Rationale	Precedent
Top-15 AI Index	Captures jurisdictions with demonstrated AI capability disproportionate to GDP	Stanford HAI methodology; Tortoise Global AI Index
Critical AI Infrastructure (>5% global share)	Captures jurisdictions controlling chokepoint infrastructure regardless of domestic AI development	Semiconductor supply chain analysis; CSET chokepoint research
Top-10 AI Investment Destination	Captures jurisdictions attracting significant capital flows indicating market confidence	PitchBook/CB Insights methodology; venture capital concentration analysis
Foundation Model Developer	Captures jurisdictions producing frontier AI capabilities regardless of other metrics	Papers with Code benchmarks; Hugging Face leaderboards

2-of-4 Threshold Rationale:

The 2-of-4 requirement balances inclusivity with rigour:

Threshold	Assessment
1 of 4	Too permissive — single-dimension outliers could qualify (e.g., jurisdiction with one well-funded AI lab but no broader ecosystem)
2 of 4	Selected — requires demonstrated significance across multiple dimensions
3 of 4	Too restrictive — would exclude jurisdictions with genuine but concentrated AI significance
4 of 4	Unnecessary — jurisdictions meeting all four likely qualify via standard J-Tier criteria

Precedent Analysis:

No direct precedent exists for AI-specific jurisdictional significance thresholds. Adjacent frameworks consulted:

- **Global Financial Centres Index (GFCI):** Private index using multi-criteria assessment of financial centre competitiveness; no comparable FSB regime exists for financial centres (FSB assesses institutions, not centres)
- **OECD Digital Economy Outlook:** Ranks jurisdictions by digital indicators but does not create qualification thresholds
- **Stanford HAI Global Vibrancy Tool:** Provides rankings but no threshold-based classification

Conclusion: J-Tier D represents genuine category creator calibration. The 2-of-4 threshold and specific criteria selection are WEO-original, informed by but not derived from external frameworks.

Validation Commitment: Annual designation review; recalibration at 100-event milestone based on classification outcomes.

4.6.3 F3 Pathway Exemption

Manual Section: §4.9, Factor 3 (F3) — pathway exemption

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: F3 filters general technology and digital economy policies where AI is mentioned incidentally (data protection law with an automated decision-making article, copyright directive with a text-mining exception). Its structural indicators — AI in policy title, AI-specific definitions, AI-specific obligations — are calibrated to regulatory and governance instruments where the legislature chooses whether to make AI the primary subject.

Two AI Connection pathways (Option F and Option G) serve event classes where the legislature structurally cannot make AI the primary subject because the instrument addresses the programme or the material — not the downstream application. Defence appropriation acts fund entire departmental budgets; critical minerals acts address mineral supply chains. In both cases, the AI infrastructure relevance is real and documentable, but the instrument's structural format guarantees F3 failure regardless of the event's genuine significance.

The pathway exemption resolves this by recognising that each pathway's own evidence requirement performs the AI centrality function more precisely than F3's generic indicators. Option F requires explicit AI designation in programme titles within the financial instrument. Option G requires documented causal chains from the controlled material to AI infrastructure applications. Each test is more demanding and more specific than F3's title-or-definitions check.

Evidence basis: Two gap assessments — one addressing Option F (April 2026) and one addressing supply-chain response events (May 2026) — separately documented the same structural barrier. The §4.9 F3 coverage note acknowledges the pattern: "Application-agnostic Policy instruments controlling physical materials or technology products... will typically fail F3 because the instrument's primary subject is the controlled material or technology, not AI." The pathway exemption implements what the coverage note identified: each qualifying pathway's own evidence standard performs the AI centrality function more precisely than F3's generic indicators.

Pass logic adjustment: For exempt events, PISA-5 is assessed on four factors (F1, F2, F4, F5). Auto-qualification requires $\geq 3/4$ factors. The F1+F2 double-fail exclusion rule is preserved — neither significant jurisdiction nor regulatory scope remains grounds for automatic disqualification regardless of F3 status. The standard PISA-5 has a 60% auto-qualify route (F1+F2+F3) that depends on F3's mandatory-gate status; with F3 removed, the lowest auto-qualify ratio rises from 60% to 75% ($\geq 3/4$), an appropriate tightening given the narrower factor set. Marginal cases (exactly 2/4 with F1+F2 not both failing) route to Human Review under the existing TB1–TB5 tiebreaker framework.

Confidence: 75%. The structural mismatch is documented in two separate gap reports. The resolution is logically sound (pathway evidence subsumes F3's function). The 65% minimum threshold (§2.3) is met. Confidence is bounded below 80% pending accumulation of F3-exempt qualification outcomes; the first applications entered the corpus under V1.4.

Validation commitment: Monitor F3-exempt events for quality. If events qualifying through pathway exemption are systematically lower quality than standard PISA-5 events, reassess whether the pathway evidence requirements are sufficiently demanding.

4.6.4 F2/F4 Assessment Object for Non-Regulatory Events

Manual Section: §4.9, Factor 2 (F2) and Factor 4 (F4) — notes on assessment object for non-regulatory events

Category: ADAPTED (HIGH 75–90%)

V1.5 specifies the assessment object for F2 and F4 in non-regulatory PISA-5 events through clarifying notes within §4.9, calibrated against factor application patterns across three non-regulatory subpopulations (institutions, spending programmes, defence programmes) within the 42-event PISA-5 corpus. Regulatory instruments — the subtype the factor definitions were written for — are assessed against the instrument's operative text: F2 evaluates mandatory language and scope, F4 evaluates penalty provisions and enforcement authority. For the three non-regulatory subtypes, two readings are possible: one assessing the legal character of the establishing or authorising instrument; the other assessing the obligations the event creates for AI ecosystem participants. The calibration analysis found that functionally identical institutions received different F2/F4 assessments depending on which reading was applied, and spending programmes of comparable scale received different outcomes based on whether programme-execution evidence was accepted as F2/F4 satisfaction.

V1.5 specifies the assessment-object rule. F2 Condition A (binding status) assesses whether the policy creates binding obligations on parties within the AI ecosystem. F4 assesses enforcement mechanisms over external parties. Factor definitions, thresholds, and pass logic are unchanged. The tiebreaker (TB1–TB5) is the designed pathway for non-regulatory events that fail F2/F4. The tiebreaker criteria — TB1 (Resource Commitment), TB2 (Institutional Permanence), TB3 (Stakeholder Engagement), TB4 (International Influence Pathway), TB5 (Novel Governance Mechanism) — assess the dimensions relevant to institutions, spending programmes, and defence programmes: resource scale, statutory basis, multi-stakeholder engagement, and international coordination mechanisms.

A four-subtype worked-example block in §4.9 illustrates the application across regulatory instruments, institutions, spending programmes, and defence programmes.

Calibration basis: Factor vectors extracted from 42 PISA-5 events; subpopulation pass-rate analysis across institutional, spending-programme, and defence event classes; factor application patterns analysed.

Confidence Assessment: HIGH. The clarification is a definitional provision specifying the assessment object for two existing factors across 42 corpus events. Factor definitions, thresholds, and pass logic are unchanged. The tiebreaker criteria (TB1–TB5) are verified as assessing dimensions relevant to the three non-regulatory subtypes routed through them.

4.7 T2 Buildout Qualification Criteria

Manual Section: 5.5 (T2 Buildout Test)

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: The four-criterion framework for T2 Buildout classification derives from analysis of explicit cross-bloc targeting language in infrastructure legislation.

Four-Criterion Framework (ONE OR MORE required for T2):

Criterion	Requirement	Evidence Standard
1. Dependency Reduction	Explicit statement reducing dependency on NAMED other-bloc	"Reduce dependence on [Country X]" in official documentation
2. National Security + Co-Mention	"National security" in same paragraph/section with NAMED other-bloc	Security language co-located with named entity
3. Operating Restrictions	Funded entities prohibited from operating in NAMED other-bloc	Binding restrictions on cross-bloc activity
4. Supply Chain Resilience + Geographic Specificity	"Supply chain resilience" co-located with NAMED other-bloc	Resilience language with geographic targeting

Criterion Strength Ranking:

- Criterion 3 (Operating Restrictions) = **STRONGEST** — creates direct legal obligation
- Criterion 4 (Supply Chain Resilience) = **WEAKEST** — often uses euphemistic language

Key Precedent: US CHIPS Act §9902 guardrails (explicit China naming); Banque de France geopolitical trade fragmentation analysis; EU Chips Act implementation criteria.

Section 5: Tier Classification Rationale

Manual Reference: Chapter 5

5.1 Three-Tier Architecture

Manual Section: 5.2

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: The T1/T2/T3 architecture captures three distinct scales of coordination dynamics:

Tier	Scope	Mirror Framework
T1	Cross-bloc (verified adoption or implementation by the principal AI actor from at least two distinct ecosystems)	Multilateral trade/security analysis
T2	Between-bloc (structural separation through restriction, buildout, or regulatory barriers)	Bilateral/adversarial analysis
T3	Intra-bloc (activity within a single bloc)	Alliance theory; bloc coordination

T1 Cross-Bloc Criteria

Category: CATEGORY CREATOR (MEDIUM-HIGH 60–75%)

Requires ALL THREE:

1. Cross-bloc participation (the principal AI actor from at least two distinct ecosystems)
2. Binding/implemented status
3. Material integration

Operational Definitions:

Criterion	Definition	Observable Marker
Cross-bloc participation	Verified cross-bloc adoption or implementation by the principal AI actor from at least two distinct ecosystems	National standards adoption, operational certification, or equivalent implementation evidence
Binding/implemented	Actual commitments with adoption mechanisms, OR operational systems with implementation evidence, OR standards with formal certification schemes	Published standards/regulations, adoption documentation, certification bodies, implementation records
Material integration	Shared infrastructure, mutual vulnerability, or enforceable agreements creating cross-bloc interdependence	Cross-border implementation, mutual recognition documentation, shared infrastructure usage, certification schemes operating across blocs

Material Integration Examples:

- Shared AI safety testing protocols
- Common model evaluation benchmarks
- Interoperable data governance standards

For full tier classification criteria: See *Manual, Chapter 5*.

5.2 Mandatory Standard Test

Manual Section: 5.5 (T2 Mandatory Standard Test)

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: The Mandatory Standard Test addresses standard-setting events that impose mandatory compliance requirements with enforcement mechanisms. The three T2 tests — Restriction, Buildout, and Mandatory Standard — capture distinct fragmentation mechanisms: explicit barriers (Restriction), parallel capacity development (Buildout), and mandatory compliance regimes (Mandatory Standard).

Two-Condition Test (both must be met for T2 classification):

Condition	Requirement	Evidence
1. Mandatory Language	Contains "shall," "must," "required"	NOT "recommended," "encouraged," "voluntary"
2. Enforcement Mechanisms	Specific consequences documented	Financial penalties, market access restrictions, licensing consequences, legal liability

Chinese Standard Rule: GB (without /T) = mandatory → test for T2; GB/T (with /T) = voluntary → automatic T3.

Routing Decision: Standard-setting events meeting both conditions route to **Restriction classification**, not to a separate category.

Rationale for Restriction Routing: Mandatory standards with enforcement are structural fragmentation tools — they impose compliance costs as condition of market access, creating non-tariff barriers. This is functionally similar to explicit restrictions.

Precedent:

- WTO TBT Agreement (mandatory vs voluntary standards distinction)
- Brussels Effect (regulatory extraterritoriality)
- Regulatory extraterritoriality doctrine

5.3 Bloc Definitions

Manual Section: 3.10 (Bloc Membership Reference)

Category: ADAPTED (HIGH 75–90%)

Rationale: Tier classification consumes the bloc architecture: T1, T2, and T3 are defined by whether a coordination event crosses, runs between, or remains within blocs, so the tier system inherits its meaning from how blocs are defined. Bloc structure derives from PAA-anchored technology ecosystems rather than political alliance frameworks — the design decision documented at §3.12 (Manual §3.10.1). The universal membership categories (Core, Integrated, Engaged, Peripheral), the bloc assignment gate, and the five-criteria Non-Aligned test are documented at §3.12.1–§3.12.2; current classifications are maintained in the WEO Bloc Membership Register (§3.12.1.2).

For tier assessment purposes, Non-Aligned nations are classified to either bloc per event on dominant partnership characteristics (Manual §3.10.3; §5.2 Scope Note). This event-level attribution is the only bloc operation Chapter 5 performs itself; all standing classifications are consumed from the Register.

5.4 Principal AI Actors

Manual Section: §5.3 (Principal AI Actors)

Category: CATEGORY CREATOR (HIGH 75–90%)

Rationale: No existing geopolitical, international relations, or technology governance framework determines "principal actor" status based specifically on AI infrastructure capability dimensions. This finding was confirmed through review of CSIS, Brookings, RAND, IISS, Chatham House, OECD, WEF, Stanford HAI, Belfer Center, Carnegie Endowment, and academic international relations literature. Within the established lexicon of international relations theory, "principal actor" remains reserved for the sovereign nation-state operating within the Westphalian state system. However, the substantive theoretical architecture underlying WEO's infrastructure-based determination is present across multiple leading institutions, which have constructed analogous full-stack technological typologies using identical supply-chain chokepoints to define sovereign AI power.

Closest Precedents:

Framework	Institution	Relevance	Divergence from WEO
IISS Cyber Capabilities Assessment (2021/2023)	International Institute for Strategic Studies	Closest methodological precedent: three-tier nation classification using qualitative multi-dimensional assessment	Uses cyber power, not AI infrastructure; AI is a sub-indicator within broader cyber assessment
"Broad Stack Leaders" typology (2026)	Brookings Institution	Closest conceptual analogue: categorises US, China, EU as full-stack AI leaders with subordinate tiers (Coordinated Stack Builders, Upstream AI Shapers, AI System Shapers)	Includes non-infrastructure variables (talent, data, education); uses descriptive categories, not threshold-based discrete classification
Cyber/Technology Power Index (2025)	Harvard Belfer Center for Science and International Affairs	Most comprehensive quantitative framework: composite index covering 4 of WEO's 7 dimensions across five technology sectors	Spans five technology sectors, not AI-specific; produces continuous numerical rankings, not discrete principal/non-principal determination
AI Building Blocks framework (2026)	Chatham House	Identifies "AI Superpowers" (US, China) using primary building blocks (compute, frontier models, data) and secondary building blocks (infrastructure, talent, energy, trust)	Bundles hardware layers under "compute"; includes talent, energy, and trust as co-equal primary/secondary dimensions
G-SIB Methodology	Basel Committee on Banking Supervision	Multi-factor threshold-based classification with discrete designation tiers — methodological precedent already used in WEO significance assessment (Basel 5-Factor)	Financial sector, not technology; the threshold-based qualification pattern is directly analogous to WEO's ≥ 3 of 7 dimensional threshold

Key Design Decisions:

1. **Hardware stack disaggregation.** No existing framework treats accelerator design, semiconductor fabrication, semiconductor equipment, and cloud compute as four separate, co-equal dimensions. Brookings bundles hardware under "compute"; Chatham House uses "compute" as a single primary building block. WEO disaggregates because supply-chain chokepoint analysis requires dimensional granularity — a nation can dominate fabrication (Taiwan) without designing accelerators, or dominate equipment (Netherlands) without fabricating chips. Bundling obscures the distribution of supply-chain power that WEO's bloc classification methodology is designed to track.
2. **Dimensional scope: sovereign control over AI infrastructure outputs.** WEO's seven SCP dimensions — AI accelerator design, advanced semiconductor fabrication, semiconductor manufacturing equipment, cloud AI compute services, frontier AI model development, AI-specific regulatory authority with extraterritorial reach, and sovereign AI infrastructure investment at scale — measure assets, instruments, and capabilities that a sovereign actor can independently control and that determine which AI infrastructure ecosystem the actor operates within. This scope is more focused than comprehensive AI power frameworks (Brookings, Chatham House, IISS, Belfer), which additionally assess variables such as talent, data, and energy. WEO's dimensional focus reflects the specific analytical question the SCP framework is designed to answer: not how much AI capability does this actor possess in aggregate, but what does this actor independently control in the AI infrastructure supply chain, and which ecosystem does that place it within? Variables that contribute to AI capability broadly but do not independently determine ecosystem position — including workforce, training data, and energy capacity — are not assessed as SCP dimensions. These variables do appear in WEO's event-level analysis where they constitute characteristics of AI infrastructure coordination activity: energy through the Energy/Infrastructure domain and Environmental Nexus Tags framework, data governance through policy events, and workforce provisions through event characteristics such as CHIPS Act award conditions. The dimensional exclusion is a scope decision at the actor-assessment level, not an analytical judgement about the importance of these variables to AI development. This focused scope, combined with the three-gate designation structure (§5.4.2), produces materially different actor classifications from broader frameworks: most notably, the EU scores 4/7 on WEO's dimensional assessment (including regulatory authority and sovereign investment) but does not qualify as a principal AI actor because its hardware dimensions cannot be sustained independently under the severance test — a structural dependency that broader typologies such as Brookings' "Broad Stack Leaders" do not assess.
3. **Discrete actor classification with binary threshold.** Most indices (Stanford HAI AI Index, Belfer CET Index, OECD.AI) produce continuous numerical rankings. WEO uses a binary threshold (≥ 3 of 7 dimensions) producing a discrete "principal actor / not principal actor" determination. This follows the methodological pattern established by the Basel G-SIB methodology (already adapted for WEO's Basel 5-Factor significance assessment): multi-factor assessment producing a discrete designation that triggers specific regulatory consequences. The ≥ 3 threshold was selected because: $N=1$ would include critical supply-chain nodes whose single-dimension dominance does not independently shape coordination dynamics; $N=2$ produces no empirically different result from $N=3$ under current assessment; $N \geq 4$ produces no empirically different result from $N=3$ but would create a threshold that structurally excludes nations with emerging but incomplete AI ecosystems from future qualification.

Evidence Basis: The two-ecosystem structure and principal actor identification are grounded in cross-referenced research from three independent AI research inputs with dedicated synthesis (March 2026), examining market share data, fabrication capability, equipment monopolies, cloud deployment, model development, regulatory frameworks, and investment scale across all candidate nations. Findings were unanimous on the bipolar hardware ecosystem structure and the US/China principal actor identification. The category creator research (principal actor as novel framework) was reviewed across two independent AI research inputs surveying CSIS, Brookings, RAND, IISS, Chatham House, OECD, WEF, Stanford HAI, Belfer Center, Carnegie Endowment, and academic international relations literature. The framework comprises a three-gate designation structure (§5.4.2), per-dimension qualification principles (§5.4.1), and is applied across 14 actors in the initial assessment cycle ('Batch 1', May 2026), producing a distribution (2 PAA, 3 AIK, 2 ACS, 7 Participant) that discriminates meaningfully across all four designations. The ≥ 3 threshold is tested at the boundary — actors scoring 3/7 (Netherlands, Taiwan) and 4/7 (Japan, South Korea, EU) span the AIK/ACS divide, supporting the threshold's discriminating power.

Confidence Assessment: 75%. The seven-dimension framework and discrete threshold classification are WEO-original with no direct precedent. Closest analogues (Brookings "Broad Stack Leaders", IISS Cyber Capabilities, Belfer CET Index) support the approach conceptually but diverge on dimensional scope and classification method. The ≥ 3 threshold is boundary-tested through Batch 1 application across 14 actors, producing meaningful discrimination at the 3/7 and 4/7 boundary.

Validation Commitment: Reassess Principal AI Actor designations at methodology version milestones. Monitor threshold sensitivity as additional nations develop AI infrastructure capability (EU hardware independence trajectory, India sovereign compute expansion). Review dimensional framework if new infrastructure categories emerge (e.g., quantum computing as a separate AI-relevant dimension). Continue applying to Batch 2 actors.

T2 Scope: Principal Actor Requirement Not Extended

Manual Section: §5.3 (T2 scope paragraph)

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

Rationale: The principal actor requirement is structurally specific to T1. T1 measures ecosystem-level integration — whether distinct AI ecosystems are operationally aligned through verified cross-bloc adoption or implementation by their respective principal AI actors. Without ecosystem controllers at the table, bilateral contacts between non-principal members from different blocs do not bridge the two ecosystems. T2 measures a different dynamic: structural separation between blocs. The significance of between-bloc fragmentation depends on the structural effect — what is disrupted, displaced, or barred — rather than the identity of the actor creating the separation.

This asymmetry is grounded in two independently established theoretical frameworks. Hegemonic Stability Theory (Kindleberger, 1973; Keohane, 1984; Gilpin, 1981) establishes that systemic cooperation requires hegemonic leadership to overcome collective action problems — Keohane's explicit disconfirmation condition makes actor identity a necessary condition for systemically significant cooperation. Weaponized Interdependence (Farrell & Newman, 2019) establishes that network fragmentation depends on structural position (control of hubs and chokepoints), not on state power or hegemonic status — the chokepoint effect is inherently actor-agnostic. Hirshleifer's aggregation technology framework (1983) provides the formal bridge: best-shot public goods (cooperation outcomes) require at least one highly capable actor, whilst weakest-link public goods (disruption vulnerability) are determined by the least capable or most disruptive node.

The asymmetry holds across all three T2 fragmentation types:

- **Restriction:** Supply-chain cuts can originate from any node with sufficient leverage. Critical supply-chain nodes command chokepoint positions (Netherlands: ~100% EUV lithography; Japan: critical deposition, etching, metrology equipment; Taiwan: ~90% advanced logic fabrication) where unilateral action by a non-principal actor produces fragmentation effects equivalent to or exceeding direct principal actor action. The T2 Restriction tests (Explicit-Targeting and PIET Impact-Demonstrated) measure the effect through named other-bloc targeting, trade flow disruption (>50%/>90%), price impacts (>20%), and government policy responses (≥ 2).
- **Buildout:** Parallel infrastructure development creates structural separation through the existence of alternative capacity, not through the identity of the builder. The Basel 5-Factor framework (\$100M floor, cross-jurisdictional significance, substitutability assessment) directly measures whether the buildout creates structurally significant parallel capacity.
- **Mandatory Standard:** Regulatory barriers create fragmentation through enforcement scope and compliance burden, not through the identity of the legislating entity. The EU currently exercises regulatory authority with extraterritorial reach without qualifying as a principal AI actor under the §5.3 dimensional assessment. The Mandatory Standard Test (mandatory language + documented enforcement) measures the structural barrier directly.

Clayton, Maggiori & Schreger (2024) provide the closest single-framework articulation: their model of economic coercion and fragmentation mathematically demonstrates that global integration is driven by hegemonic coordination of externalities (supporting T1's actor requirement), whilst fragmentation is driven by uncoordinated, structurally-determined pursuit of economic security by actors at any scale (supporting T2's effect-based approach).

Evidence Chain:

Precedent	Type	Contribution
Keohane (1984) <i>After Hegemony</i>	Direct	Cooperation requires great-power leadership; disconfirmation condition makes actor identity necessary for systemically significant cooperation
Kindleberger (1973) <i>The World in Depression</i>	Direct	Systemic integration requires a hegemon willing to absorb asymmetric costs; absence of leadership produces fragmentation
Farrell & Newman (2019) "Weaponized Interdependence," <i>International Security</i> 44(1)	Direct	Coercive/disruptive power derives from structural position in network topology, not state size; chokepoint effect is actor-agnostic
Clayton, Maggiori & Schreger (2024) "A Theory of Economic Coercion and Fragmentation," NBER WP 33309 / BIS WP 1224	Structural	Only identified work modelling both hegemonic integration and structurally-driven fragmentation within a single framework
Hirshleifer (1983) "From weakest-link to best-shot"; Sandler (2004) <i>Global Collective Action</i>	Structural	Best-shot/weakest-link distinction provides formal basis for why cooperation is actor-dependent and disruption is actor-agnostic
Olson (1965) <i>The Logic of Collective Action</i>	Structural	"Privileged groups" provide foundational logic beneath HST; defection as rational default grounds the asymmetry
Strange (1988) <i>States and Markets</i>	Structural	Relational power (actor-centred) versus structural power (position-based) directly maps to T1/T2 actor-identity logic
Basel Committee G-SIB Assessment Methodology	Empirical	Actor-agnostic systemic significance assessment via structural factors; directly adapted by WEO for T2

Empirical observation: All 17 T2 events in the current database (as of May 2026) involve at least one principal AI actor as either the initiating entity or the targeted party. This reflects the current geopolitical landscape where US-China strategic competition drives the majority of between-bloc fragmentation activity. It is an empirical feature of the current database population, not a structural requirement of the methodology.

Confidence Assessment: WEO SYNTHESIS, 70%. Both T1 and T2 components are independently grounded at very high confidence (90%+). The connecting logic has formal precedent in Hirshleifer's aggregation technology and single-framework articulation in Clayton-Maggiori-Schreger. Confidence is bounded below 75% because no framework explicitly articulates the differential actor-identity requirement as a classification principle, and the Middle Power Theory challenge is genuine — if minilateral coalitions achieve significant cross-bloc AI coordination without US/China involvement, the T1 requirement faces empirical pressure. Confidence is above 65% because both components are independently strong, the connecting logic has formal precedent, and no framework was found arguing for uniform actor-identity requirements across both cooperation and fragmentation.

Validation Commitment: Monitor for edge cases where a T2 event qualifies without any principal actor involvement on either side. If such cases arise, assess whether the qualification frameworks correctly identified structural significance without the principal actor signal. Monitor for T1 empirical pressure from minilateral cross-bloc coordination. Review at each database doubling milestone.

5.4.1 Dimension Qualification Principles

Manual Section: §5.3.2 (Dimension Qualification Principles)

Category: CATEGORY CREATOR (HIGH 75–90%)

Rationale: The parent PAA framework (§5.3, 7-dimension, ≥ 3 threshold) is documented as CATEGORY CREATOR at 75% confidence in Rationale §5.4. The per-dimension qualification principles operationalise the existing "dominant or independently significant capability" language with specific definitions that resolve edge cases at the boundary of each dimension. No existing geopolitical or technology governance framework publishes explicit per-dimension qualification thresholds for AI infrastructure capability assessment. The frameworks surveyed for §5.4 (IISS Cyber Capabilities, Brookings "Broad Stack Leaders," Belfer CET Index, Chatham House AI Building Blocks, Basel G-SIB) all assess analogous dimensions but none provide the definitional specificity required for reproducible binary classification.

Key Design Decisions:

1. **Qualification principles as sentences, not quantitative thresholds.** Each dimension's qualification principle is expressed as a definitional sentence rather than a numerical threshold (e.g., market share percentage, revenue floor, benchmark percentile). This choice trades precision for durability: the AI infrastructure landscape evolves rapidly, and quantitative thresholds calibrated to 2026 market conditions would require constant recalibration. Definitional sentences permit consistent application across assessment cycles whilst accommodating landscape evolution.

2. **Subsidiary clause harmonisation.** A uniform subsidiary/acquisition clause is applied across Dimensions 1, 3, 4, and 5 — the four dimensions where corporate entities may be controlled from a different territory than their physical operations. This addresses edge cases (Habana Labs/Israel, Graphcore/UK, DeepMind/UK) where corporate control and physical operations diverge. The clause follows the same logic as Dimension 5's existing subsidiary treatment, extended for cross-dimensional consistency.

Sensitivity note: The subsidiary and acquisition clause affects the distribution of PAA scores across actors. Attribution of subsidiary lab outputs to the parent entity's territory (e.g., DeepMind → United States via Alphabet) concentrates Dimension 5 credits in actors hosting parent companies rather than those hosting research operations. As of V1.4 publication (May 2026), no actor would cross the $\geq 3/\geq 2$ -categories threshold under an alternative attribution rule granting strategic autonomy to subsidiary labs — this finding should be reassessed at each methodology version milestone, particularly as new actors develop frontier AI capabilities through subsidiary or acquisition structures.

3. **Subsystem inclusion in Dimension 3.** The equipment dimension extends to "sole-sourced subsystems and components essential to the functioning" of critical equipment, not only complete equipment systems. This covers the Zeiss SMT case (sole-source EUV lithography optics, without which no advanced fab can operate) that four independent AI research inputs unanimously identified as structurally necessary. The "sole-sourced" and "essential to functioning" qualifiers prevent the inclusion of commodity components.
4. **Sovereign control principle generalisation (Dimension 2 → all physical infrastructure dimensions).** The sovereign control principle applies territorial presence as the primary test for all physical infrastructure dimensions (D2 fabrication, D4 compute), reflecting the physical reality that these capabilities are geographically bound — a fab cannot be relocated, and its output serves the territory where it operates regardless of corporate ownership. D4 additionally requires both domestic HQ and territorial compute presence (see key design decision 8 below). Corporate capability dimensions (D1 accelerator design, D5 model development) are assessed by ownership chain — the corporate parent's territory has effective control through strategic direction and IP ownership. D3 (equipment) applies both tests. See key design decisions 7–8 below for the sovereign control principle's full specification and the D4 territorial compute requirement.

5. **Dimension 2/3 process-node threshold recalibration.** Dimension 2 uses a quantitative process-node threshold ($\leq 7\text{nm}$) as the boundary for "advanced" fabrication — an exception to the general design preference for definitional sentences over quantitative thresholds (see point 1 above). This threshold reflects the assessment-date definition of advanced semiconductor manufacturing and is subject to periodic recalibration as the global fabrication frontier advances. The threshold should be reviewed at each methodology version milestone against the current state of leading-edge production. The recalibration trigger is when the specified threshold ceases to discriminate between actors with frontier or near-frontier fabrication capability and those operating at mature commodity nodes — operationally, when ≥ 4 register actors sustain active volume manufacturing at the threshold node, as assessed under the Dimension 2 qualification principle. Equipment-dependency profiles form no part of the trigger condition: reliance on imported equipment is documented as a constraint in the actor's profile per the Dimension 2 text, and no fabricating actor operates without structural equipment dependencies. A recalibration proposal is checked for designation stability — the effect of the candidate threshold on every current register designation — before adoption. Recalibration adjusts the threshold downward (e.g., from $\leq 7\text{nm}$ to $\leq 5\text{nm}$ or $\leq 3\text{nm}$) to maintain analytical discrimination. This recalibration applies equally to Dimension 3, which references $\leq 7\text{nm}$ in its equipment substitutability test — when the Dimension 2 threshold is recalibrated, Dimension 3 must follow to maintain cross-dimensional consistency.
6. **Dimension 5 benchmark saturation monitoring.** The Dimension 5 qualification principle relies on "broadly recognised evaluation benchmarks" as external instruments to discriminate between frontier and non-frontier models. These benchmarks saturate over time — a benchmark on which the top 20 models score within 1–2% of each other no longer discriminates meaningfully. Unlike the $\leq 7\text{nm}$ threshold, this is not a quantitative anchor requiring recalibration but a dependency on external measurement instruments whose discriminating power degrades. The methodology does not prescribe specific benchmarks (this is deliberate — benchmark relevance evolves faster than methodology versions), but assessors should document which benchmarks informed each assessment cycle's D5 verdicts. When the benchmarks available at an assessment date can no longer clearly separate "top tier" from the broader field, this should be flagged as a reduced-confidence assessment rather than treated as an expansion of the qualifying set. From V1.5, this documentation is a mandatory register metadata field — the per-cycle benchmark declaration defined in Manual §5.3.3 — rather than assessor guidance.

Additional Qualification Principles

7. **Sovereign control principle (governing rule).** Physical infrastructure dimensions (D2 fabrication, D4 compute) are assessed by territorial presence — the sovereign territory where infrastructure physically operates has effective control. Corporate capability dimensions (D1 accelerator design, D5 model development) are assessed by ownership chain — the corporate parent's territory has effective control through strategic direction and IP ownership. D3 (equipment) applies both tests: the entity must be headquartered in the actor's territory AND control the qualifying technology. D6–D7 (governance) are inherently sovereign. *Precedent basis:* International investment law (territorial jurisdiction vs nationality of investment in BIT framework), OECD permanent establishment doctrine, WTO rules of origin, CFIUS screening rationale. *Establishing cases:* Netherlands D4 (Nebius — Dutch corporate domicile, zero Dutch-territory compute → NOT MET); Taiwan D1 (MediaTek — Taiwanese firm, client retains architectural IP under design-service model → NOT MET under ownership chain test); US D2 (TSMC Arizona — Taiwanese-owned fab on US soil → MET under territorial test).
8. **D4 territorial compute requirement.** "Domestically controlled" in D4 requires operational compute on the actor's territory or under the actor's direct sovereign jurisdiction, not merely corporate domicile of the parent entity. Corporate ownership of compute infrastructure entirely abroad does not satisfy D4. Both D2 and D4 assess physical infrastructure; the sovereign control principle applies the same territorial presence test to both. *Establishing case:* Netherlands / Nebius Group N.V. — Dutch-domiciled company with zero qualifying compute on Dutch soil; all operational compute located in Finland (Mäntsälä). D4 NOT MET for the Netherlands.
9. **D6 dual-pathway clarification.** The Dimension 6 qualification principle covers both AI-specific legislation and general-purpose regulatory or trade control frameworks whose primary documented application targets AI systems, AI compute hardware, AI models, or AI-relevant manufacturing equipment. Export control regimes with documented extraterritorial enforcement qualify alongside dedicated AI-specific regulatory frameworks. This clarification ensures that actors exercising AI-relevant extraterritorial authority through trade controls (e.g., US export controls on advanced semiconductors and AI chips) are assessed equivalently to actors exercising authority through dedicated AI legislation (e.g., EU AI Act).
10. **D7 national proportionality standard.** "Nationally transformative" in Dimension 7 is assessed relative to the actor's own GDP, fiscal capacity, and industrial base — not against a global spending threshold. A modest absolute spend that represents a structural reorientation of a smaller actor's industrial policy may satisfy the dimension, whilst a large absolute spend by a major economy that represents marginal budgetary commitment may not. *Establishing case:* Netherlands — modest absolute spend relative to US or China, but representing fundamental reorientation of Dutch industrial and technology policy towards sovereign AI infrastructure capability.

11. **D4 assessment-date-relative threshold.** The D4 hyperscale threshold ("sufficient to train a current-generation frontier foundation model") is assessment-date-relative. "Current-generation frontier" shifts as frontier training scales up; no fixed GPU count is prescribed. At each assessment cycle, the threshold is calibrated against the most recent verified frontier training runs. Clusters significantly below the prevailing frontier training scale are clearly NOT MET; clusters at or above are clearly MET; actors between the two require case-specific evidence (empirical frontier-class training run, or demonstrable technical configuration showing frontier training feasibility). *Establishing cases:* South Korea NOT MET at ~4,000 B200 GPUs, Taiwan NOT MET at ~4,608 GB200, Germany NOT MET at ~10,000 Blackwell (~0.5 ExaFLOPS vs ~5+ ExaFLOPS required), India NOT MET at 16,384 H100 (empirical training run produced sub-frontier model).
12. **D7 general-infrastructure-fund exclusion.** General infrastructure funds not earmarked for AI or technology do not satisfy Dimension 7. The dimension assesses sovereign AI infrastructure investment, not aggregate government capital expenditure. Attribution of a country's entire infrastructure or defence budget to D7 inflates the figure and misrepresents the sovereign AI commitment. *Establishing case:* Germany — €500B Special Fund for Infrastructure and Climate Neutrality is a general infrastructure instrument, not creditable to D7.
13. **D7 announced-vs-executed investment.** Government adoption of committee recommendations or announcement of national project budgets does not satisfy D7 if execution data shows systematic under-delivery. The dimension requires evidence of actually appropriated and disbursed capital, not aspirational targets. Announced envelopes that have been systematically under-executed do not meet the "at scale" threshold. *Establishing case:* Russia — announced budgets exceeded executed spending by 5–20× across multiple budget cycles; Data Economy national project 11.2% executed by March 2025.
14. **D5 single-domain exclusion.** Frontier-competitive performance in a single capability domain (e.g., image generation, speech recognition, code) does not satisfy D5's "≥3 benchmarks across capability domains" requirement. The dimension tests general-purpose frontier capability, not domain-specific excellence. *Establishing case:* Germany — Black Forest Labs FLUX.2 is frontier in image generation but single-domain only.
15. **D2 marketing-node exclusion.** Semiconductor process nodes are assessed by their lithographic and transistor-density characteristics, not marketing labels. Intel's rebranded node names (Intel 7, Intel 4, Intel 3) are mapped to their actual technology class for assessment purposes. Intel 7 is assessed as 10nm-class (DUV, no EUV) and does not qualify for the ≤7nm threshold. *Establishing case:* Israel — Intel Fab 28 at Kiryat Gat operates Intel 7 / 10nm Enhanced SuperFin.
16. **Synthesis majority override rule.** A nominal input majority in the synthesis process can be overridden when the majority's interpretive framework contradicts its own cited evidence. The factual base must be agreed across inputs; the override applies on interpretive grounds only, where the majority draws a conclusion inconsistent with the evidence those same inputs cite. *Establishing cases:* France D5 (two of three inputs concluded MET; overridden — 44% GPQA Diamond vs 91–94% frontier is not "top tier" by any defensible reading); Russia D7 (two of three inputs concluded MET; overridden — announced budgets under-executed by 5–20×).

17. **D3 subsidiary clause.** The D3 dual test (HQ in territory + technology control) means foreign-parented subsidiaries with local manufacturing are assessed under the parent's territory, consistent with the subsidiary clause. This is distinct from D2, where the territorial presence exception credits any fab on an actor's soil regardless of ownership. D3 requires the qualifying entity to be headquartered in the actor's territory; a US-parented subsidiary manufacturing in Canada credits to the US, not Canada. *Establishing case:* Canada — Teledyne DALSA (Teledyne Technologies subsidiary, US parent) manufactures EUV pellicle membranes at Bromont, QC; credits to US D3, not Canadian D3. Consistent with exclusion of Applied Materials Israel and KLA Israel from Israel's D3.
18. **Actor Designation Framework temporal disclaimer.** Designation scores are assessment-date-stamped snapshots of current capabilities. They are not retrospective assessments of capabilities at the time each event in the database occurred. Bloc classifications for individual events were assigned under the methodology version current at the time of entry. The Actor Designation Framework does not retroactively alter event-level classifications made under prior methodology versions. This distinction is documented in the platform interface alongside the designation display.

Evidence Basis: Dimension qualification principles were calibrated through multi-agent independent AI research synthesis — a methodology necessitated by the absence of historical empirical data or established expert consensus for sovereign AI infrastructure capability assessment at dimension-level granularity. Multiple independent AI research inputs assessed each actor against each dimension independently; a dedicated synthesis process consolidated convergences, resolved divergences, and applied the majority override rule (addition 16 above) where interpretive disagreement conflicted with cited evidence.

Edge-case stress testing was conducted across four independent AI research inputs with dedicated synthesis (May 2026), producing 84 edge cases (3 per dimension × 7 dimensions × 4 inputs). Three dimensions required structural revision (D1, D3, D6); four required targeted additions (D2, D4, D5, D7). The cross-input convergence on the Zeiss/D3 subsystem gap (4/4 inputs, unanimous) and the subsidiary clause inconsistency (4/4 inputs in summary sections) provided the highest-confidence findings informing the initial design. Full Batch 1 actor assessments (14 actors across 3–4 independent AI research inputs per actor, with dedicated synthesis) produced 34 calibration precedents, 18 methodology additions, and 2 instances where nominal input majorities were overridden on evidence-quality grounds (France D5, Russia D7).

Confidence Assessment: 75%. Inherits from the parent PAA framework (75%) with calibration refinements from Batch 1 actor assessments. The edge-case stress test across four independent AI research inputs provided initial validation, and the qualification principles are applied across fourteen actors in the Batch 1 assessment cycle — producing a distribution (2 PAA, 3 AIK, 2 ACS, 7 Participant) that discriminates meaningfully across all four designations. The sovereign control principle, D4 territorial compute requirement, and D7 national proportionality standard are grounded in establishing cases across multiple actors. Thirty-four calibration precedents are established across all seven dimensions, covering the principal boundary conditions (inference-only exclusion, subsystem scope, frontier threshold calibration, operational-at-assessment-date, AI-specificity, foreign-base fine-tune exclusion, national proportionality, general-fund exclusion, announced-vs-executed investment, marketing-node exclusion, subsidiary clause in D3). Residual uncertainty remains at the boundary — the India/D5 (Sarvam-105B self-reported benchmarks) and France/D5 (Mistral ~35% frontier deficit) edge cases set the current D5 threshold but may shift with model releases.

Validation Commitment: Batch 1 assessment complete (14 actors). Continue reviewing qualification principles against Batch 2 actor assessment data. Particular attention to edge cases at the D5 frontier threshold (~35% deficit precedent from France/Mistral) and the D4 assessment-date-relative threshold (which will shift as frontier training compute increases). Threshold sensitivity review when any actor's score changes at the ≥ 3 boundary. Review the Dimension 2/3 process-node threshold ($\leq 7\text{nm}$) against the current fabrication frontier at each methodology version milestone; recalibrate downward when the threshold ceases to discriminate (see key design decision 5). Monitor Dimension 5 benchmark saturation and document which benchmarks informed each assessment cycle's D5 verdicts (see key design decision 6). Reassess subsidiary clause sensitivity at each methodology version milestone (see sensitivity note under key design decision 2). Review sovereign control principle test assignments if new dimensions are introduced (see key design decision 7). Monitor D7 national proportionality assessments for cross-actor consistency (see key design decision 10). Verify disbursement data for D7 where execution rates are flagged (see addition 13).

5.4.2 Three-Gate PAA Designation, Severance Test, and AIK Classification

Manual Section: §5.3.1 (Principal AI Actor Designation: Three-Gate Structure)

Category: CATEGORY CREATOR (MEDIUM-HIGH 60–75%)

Rationale: A capability threshold alone (≥ 3 of 7 dimensions) would conflate capability breadth with ecosystem independence. As additional actors accumulate capability breadth whilst remaining structurally dependent on an existing PAA's ecosystem for critical inputs, the threshold must be supplemented by a resilience assessment. Without the severance test, the designation would potentially classify Japan (deeply embedded in the US-centred ecosystem despite significant sovereign capabilities) as an ecosystem anchor equivalent to the United States. The three-gate structure separates capability (Gate 1), independence (Gate 2), and ecosystem provision (Gate 3) — three analytically distinct properties that a single threshold cannot discriminate.

Closest Precedents:

Framework	Institution	Relevance	Divergence from WEO
Basel III Stress Testing	Basel Committee on Banking Supervision (BIS, 2010/2017)	Closest methodological analogue: institutions must maintain capital ratios (quantitative threshold) AND survive hypothetical adverse scenarios (stress test). Dual-gate structure: quantitative floor + resilience assessment	Financial sector, not technology infrastructure; stress scenarios are macroeconomic, not supply-chain severance; assessment is continuous (capital ratio), not binary
Weaponised Interdependence: chokepoint vulnerability	Farrell & Newman (2019)	Establishes that network fragmentation depends on structural position and that chokepoint control creates asymmetric dependencies — the theoretical basis for why capability breadth alone does not imply ecosystem independence	Descriptive theory, not an assessment methodology; does not produce actor classifications
Strategic Autonomy discourse	European Commission (2021 Strategic Compass; 2023 Economic Security Strategy)	Policy framework asserting that capability without independence is insufficient for sovereign action — conceptual validation of the dual-gate principle	Political discourse, not a codified assessment methodology; "strategic autonomy" is not operationally defined with testable criteria
IISS Cyber Capabilities Assessment	IISS (2021/2023)	Tier 1 designation requires both capability breadth AND the ability to exercise cyber power independently; lower tiers explicitly note dependency limitations	Qualitative tiers without explicit severance criteria; cyber, not AI infrastructure

Key Design Decisions:

1. **Hypothetical withdrawal, not observed disruption.** The severance test uses a hypothetical scenario rather than waiting for actual supply-chain disruption. This follows Basel III stress testing methodology: the test assesses structural resilience, not historical experience. Waiting for actual ecosystem severance before classifying would provide accurate data only after catastrophic events.
2. **24–36 month sustainability window.** The test asks whether met dimensions would remain operationally sustainable within 24–36 months, not indefinitely. This window reflects the approximate timeline for emergency substitution of critical AI infrastructure inputs (alternative chip supply, equipment qualification, model retraining on different compute). A shorter window would over-credit actors with large existing inventories; a longer window would under-credit actors with genuine but time-limited domestic capability.
3. **Three capability categories with ≥ 2 spanning requirement.** The hardware/platform/governance trichotomy prevents an actor from qualifying as a PAA based solely on regulatory authority and investment (governance dimensions) without any hardware or platform independence. An actor exercising extraterritorial regulation and directing sovereign investment but possessing no domestic accelerator design, fabrication, equipment, cloud compute, or frontier models is not an ecosystem anchor — it is a regulatory power operating within another actor's technological ecosystem.
4. **Advanced Capability State as a distinct classification.** Rather than treating the severance test as a simple pass/fail that leaves non-qualifying actors undifferentiated, the methodology introduces a distinct classification tier. This captures the analytically significant difference between an actor scoring 4/7 with ecosystem dependence (e.g., the EU, which fails the severance test due to Intel ownership and Cymer dependency) and an actor scoring 0/7 (e.g., UAE). The "advanced capability state" classification is informative for users assessing supply-chain permission risk.
5. **AI Infrastructure Keystone (AIK) as a distinct classification.** The three-gate structure produces a classification tier between PAA and ACS: actors that score ≥ 3 , pass the severance test (demonstrating genuine ecosystem independence), but hold no sustainable platform dimension (D4 or D5). These actors exercise structural chokepoint leverage within an ecosystem — through equipment monopolies, fabrication capacity, or governance authority — without providing the ecosystem substrate (compute or model development platforms) upon which other actors operate. The AIK concept may be inherently asymmetric: it characterises the distributed Western supply chain (where equipment, fabrication, and governance are dispersed across allied actors) rather than vertically integrated ecosystems (where a single actor controls the full stack by necessity). *Establishing cases:* Japan (4/7, D3/D4/D6/D7, severance PASS; D4 met at assessment date but degraded-but-sustained under severance due to US-designed accelerator dependency), South Korea (4/7, D2/D3/D6/D7, severance PASS), Netherlands (3/7, D3/D6/D7, severance PASS).

Evidence Basis: The gate structure addresses the case where actors accumulate capability breadth whilst remaining ecosystem-dependent — Japan at 4/7 is the defining example. Analysis of Japan's dependency chain demonstrates that its met dimensions (equipment, compute, regulation, investment) include structural dependencies on US-ecosystem inputs (accelerator design, frontier models) — a clear case where capability breadth does not automatically imply ecosystem equivalence with the two PAAs. Basel III stress testing provides the strongest structural precedent for combining a quantitative threshold with a resilience assessment. The third gate (sustainable platform dimension) addresses actors that pass the severance test but hold no sustainable platform dimension (Japan, South Korea, Netherlands) — distinguishing chokepoint leverage from ecosystem anchoring. The AIK classification resolves this by distinguishing actors that shape supply chains from actors that provide the ecosystem substrate. The severance test is applied across seven actors with two distinct outcomes: PASS (US 7/7 sustainable, China D1/D5/D6/D7 sustainable with D2 and D4 degraded-but-sustained, Japan D3/D6/D7 sustainable with D4 degraded-but-sustained, South Korea D3/D6/D7 sustainable with D2 degraded-but-sustained, Netherlands D3/D6/D7 with D3 sustained under stress with multi-vector exposure from Cymer jurisdiction plus five additional US leverage vectors identified in FDPR sub-tier research) and FAIL (EU — D2 collapses on Intel ownership, D3 at structural risk from Cymer and broader US leverage; Taiwan — all three MET dimensions structurally coupled to US ecosystem). The severance test's operational components — ecosystem inputs definition, per-PAA testing, deterrence principle, cascade boundary, and dimension sustainability classifications — are documented in §5.4.3–§5.4.7.

Confidence Assessment: 70–75%. The three-gate structure addresses a genuine analytical gap in the parent framework, and the Basel III stress testing analogy provides strong methodological precedent for the approach. The severance test is applied across seven actors (US, China, Japan, South Korea, Netherlands, EU, Taiwan), producing two distinct outcomes (PASS: US, China, Japan, South Korea, Netherlands; FAIL: EU, Taiwan) that are internally consistent and analytically meaningful. The three-gate structure produces a distribution (2 PAA, 3 AIK, 2 ACS) that discriminates meaningfully among actors scoring ≥ 3 . Confidence is bounded below 80% because: (a) the 24–36 month sustainability window remains a calibration choice without direct precedent in technology infrastructure assessment, though its application across seven actors has not produced anomalous results; (b) the AIK classification is a novel construct whose long-term analytical utility will depend on whether the structural pattern it captures (chokepoint leverage without platform provision) remains stable across assessment cycles and Batch 2 actors; and (c) the three-category spanning requirement (hardware/platform/governance) is a novel analytical construct. Confidence is above 70% because: the structural analogy to Basel stress testing is well-grounded; the underlying theoretical basis (Farrell & Newman's weaponised interdependence) is established at very high confidence; the seven-actor application has produced a distribution that discriminates meaningfully; and the alternative (single- or dual-gate structures) produces demonstrably less useful classifications.

Validation Commitment: Continue applying the three-gate structure to Batch 2 actors. Review the 24–36 month window and category spanning requirement against cumulative results. Monitor for actors that transition between designations as their dependency chains evolve (Dimension Watch items tracked in the register). Assess whether the AIK classification produces stable, useful discrimination or converges with ACS over time. Review at each methodology version milestone.

5.4.3 Ecosystem Inputs Definition

Manual Section: §5.3.1 (Severance Test Criteria — "Ecosystem inputs defined")

Category: CATEGORY CREATOR (MEDIUM-HIGH 60–75%) — refined, inherits 70–75% from §5.4.2

Rationale: The ecosystem inputs definition uses a two-part (a)/(b) structure to draw a precise boundary around what the severance test withdraws: (a) inputs directly produced or controlled by entities under the PAA's sovereign jurisdiction, and (b) inputs produced by entities outside the PAA's jurisdiction but subject to denial through the PAA's extraterritorial regulatory instruments (FDPR, direct EAR jurisdiction). This boundary follows the PAA's sovereign denial capability — its ability to prevent an input from reaching the actor under test — rather than its manufacturing footprint or its geopolitical bloc membership. The China severance assessment established this boundary: Korean HBM blocked by FDPR is treated as a US ecosystem input (the US controls access through FDPR), whilst Zeiss and TRUMPF (both German and Western Bloc members) are treated as unaffected by US withdrawal (the US cannot deny their output).

The definition explicitly excludes bloc-wide withdrawal — "The severance test withdraws ecosystem inputs as defined above; it does not withdraw all inputs originating from actors aligned to the PAA's geopolitical bloc" — because ecosystem inputs and bloc membership are distinct concepts. An actor can be in the same geopolitical bloc as the PAA without its inputs being subject to the PAA's denial capability. The boundary is drawn at what the PAA can deny, not at who the PAA is aligned with.

The (b) formulation including direct EAR jurisdiction was supported by the FDPR sub-tier research (May 2026), which found that ASM International's epitaxy product line is designed at ASM America in Phoenix, AZ — making it subject to direct EAR jurisdiction, not merely FDPR-eligible. The definition's reference to "inputs subject to direct Export Administration Regulations jurisdiction from US-based design or manufacturing activity" captures this finding.

Key Design Decisions:

1. **Boundary drawn at denial capability, not production.** A non-PAA actor's inputs blocked by the PAA's extraterritorial instruments (e.g., Korean HBM blocked by FDPR) are PAA ecosystem inputs because the PAA controls access to them. Inputs from the same non-PAA actor that are not blocked remain available. The boundary follows the PAA's sovereign reach, not its manufacturing footprint.
2. **Explicit bloc exclusion.** The final sentence of the definition — "it does not withdraw all inputs originating from actors aligned to the PAA's geopolitical bloc" — is operationally necessary because without it, an assessor could conflate ecosystem withdrawal with bloc withdrawal, producing trivially incorrect FAIL verdicts for every Western actor.

Evidence Basis: China severance assessment (FDPR ruling on Korean HBM and ASML equipment); Netherlands severance assessment (Zeiss/TRUMPF treated as non-US, unaffected — establishing the ecosystem ≠ bloc boundary); FDPR sub-tier multi-agent AI research synthesis (9 entities across Japan, South Korea, Netherlands; May 2026) confirming the ASM International direct-EAR finding.

Establishing Cases:

- China D1/D2: Korean HBM blocked by FDPR → US ecosystem input (withdrawn under severance)
- Netherlands D3: Zeiss (German) and TRUMPF (German) → not US ecosystem inputs (available under US severance, despite being Western Bloc members)
- Netherlands D3: ASM International epitaxy line designed at ASM America, Phoenix → direct EAR jurisdiction (US ecosystem input regardless of de minimis threshold)

Confidence Assessment: Inherits 70–75% from §5.4.2. The (a)/(b) boundary is grounded in the PAA's sovereign denial capability — a precise, testable criterion supported by the FDPR sub-tier research confirming direct EAR jurisdiction findings.

Validation Commitment: Review if new extraterritorial regulatory instruments emerge beyond FDPR/EAR. Monitor for cases where the ecosystem/bloc boundary produces anomalous results. Reassess if a third PAA emerges, as the definition's applicability to non-US extraterritorial instruments would need testing.

5.4.4 Per-PAA Severance Testing (Option B)

Manual Section: §5.3.1 (Severance Test Criteria — per-PAA testing)

Category: CATEGORY CREATOR (MEDIUM-HIGH 60–75%) — refined, inherits 70–75% from §5.4.2

Rationale: The choice of individual-PAA testing (each PAA tested separately, actor must pass each) over simultaneous-PAA testing (all PAAs withdrawn at once) follows established institutional practice for structural dependency assessment. Individual-scenario testing is the institutional default for classification and certification across financial, infrastructure, and supply-chain frameworks. Combined worst-case scenarios are reserved for non-binding strategic planning and resilience exercises. The methodological principle is that single-source dependency is a structural property of the assessed actor, whilst systemic fragility is a property of the global environment. Conflating them in one test produces a result that cannot be attributed to either cause, destroying the diagnostic value that classification is designed to provide. The severance test asks whether an actor's capability stack is structurally contingent on another actor's permission — a question that requires isolating each potential source of contingency.

Closest Precedents:

Framework	Institution	Scenario Application	Relevance to WEO
NERC Reliability Standard TPL-001-4/5.1	North American Electric Reliability Corporation	Planning Events (P1–P7) tested individually, binding; Extreme Events (cascading) evaluative only	Closest formal codification: Planning Events ↔ WEO Option B (binding individual test); Extreme Events ↔ Option C (evaluative cascading study)
Critical Raw Materials Act, Article 5	European Union (Regulation 2024/1252)	65% single-third-country cap as binding law	Legislatively binding single-source dependency threshold — cleanest legislative formalisation of the principle WEO applies
EBA Guidelines on Recovery Plan Scenarios (EBA-GL-2014-06)	European Banking Authority / FSB	Idiosyncratic, systemic, and combined shocks tested as separate parallel exercises	Explicit principled statement that different shock types must be tested separately because they answer different methodological questions
EU Strategic Dependencies (SWD 2021/352)	European Commission	HHI concentration metrics across ~5,000 products; single-supplier shock modelling	Operationalises single-source dependency through product-level concentration analysis
Basel Large Exposures Framework (BCBS d283)	Basel Committee	25% of Tier 1 capital cap on single-name exposure	Regulatory codification of single-counterparty dependency limit

Key Design Decisions:

1. **Per-PAA individual testing for structural classification.** The severance test isolates each PAA's ecosystem inputs and tests withdrawal individually because the framework identifies structural dependency on specific ecosystems — a property of the actor being assessed. Every institutional precedent surveyed applies individual-scenario testing for classification and certification: NERC tests Planning Events individually (P1–P7), EBA mandates separate idiosyncratic and systemic exercises, and the EU Critical Raw Materials Act sets single-country concentration thresholds. Simultaneous multi-source withdrawal addresses a different analytical question (systemic environmental fragility) and is used in planning contexts, not classification.

Evidence Basis: Deep research across two independent AI research inputs (May 2026) surveying Basel III, IISS, NATO, NIST, NIS2, G-SIB, EU Strategic Dependencies, EU CRM Act, and IMF/OECD trade vulnerability frameworks. Finding: individual-scenario testing is the consensus institutional default for classification. The NERC TPL-001 Planning Events / Extreme Events bifurcation provides the single strongest formal precedent.

Confidence Assessment: Inherits 70–75% from §5.4.2. The per-PAA choice is strongly supported by institutional precedent.

Validation Commitment: Monitor for the emergence of a third PAA, at which point secondary and tertiary PAA testing becomes operationally significant. Review the per-PAA vs simultaneous distinction if any actor's designation depends on which approach is used.

5.4.5 Deterrence Principle

Manual Section: §5.3.1 (Severance Test Criteria — "Deterrence principle")

Category: CATEGORY CREATOR (MEDIUM-HIGH 60–75%) — refined, inherits 70–75% from §5.4.2

Rationale: The deterrence principle classifies a dependency as "sustained under stress" rather than "collapsed" when the withdrawn PAA holds jurisdiction over a subsidiary or facility whose denial would be self-destructive for the PAA. This addresses the analytical problem that some dependencies are theoretically severable but practically unseverable because the PAA cannot exercise its denial capability without harming its own infrastructure capability. The three-part documentation requirement ensures assessor consistency by requiring explicit evidence for each deterrence claim.

Key Design Decisions:

1. **Three-part documentation requirement.** Each deterrence assessment must document: (a) the specific facility at risk with jurisdictional basis, (b) the denial mechanism, (c) the specific cost to the PAA. The requirement ensures consistency across assessments — the Cymer dependency affects both the Netherlands and the EU, requiring explicit documentation of why deterrence applies differently for each (the Netherlands holds direct sovereign jurisdiction over ASML; the EU does not). The three-part structure makes each assessment explicit and auditable.
2. **Multi-vector leverage assessment.** The FDPR sub-tier research (May 2026) revealed that the US holds five independent leverage vectors over Netherlands D3 beyond Cymer: ASML Wilton reticle top module (confirmed physical hardware), Brion computational lithography software (US-developed), HMI e-beam R&D, ASM International epitaxy line (direct EAR jurisdiction), and Besi Kinex cluster (Applied Materials co-development). Multiple leverage points weaken the deterrence argument because the PAA can apply graduated pressure through secondary vectors without triggering the self-harming consequences of disrupting the primary entity. The "sustained under stress with multi-vector exposure" classification captures this narrower margin of sustainability.
3. **Supranational entity clause.** Deterrence based on a member state's sovereign leverage over a headquartered entity does not automatically transfer to the supranational level. ASML is headquartered in the Netherlands (an EU member state), but the EU as a supranational entity does not hold the direct corporate-law jurisdiction, nationalisation authority, or operational sovereignty over ASML that the Netherlands holds as ASML's home state. The EU can regulate ASML (competition law, Foreign Subsidies Regulation) but cannot ensure ASML's continued operation against PAA pressure in the way the Netherlands can. The clause ensures that each actor's deterrence assessment is based on its own sovereign jurisdiction over the relevant entity, producing the correct result: deterrence applies for the Netherlands (home state) but not for the EU (supranational entity).

Establishing Cases:

- Netherlands D3 / Cymer: deterrence applied → "sustained under stress with multi-vector exposure" (five additional US leverage vectors beyond Cymer)
- EU D3 / Cymer: deterrence NOT applied → "at structural risk" (supranational entity lacks direct sovereign jurisdiction over ASML; EU does not satisfy the supranational entity clause)

Evidence Basis: Netherlands severance assessment (Cymer deterrence precedent); EU severance assessment (supranational entity exclusion precedent); FDPR sub-tier multi-agent AI research synthesis (five US leverage vectors beyond Cymer confirmed: Wilton reticle module, Brion software, HMI R&D, ASM International direct EAR, Besi/Applied Materials partnership).

Confidence Assessment: Inherits 70–75% from §5.4.2. The deterrence principle is grounded in standard deterrence logic (mutual assured destruction applied to supply-chain dependencies). The multi-vector extension and supranational entity clause are grounded in deterrence logic and supported by establishing cases.

Validation Commitment: Monitor for changes in PAA leverage vectors (e.g., if Cymer operations are relocated to Europe, if ASML reduces US-origin content). Review supranational entity clause if EU institutional authority over corporate operational decisions strengthens. Monitor for new deterrence cases in Batch 2 assessments.

5.4.6 Cascade Analysis Boundary

Manual Section: §5.3.1 (Severance Test Criteria — "Cascade analysis boundary")

Category: CATEGORY CREATOR (MEDIUM-HIGH 60–75%) — refined, inherits 70–75% from §5.4.2

Rationale: The single-level cascade boundary limits the depth of dependency analysis in the severance test. The assessor checks whether the actor's primary capability-sustaining entities directly depend on the withdrawn PAA's ecosystem inputs. Sub-tier dependencies (e.g., whether a Japanese equipment company's sub-components contain US-origin FPGAs) are documented as constraints but do not determine the severance verdict. This boundary balances analytical rigour with practical tractability: full sub-tier supply chain mapping for every entity in every MET dimension is intractable from public sources, whilst ignoring dependencies entirely would produce unreliable verdicts.

Key Design Decisions:

1. **Single-level analysis is sufficient.** The FDPR sub-tier research (May 2026) supported the single-level standard by demonstrating that entity-level analysis — examining US-origin design centres, US-acquired IP, US-jurisdiction subsidiaries, and US-dominated subsystem categories — identified the structural exposure patterns that determine severance verdicts. The research identified material findings at the single-level (ASM International direct EAR jurisdiction, ASML Wilton reticle module, Hanmi HIGH FDPR exposure) without requiring intractable component-level BOM analysis.
2. **Sub-tier findings documented as constraints, not verdicts.** Sub-tier FDPR/EAR exposure is real but unverifiable from public sources at the component level. Incorporating unverifiable findings into the primary severance verdict would degrade the methodology's evidence standard (§5.3.4). The constraint register captures the finding without degrading the methodology — creating a structured record for future investigation without changing current designations.
3. **Single-level boundary extends to non-PAA supply chain assessment.** When criterion 1 requires assessing whether non-PAA actors' supply chains would remain functional under severance, the same single-level boundary applies: the assessor checks whether the non-PAA actor's primary entities depend on the withdrawn PAA's ecosystem inputs, without tracing sub-tier dependencies.

Evidence Basis: FDPR sub-tier multi-agent AI research synthesis (9 entities across Japan, South Korea, Netherlands). Entity-level (single-level) analysis identified: TEL MODERATE FDPR exposure, Lasertec LOW-MODERATE, SCREEN MODERATE-LOW, HPSP MODERATE, Hanmi HIGH, ASML HIGH (non-Cymer), ASM International HIGH (direct EAR), Besi HIGH (Kinex) / MODERATE (standalone). These findings were consequential for constraint documentation (South Korea and Netherlands flagged for review) without requiring sub-tier BOM analysis.

Confidence Assessment: Inherits 70–75% from §5.4.2. The single-level boundary is supported by the FDPR research demonstrating that material findings emerge at entity level.

Validation Commitment: Monitor whether sub-tier findings (documented as constraints) produce designation-changing evidence in future assessment cycles. If sub-tier FDPR exposure is confirmed to change any entity's sustainability classification, reassess whether the single-level boundary should be tightened for specific tool categories.

5.4.7 Dimension Sustainability Taxonomy

Manual Section: §5.3.1 (Severance Test Criteria — "Dimension sustainability classifications")

Category: CATEGORY CREATOR (MEDIUM-HIGH 60–75%) — refined, inherits 70–75% from §5.4.2

Rationale: The five-level dimension sustainability taxonomy provides a closed set of classifications for assessing each met dimension under each per-PAA severance scenario. The five levels — sustainable, sustained under stress, sustained under stress with multi-vector exposure, degraded-but-sustained, and at structural risk — capture the analytically significant gradations between full independence and structural collapse. The taxonomy includes an explicit counting rule: only the first four levels count towards the ≥ 3 threshold; "at structural risk" does not.

Key Design Decisions:

1. **Five levels rather than binary (pass/fail).** A binary dimension-level classification (sustainable / not sustainable) would lose analytically significant information. The difference between "sustainable" (Japan D3 — no US dependency in equipment itself) and "sustained under stress with multi-vector exposure" (Netherlands D3 — five US leverage vectors but deterrence holds) is material for risk assessment, even though both count towards the ≥ 3 threshold. The taxonomy preserves this gradation.
2. **"Degraded-but-sustained" as a counting level.** A dimension that degrades under severance but remains above the §5.3.2 qualification threshold counts towards the ≥ 3 criterion. This is analytically correct because the severance test asks whether the actor can sustain its capability stack, not whether it can sustain it without any degradation. China's D2 (SMIC on installed DUV equipment, no new EUV access) degrades significantly but retains $\leq 7\text{nm}$ production within the assessment window — this is a meaningful capability, not an analytical fiction.

Establishing Cases:

Classification	Actor/Dimension	Basis
Sustainable	Japan D3 (TEL, Lasertec, SCREEN)	No material dependency on US ecosystem inputs for core product lines
Sustained under stress	(Pre-FDPR) Netherlands D3	Cymer dependency with deterrence applied
Sustained under stress with multi-vector exposure	Netherlands D3	Five additional US leverage vectors beyond Cymer weaken deterrence calculus
Degraded-but-sustained	China D2 (SMIC)	Existing DUV production continues but cannot advance to new nodes
Degraded-but-sustained	China D4 (Ascend + stockpiled NVIDIA)	Existing compute degrades but retains frontier training capacity within window
Degraded-but-sustained	Japan D4 (SoftBank cluster)	Existing NVIDIA GPUs operational, no replenishment under US severance
Degraded-but-sustained	South Korea D2 (Samsung Foundry)	Existing ASML equipment operational, no resupply under US severance
At structural risk	EU D2 (Intel Fab 34)	US-owned facility; Intel could withdraw
At structural risk	EU D3 (ASML)	Cymer + multi-vector exposure; supranational entity clause blocks deterrence
At structural risk	Taiwan D2 (TSMC)	Requires US equipment and EDA tools
At structural risk	Taiwan D3 (Gudeng, GMM, GPTC)	Subsystem value contingent on fab ecosystem; D2 collapse renders subsystem capability moot
At structural risk	Taiwan D7 (NCHC/TSMC programme)	Programme procures NVIDIA GPUs; compute delivery dependent on US ecosystem

Confidence Assessment: Inherits 70–75% from §5.4.2. The five classifications are grounded in the analytical distinctions required by the severance test, with establishing cases across all five levels.

Validation Commitment: Monitor whether the five levels remain sufficient or whether Batch 2 assessments reveal edge cases requiring additional classifications. Review the counting rule (first four count, "at structural risk" does not) if any actor's designation depends on the boundary between "degraded-but-sustained" and "at structural risk."

5.4.8 AI Infrastructure Keystone (AIK) Designation

Manual Section: §5.3.1 (Actor Designation Framework — Gate 3)

Category: ADAPTED (HIGH 75-90%) — with synthesis elements

Rationale: The AIK designation classifies nation-state actors scoring ≥ 3 dimensions, passing the per-PAA severance test, but holding no sustainable platform dimension (D4 Cloud AI Compute or D5 Frontier AI Model Development). These actors hold structural chokepoint leverage within an AI ecosystem without providing the ecosystem itself. The structural distinction between ecosystem-providing power and chokepoint-leveraging power is not novel to WEO — it has been independently theorised across international relations, ecology, industrial organisation, network science, and documented in historical cases. WEO's contribution is the bounded application of this well-established structural distinction to sovereign AI infrastructure capability assessment, combined with operational calibration ($\geq 3/7$ dimensional threshold, 24-36 month severance test, Gate 3 platform-dimension test).

The designation was supported by cross-disciplinary deep research across four independent AI research inputs with dedicated synthesis (May 2026), surveying frameworks across five domains. Eight independent intellectual traditions converge on essentially the same bifurcation between ecosystem-providing/dominant/hub/platform power and chokepoint-leveraging/keystone/bridge/bottleneck power, suggesting the distinction tracks a real structural feature of complex networked systems.

Closest Precedents:

Framework	Author/Institution	Year	Relevance	Divergence from WEO
Vertical vs Horizontal Foreclosure	Rey & Tirole	2007	Primary theoretical anchor. Formally separates bottleneck-input power (AIK) from end-market dominance (PAA). Near-isomorphic. Embedded in peer-reviewed IO theory and active antitrust case law (US Section 2; EU Art. 102).	IO/competition framework, not geopolitical; treats firms, not nation-states
Structural vs Relational Power	Strange	1988	Cleanest IR-native binary. PAA = structural power (shapes the game); AIK = relational power (leverages position within the game).	Strange's four structures (security, production, finance, knowledge) do not map precisely to WEO's seven dimensions
Weakest-Link vs Best-Shot Aggregation	Hirshleifer	1983	Most elegant formal encoding. AIK = weakest-link gatekeeper whose removal degrades the system; PAA = best-shot provider whose contribution defines the frontier.	Public goods model; requires adaptation to supply-chain context
Keystone vs Dominant Species	Paine (1966); Power et al. (1996)	1966/1996	Strongest ecological analogue. AIK = keystone species (disproportionate systemic impact relative to abundance); PAA = dominant species (proportionate impact, high abundance). Power et al.'s canonical abundance-impact framework formalises the keystone concept.	Ecological metaphor requires disambiguation from Iansiti & Levien (2004) who use "keystone" for the PAA/hub position
Degree vs Betweenness Centrality	Freeman	1977/1979	Cleanest mathematical formalisation. PAA = high-degree hub (many connections); AIK = high-betweenness bridge (controls flow between clusters).	Pure scale-free networks correlate hub and bridge positions; distinction requires modular network structure (empirically satisfied in semiconductor GVCs)
DMA Gatekeeper vs Art. 102 Essential Facility	EU regulatory architecture	2022	Institutional embodiment: DMA governs PAA-type platform gatekeepers; Art. 102/Bronner governs AIK-type essential facility holders. Distinction embedded in positive law.	Regulates firms within single market, not nation-states across blocs
Weaponised Interdependence: Chokepoint vs Panopticon	Farrell & Newman	2019	Most cited contemporary geopolitical vocabulary. Chokepoint effect (denial = AIK) vs panopticon effect (surveillance/ecosystem provision = PAA).	Under-theorises non-hegemonic chokepoint case: F&N locate both effects in the same state (US), whereas WEO separates ecosystem providers from small-state chokepoint holders
Geopolitical Pivots vs Geostrategic Players	Brzezinski	1997	AIK = geopolitical pivot (importance derives from position, not intrinsic power); PAA = geostrategic player (importance derives from power and strategic vision).	Cold War framing; geographic rather than supply-chain positioning

Key Design Decisions:

1. **ADAPTED classification, not CATEGORY CREATOR.** The PAA/AIK structural distinction is pre-existing across five disciplines. Someone familiar with weaponised interdependence, keystone ecology, or vertical foreclosure theory could independently arrive at a similar classification. WEO's contribution is the bounded application to sovereign AI infrastructure capability and the operational calibration ($\geq 3/7$ threshold, severance test, Gate 3), not the underlying typology.
2. **Rey & Tirole (2007) as primary theoretical anchor.** Rey & Tirole's vertical/horizontal foreclosure bifurcation provides the strongest single formal precedent for three reasons: (a) formal isomorphism — bottleneck-input power vs end-market dominance IS the AIK/PAA distinction in IO theory; (b) operational foundation — embedded in peer-reviewed theory, active case law, and positive-law regulatory architecture; (c) predictive content — generates testable hypotheses about different strategic behaviours for vertical and horizontal power holders.
3. **PAA/AIK as ideal-type endpoints, not mutually exclusive categories.** All five researched domains warn that ecosystem-provision and chokepoint-leverage co-occur in many actors. The US is both PAA (frontier models, cloud compute) and AIK-adjacent (CUDA software stack, EDA tools). The WEO $\geq 3/7$ dimensional threshold and severance test handle hybrid cases procedurally through the Gate 3 platform-dimension test, but the designations capture the dominant mode of structural leverage, not the exclusive mode.
4. **Iansiti-Levien disambiguation required.** Iansiti & Levien's *The Keystone Advantage* (2004) uses "keystone" for platform hub orchestrators (Microsoft, Walmart) — aligning with PAA, not AIK. WEO's "AI Infrastructure Keystone" aligns with Paine's (1966) ecological meaning — disproportionate systemic impact relative to abundance — not the Iansiti-Levien business-strategy meaning. Manual §5.3.1 includes the disambiguation parenthetical.
5. **AIK fragility acknowledged as structural feature, not classification defect.** Historical precedent demonstrates non-hegemonic chokepoint power erodes through substitution (OPEC post-1980), hegemonic coercion (Switzerland post-2008), or technological bypass (canal states). The 24–36 month severance test incorporates temporal bounds. AIK status is inherently provisional and subject to reassessment at each methodology version milestone, distinguishing it from PAA status which is more structurally durable. AIK leverage in the semiconductor supply chain is structurally deeper than commodity leverage — protected by cumulative process knowledge, IP portfolios, and capital barriers measured in decades — but it is not permanent.
6. **Österblom operational criteria adopted for AIK identification.** To inoculate against the keystone-concept vagueness critique (Mills, Soulé & Doak 1993; Cottee-Jones & Whittaker 2012), the designation adopts Österblom et al.'s (2015) four-criteria approach adapted for AI infrastructure: (i) production concentration, (ii) segment control, (iii) network reach, (iv) governance influence.
7. **Formalising the non-hegemonic chokepoint gap in weaponised-interdependence theory.** Farrell & Newman (2019) identified chokepoint and panopticon effects but located both in the same hegemon (US), under-theorising the case of small states hosting non-substitutable chokepoints within a hegemon's broader ecosystem. WEO separates hegemon-hosted chokepoints from small-state-hosted chokepoints that F&N gesture at but do not formalise.

Evidence Chain:

Source	Type	Contribution
Rey & Tirole (2007) "A Primer on Foreclosure," <i>Handbook of Industrial Organization</i> Vol. 3	Direct	Formally separates vertical foreclosure (AIK) from horizontal foreclosure (PAA). Near-isomorphic.
Strange (1988) <i>States and Markets</i>	Direct	Structural power (PAA) vs relational power (AIK). IR-native binary.
Hirshleifer (1983) "From Weakest-Link to Best-Shot," <i>Public Choice</i> 41(3)	Direct	Best-shot (PAA as frontier provider) vs weakest-link (AIK as supply-chain bottleneck).
Paine (1966) "Food Web Complexity and Species Diversity," <i>American Naturalist</i> 100(910)	Direct	Keystone species concept — disproportionate impact relative to abundance = AIK.
Power et al. (1996) "Challenges in the Quest for Keystones," <i>BioScience</i> 46(8)	Direct	Formalised keystone-vs-dominant distinction.
Freeman (1977/1979) centrality measures, <i>Sociometry</i> 40(1); <i>Social Networks</i> 1(3)	Direct	Degree centrality (hub = PAA) vs betweenness centrality (bridge = AIK).
Farrell & Newman (2019) "Weaponized Interdependence," <i>International Security</i> 44(1)	Structural	Chokepoint (AIK) vs panopticon (PAA). Under-formalises non-hegemonic case.
EU DMA Reg. 2022/1925; Case C-7/97 <i>Bronner</i> (1998)	Structural	DMA governs PAA-type gatekeepers; Art. 102/Bronner governs AIK-type essential facilities.
Österblom et al. (2015) "Transnational Corporations as Keystone Actors," <i>PLoS ONE</i> 10(5)	Structural	Peer-reviewed biology-to-political-economy transfer. Four operational criteria.
Netherlands/ASML; CSET (2022) "Preserving the Chokepoints"	Empirical	Contemporary single-state chokepoint precedent.

Establishing Cases (Batch 1, May 2026):

Actor	Score	AIK Profile	Chokepoint Nature
Japan	4/7 (D3, D4, D6, D7)	Front-end equipment monopolies + sovereign compute (D4 degraded-but-sustained under severance)	TEL coater/developer ~90–100% EUV share; Lasertec 100% sole-source actinic mask inspection; SCREEN single-wafer cleans
South Korea	4/7 (D2, D3, D6, D7)	Memory packaging + fabrication + regulation	HPSP sole global source for high-pressure annealing; Hanmi ~71% TC bonder share; Samsung Foundry HVM at 2nm GAA
Netherlands	3/7 (D3, D6, D7)	Core lithography monopoly + export control leverage	ASML 100% EUV monopoly; ASM International ALD/epi; Besi hybrid bonding; Strategic Goods Decree enforcement

Confidence Assessment: 82% (bounded range: 72–90%). The structural distinction is well-grounded across five independent disciplines, with at least one framework (Rey & Tirole 2007) providing near-isomorphic formal correspondence. Institutional embodiment in positive law (EU DMA vs Art. 102/Bronner) provides additional validation. Four independent AI research inputs converge on the distinction's validity. Confidence exceeds the parent three-gate framework (§5.4.2, 70–75%) because the AIK/PAA structural distinction is independently grounded across five disciplines — it does not depend on the three-gate calibration choices (24–36 month window, ≥ 2 category spanning) that bound the parent framework's confidence. The cross-disciplinary convergence provides validation from outside the WEO methodology. Confidence is capped below 90% because: (a) PAA and AIK traits empirically co-occur in some actors; (b) historical chokepoint power erodes through substitution, diversification, or hegemonic coercion; (c) the Iansiti-Levien terminological collision requires active disambiguation. Confidence is held above 72% because: (a) cross-disciplinary convergence across five independent domains is unusually strong; (b) Rey & Tirole provides formal near-isomorphism; (c) the distinction is institutionally embodied in positive law; (d) three establishing cases from Batch 1 application produce a coherent structural pattern.

Validation Commitment: Monitor AIK designation stability across Batch 2 actor assessments. Review whether additional actors qualify and whether the structural pattern (hardware chokepoints without platform provision) remains stable. Monitor for technological developments that could erode AIK positions. Reassess at each methodology version milestone.

5.4.9 Advanced Capability State (ACS) Designation

Manual Section: §5.3.1 (Actor Designation Framework — Gate 2 failure)

Category: ADAPTED (HIGH 75–90%)

Rationale: The ACS designation classifies nation-state actors scoring ≥ 3 dimensions but failing the per-PAA severance test — actors with significant sovereign capabilities that nonetheless cannot sustain those capabilities autonomously under withdrawal from the dominant ecosystem. ACS is the structural residual of the severance test: actors that pass the capability gate (Gate 1) but fail the resilience gate (Gate 2). The concept of "capable but dependent" actors is well-established in international relations and technology governance literature.

Closest Precedents:

Framework	Author/Institution	Year	Relevance	Divergence from WEO
IISS Cyber Capabilities Tier 2	IISS	2021/2023	Tier 2 states possess "significant" cyber capability but lack the ability to exercise it independently across all domains	Cyber, not AI infrastructure; qualitative assessment without binary severance test
"Coordinated Stack Builders"	Brookings Institution	2026	Nations with multi-layer AI capability that coordinate within (rather than define) an ecosystem	Includes non-infrastructure variables; descriptive, not threshold-based
Basel III Stress Test Failures	Basel Committee	2010+	Banks meeting capital adequacy thresholds but failing stress scenarios	Financial sector; institutional stress, not geopolitical severance
NATO Interoperability Tiers	NATO	Various	Alliance members with capable military forces that cannot operate independently of the US logistics, intelligence, and command infrastructure	Military, not technology; alliance framework

Key Design Decisions:

1. **ACS is a severance-test outcome, not an independent structural concept.** Unlike AIK (which captures a distinct mode of structural power), ACS captures a dependency condition. The designation exists because the severance test produces a meaningful binary: actors whose capabilities survive ecosystem withdrawal (PAA/AIK) vs actors whose capabilities collapse (ACS).
2. **ACS does not imply weakness.** Both current ACS actors (EU 4/7, Taiwan 3/7) hold globally significant capabilities. The EU enforces the AI Act with extraterritorial reach; Taiwan hosts TSMC. ACS flags structural dependency, not capability deficit.
3. **ACS actors are attributed to a specific PAA ecosystem.** The severance test identifies the primary PAA dependency (for both current ACS actors: the United States). This attribution is documented in the register.

Establishing Cases (Batch 1, May 2026):

Actor	Score	Severance Failure Mechanism
European Union	4/7 (D2, D3, D6, D7)	D2 collapses (Intel Fab 34 — US-owned); D3 at structural risk (Cymer + multi-vector US leverage; supranational entity clause blocks deterrence)
Taiwan	3/7 (D2, D3, D7)	All three MET dimensions structurally coupled to US ecosystem: D2 requires US equipment and EDA tools; D3 subsystem value contingent on fab ecosystem; D7 programme procures NVIDIA GPUs

Confidence Assessment: 78%. The "capable but dependent" concept is well-established. The application via the severance test is a bounded domain transfer. The two establishing cases produce intuitively correct results.

Validation Commitment: Monitor whether ACS actors transition to AIK (if severance resilience improves) or to Participant (if capabilities degrade). The EU's severance profile may shift if Cymer operations are relocated or if European EDA alternatives emerge. Taiwan's profile is structurally stable absent fundamental changes to TSMC's equipment dependency. Review at each methodology version milestone.

5.4.10 Participant Designation

Manual Section: §5.3.1 (Actor Designation Framework — below Gate 1)

Category: ADAPTED (HIGH 75–90%)

Rationale: The Participant designation classifies all actors scoring fewer than 3 dimensions. These actors operate within ecosystem(s) shaped by PAAs and AIKs. No further subdivision is applied below the ≥ 3 threshold because the structural distinctions at this level are better captured by per-dimension data (which specific dimensions are met) than by additional designation categories.

Key Design Decisions:

1. **Single designation below ≥ 3 .** A single "Participant" designation covers all actors scoring fewer than 3 dimensions because: (a) the structural distinction between a 1/7 actor with a chokepoint and one without is already captured by which dimension is met; (b) subdividing below ≥ 3 introduces "designation competition" consuming attention disproportionate to the structural significance; (c) the G-SIB methodology provides precedent: banks below the systemic threshold are simply "non-G-SIB."
2. **Dimension profiles carry the analytical detail.** A Participant scoring D3 only (Germany — Zeiss/TRUMPF) has a fundamentally different structural profile from a Participant scoring D7 only (India — IndiaAI Mission). This difference is visible in the dimension data without requiring separate designations.

Establishing Cases (Batch 1, May 2026):

Actor	Score	Profile
United Kingdom	1/7 (D7)	Sovereign investment only
France	2/7 (D6, D7)	Governance dimensions only
Germany	2/7 (D3, D6)	Equipment chokepoints (Zeiss, TRUMPF) + export controls
India	1/7 (D7)	Sovereign investment only
Israel	1/7 (D3)	Inspection/metrology equipment only (Nova, Camtek)
Russia	0/7	No qualifying dimensions under sanctions
Canada	0/7	No qualifying dimensions; Tenstorrent redomiciled, Teledyne DALSA US-parented

Confidence Assessment: 85%. The concept of a residual category below a structural threshold is universal across comparable frameworks. The decision to use a single designation is conservative and well-supported.

Validation Commitment: Monitor whether per-dimension data alone provides sufficient analytical detail. If demand for subdivision emerges from user feedback, reassess. Review at each methodology version milestone.

5.4.11 Sovereign Capability Profiles Register

Manual Section: §5.3.3 (Sovereign Capability Profiles Register)

Category: ADAPTED (HIGH 75–90%)

Rationale: The register format directly adapts the operational pattern established by the WEO Bloc Membership Register (blocs.html), extending it to actor-level capability assessment. The versioning, dating, living-document status, and periodic reassessment mechanisms are carried over without modification. The addition of per-dimension source citations, constraint documentation, and the severance test result extends the register to accommodate the richer analytical content of capability assessment versus bloc membership classification.

Key Design Decisions:

1. **Dual-pathway register inclusion prerequisite.** The register inclusion prerequisite uses a dual pathway: event presence in the WEO Events Database, or ≥ 1 met dimension as defined in §5.3.2. The dual pathway ensures that actors with demonstrated AI infrastructure capability are eligible for inclusion regardless of database processing sequence. An actor that meets one or more AI infrastructure dimensions but has not yet had a verified event processed into the Events Database should not be excluded from the register solely due to event-processing timing — the capability assessment is independently grounded in the §5.3.2 qualification principles and §5.3.4 evidence standard. The dual pathway removes the sequencing dependency between event verification and register inclusion, allowing both analytical processes to operate independently.

Confidence Assessment: 80%. Direct adaptation of an established WEO operational pattern. The dual-pathway prerequisite removes the sequencing dependency between event verification and register inclusion.

Validation Commitment: N/A — operational format, not analytical methodology.

5.4.12 Evidence Standard for Dimension Qualification

Manual Section: §5.3.4 (Evidence Standard for Dimension Qualification)

Category: ADAPTED (MEDIUM-HIGH 60–75%)

Rationale: WEO's event verification and coordination connection standards (1 primary + 2 corroborating) are grounded in journalism and research best practice (Reuters triangulation norm, AP verification standards). The dimension qualification evidence standard required separate research because structural capability claims are epistemically distinct from event claims — a difference recognised across the frameworks surveyed but formalised only by ICD 203 (US Intelligence Community Directive on Analytic Standards), which doctrinally separates confidence levels (for structural judgments) from likelihood language (for events).

Closest Precedents:

Framework	Institution	Evidence standard	Relevance
ICD 203 Analytic Standards	US ODNI	Quality-over-quantity; confidence tied to source quality for structural judgments; likelihood for events	Only framework with formal structural/event doctrinal separation
Transparency International CPI	TI	≥3 of 13 vetted sources	Strongest published mathematical rationale ($\sigma/\bar{n}$); applies to perception composites, not binary structural claims
TechInsights market research	TechInsights	≥3 independent data vectors	Triangulation for commercial data where single-vendor self-reporting is unreliable
Reuters Handbook of Journalism	Reuters	≥2 sources	Century-old convergent journalism norm; strongest for event claims
ITU Global Cybersecurity Index	ITU	1 government submission + expert verification	Closest operational model to 1+1 for structural state-level capability
Sovereign credit rating agencies (S&P, Moody's, Fitch)	S&P/Moody's/Fitch	No fixed per-claim minimum; standardised datasets + committee judgement	Structural assessment anchored to authoritative primary data

Key Design Decisions:

1. **1+1 base standard.** The cross-framework finding is that structural claims are evidenced by fewer, higher-authority sources than event claims. A MET dimension anchored to a Grade 1 canonical instrument (statute, benchmark output, audited filing) carries its own verification — the primary instrument is durable, unambiguous, and publicly accessible. One independent corroboration confirms the claim is not misinterpreted. This standard is consistent with the operational practice of the most analogous frameworks (ITU Global Cybersecurity Index, sovereign credit rating agencies).
2. **Rule-based escalation rather than analyst discretion.** The four escalation conditions (opaque-regime self-reporting, contested claim, unaudited vendor disclosure, no canonical instrument) are defined upfront. This preserves predictability whilst capturing the cases where additional corroboration is genuinely needed.
3. **Mandatory source descriptors (from ICD 203).** Per-source quality annotation — source type, publisher, date, credibility caveats — ensures that each cited source's authority and limitations are transparent to reviewers. The combination of quantity (1+1), hierarchy (tiered authority), and descriptor (per-source quality annotation) provides a layered evidence architecture where source quality is assessed alongside source count.
4. **Structural/event asymmetry is deliberate.** The dimension standard (1+1 base) is lower than the event standard (1+2) because the evidence types are structurally different. WEO applies the same principle as CPI, S&P, and ICD 203: structural capability claims and discrete event claims require different evidence architectures.

Evidence Basis: Deep research conducted across three independent AI research inputs, surveying 23 frameworks across four categories (sovereign/institutional assessments, technology/infrastructure assessments, regulatory/legal assessments, journalism/intelligence best practice). The cross-framework consensus supported a quality-tiered base standard with rule-based escalation.

Confidence Assessment: 75%. The standard adapts well-established precedents to a novel application. Confidence is bounded below 80% because: (a) no civilian assessment framework publishes per-source quality descriptors at the level WEO now requires; (b) the escalation triggers are untested. Confidence is above 70% because: the structural/event distinction is well-grounded; the source hierarchy is consensus; the base standard matches analogous frameworks.

Validation Commitment: Monitor escalation trigger frequency across Batch 2 assessments. If >50% of MET dimensions trigger escalation, review whether the base standard is set too low. Track whether any dimension qualification is subsequently challenged on evidence grounds. Review source descriptor implementation for consistency across assessors.

Standing Review Positions (adopted 3 July 2026; reassess at each methodology version milestone)

1. **No eighth dimension** is admitted without a demonstrated chokepoint that no existing dimension can absorb. Candidate additions assessed and declined on this test: AI safety infrastructure (folds into D6 if certification becomes market-access-binding), data governance (D6 territory where extraterritorial; event-level otherwise), synthetic data generation (no chokepoint structure), energy (ecosystem-agnostic; routed through the Energy/Infrastructure domain and ENT framework).
 2. **No SCP influence on event tier classification** beyond the existing T1 PAA gate. Events answer what is occurring; profiles answer what each actor controls. Register narratives may cite linked-event tier profiles as context; this is rendering, not methodology.
 3. **The D7 threshold does not inflate with global routinisation.** The qualification's second disjunct ("or the actor's own structural position within it") and the national proportionality standard (calibration precedent 10) exclude threshold inflation by design; the dimension's discriminating work sits in the operational tests (announced-vs-executed, foreign-ecosystem exclusion, general-fund exclusion). Register-population pressure is managed at the inclusion rule (§5.3.3), not the D7 bar.
 4. **Recalibration decisions are designation-stability-checked before adoption.** The effect of any candidate threshold on every current register designation is computed and recorded before the recalibration is taken. The $\leq 5\text{nm}$ D2 check is on record: no designation changes (CN 6/7→5/7 remains PAA; EU 4/7→3/7 remains ACS); the same holds at $\leq 3\text{nm}$.
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Section 6: Domain Classification Rationale

Manual Reference: Chapter 6

6.1 Four-Domain Framework

Manual Section: 6.1

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%) — Domain categories follow established sectoral frameworks; durability ranges derive from empirical sources (IRS depreciation, NRC licensing, insurance tenors, WTO principles); durability-based tiebreaker hierarchy is WEO-original, operationalising implicit market judgements and path dependence theory into explicit classification methodology.

Rationale: The four domains capture distinct coordination dynamics with materially different durability profiles:

Domain	Durability	Empirical Basis
Energy/Infrastructure	20-60 years	Infrastructure asset lifespan data (IRS depreciation schedules, NRC licensing, TSMC operational evidence and industry analysis)
Security	15+ years effective horizon	Institutionalised alliance persistence; political risk insurance tenors (MIGA 20-year maximum; DFC political risk insurance up to 20 years); implementing agreement cycles (DCAs, arms control)
Technology	2-5 years	Technology lifecycle analysis; Moore's Law adaptation cycles
Trade	3-10 years	Trade agreement lifecycle data (WTO, bilateral FTAs)

Domain Durability Clarification

Technology Domain uses two overlapping ranges:

Context	Range	Typical Events
General classification	2-5 years	Standards, protocols, algorithms
AI Software events	2-7 years	Model releases, platforms, APIs

Rationale: AI software durability is more variable than general technology standards. Model releases (e.g., GPT-4 → GPT-5) have shorter lifecycles than protocol standards, whilst foundational AI platforms may persist longer.

Edge Cases: Foundational protocols (TCP/IP: 44+ years, IPv4: 40+ years) achieved durability through network effects, not design intent. These are exceptional and do not change typical domain classification. Original designers expected replacement.

Security Domain Durability Clarification

Security durability reflects *effective horizon* rather than formal treaty terms. Framework alliances (NATO, US-Japan, US-ROK) are formally perpetual with one-year withdrawal provisions, but institutionalised alliances create substantial practical lock-in through reputational costs, infrastructure entanglement, and institutional path dependence.

The "15+ years effective horizon" framing derives from convergent evidence:

Source	Finding	Implication
Political risk insurance	MIGA offers 20-year maximum coverage; DFC political risk insurance up to 20 years	Institutional investor consensus on reliable forecasting horizon for security commitments
Historical alliance persistence	Major post-WWII alliances (NATO, US bilateral treaties) have persisted 70+ years	Institutionalised alliances substantially exceed formal term structures
ATOP dataset	Historical bilateral alliance duration data (1815-2018, v5.1)	Historical volatility confirms alliances are not structurally perpetual
Withdrawal precedents	No NATO member has formally exited in 75 years; partial exits (France 1966-2009) and reversal of withdrawal attempts (Philippines VFA 2020-2021) confirm exit remains practically possible	High barriers but not absolute lock-in

Hierarchy Rationale: Security ranks below Energy/Infrastructure because the lock-in mechanisms differ fundamentally. Physical infrastructure creates *irreversible* sunk costs — a data centre, fab, or power plant cannot be "un-built." Security commitments create *substantial but reversible* reputational and legal sunk costs — a determined government can exit an alliance, however costly, whereas it cannot dismantle physical infrastructure retroactively. This irreversibility asymmetry, rather than formal duration, justifies the domain hierarchy.

6.2 Technology vs Energy/Infrastructure Distinction

Manual Section: 6.2

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: This distinction is material for classification. Physical AI infrastructure events carry stronger coordination signals than software events due to:

Characteristic	Energy/Infrastructure	Technology
Durability	20-60 years	2-5 years
Capital intensity	\$500M-\$50B+	\$10M-\$100M
Reversibility	Low (cannot relocate fabs)	High (software updates)
Coordination mechanism	Irreversible	Malleable

Government Framework Alignment:

- CFIUS reviews semiconductor fabs under national security authority
- CISA classifies data centres under the Information Technology and Communications critical infrastructure sectors
- Export controls target production-enabling capabilities, including physical equipment, technical data, software, and know-how transfers ("deemed exports" under EAR/ITAR). The regulatory focus is on items and knowledge that enable manufacturing and operational capabilities.

6.3 Domain Tiebreaker Hierarchy

Manual Section: 6.7

Category: WEO-ORIGINAL (MEDIUM-HIGH 60–75%) — derivative of durability ranges

Rationale: The tiebreaker hierarchy follows durability order — domains creating longer-lasting structural commitments take classification precedence in genuinely ambiguous cases:

1. **Energy/Infrastructure** (20-60 years) — Physical buildouts create highest-durability coordination
2. **Security** (15+ years effective horizon) — Treaties and alliances persist across political cycles
3. **Technology** (2-5 years) — Algorithmic/capability events take precedence over economic exchange
4. **Trade** (3-10 years) — Economic agreements

Derivation: This hierarchy is derivative of the empirically-grounded Domain Durability Ranges (*Manual Section 6.6*).

Note: True 50/50 splits are rare. Most events have identifiable primary domains; tiebreaker application is exceptional.

Technology > Trade Ordering Rationale:

Technology ranks above Trade in the tiebreaker hierarchy despite shorter nominal durability (2-5 years vs 3-10 years) based on mechanism analysis. Technology standards and capability events exhibit stronger path dependence and network effects than trade agreements (Arthur, 1989; David, 1985). Technology choices create self-reinforcing lock-in through increasing returns and switching costs. Trade agreements, by contrast, include explicit modification provisions and demonstrate higher practical reversibility.

Citations:

- Arthur, W. B. (1989). "Competing Technologies, Increasing Returns, and Lock-In by Historical Events." *The Economic Journal*, 99(394), 116-131.
 - David, P. A. (1985). "Clio and the Economics of QWERTY." *American Economic Review*, 75(2), 332-337.
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Section 7: Industry Tags Rationale

Manual Reference: Chapter 7

7.1 11-Category Industry Taxonomy with GICS Mapping

Manual Section: 7.3

Category: ADAPTED (HIGH 75–90%)

Rationale: The 11-category taxonomy adapts the Global Industry Classification Standard (GICS, maintained by S&P/MSCI since 1999) as its primary institutional reference, supplemented by NAICS (2022 revision) and ICB (FTSE Russell) where GICS does not capture Eastern Bloc industries adequately. Eight of eleven WEO categories map to specific GICS sub-industry codes; Financial Services & Investment maps at GICS sector level; Trade & Logistics and Quantum Computing Infrastructure have no direct GICS equivalent. The correspondence provides institutional investor alignment where mappings exist and documents gaps transparently where they do not.

The practice of mapping geopolitical events to industries they address is established. Caldara & Iacoviello (2022, *American Economic Review*) map geopolitical risk events to Fama-French 49-industry portfolios (based on SIC codes). The IMF Global Financial Stability Report (April 2025, Chapter 2) and Shen (2025, Federal Reserve Bank of Boston) conduct sector-differentiated geopolitical analysis using GICS-comparable groupings. Supply chain intelligence platforms (Resilinc EventWatchAI, Everstream Analytics) map events to industries using proprietary taxonomies.

Adaptation Gap: These precedents use industry groupings analytically — mapping events to industries for downstream risk analysis. WEO formalises this as a structured tagging operation within an event database, producing reproducible per-event industry metadata. No formal adaptation of GICS for classifying geopolitical events by addressed industry has been documented. This gap is real but bounded: the measurement concept is established; the operationalisation as structured event metadata is the contribution.

Evidence Chain:

Precedent	Type	Contribution
GICS (S&P/MSCI, 1999)	Direct (source framework)	Primary institutional reference for taxonomy design; 8 of 11 categories map to specific GICS sub-industry codes
Caldara & Iacoviello (2022, <i>AER</i>) — Fama-French 49-industry mapping (based on SIC codes)	Direct	Geopolitical event-to-industry mapping precedent
NAICS (2022 revision) / ICB (FTSE Russell)	Direct (supplementary reference)	Eastern Bloc coverage and cross-referencing
IMF GFSR (April 2025, Ch. 2) + Shen (2025, Fed Boston)	Direct	Sector-differentiated geopolitical analysis using GICS-comparable groupings
GB/T 4754-2017 (China) — 20-division national classification aligned to ISIC Rev. 4	Analogue	National taxonomy adaptation within international frameworks
SASB SICS — taxonomy analogous to GICS in structure, designed for sustainability risk	Analogue	Formal GICS-analogous taxonomy for non-traditional purposes

Key Design Decisions:

GICS was selected over Fama-French classifications (based on SIC codes) despite the latter having stronger geopolitical-event precedent (Caldara & Iacoviello). GICS provides institutional investor alignment — WEO's primary audience uses GICS as its standard reference. The correspondence is documented: a complete GICS-to-WEO mapping table is maintained in the companion *Industry Tagging Reference*.

The 11-category count, category boundaries, structural decisions (Telecoms as secondary-only; Quantum Computing Infrastructure as a separate category with no GICS equivalent; Trade & Logistics defined outside GICS entirely), and the formal adaptation for event-level tagging represent the novel portion (~25%). Category design rationale for each structural decision is documented in Manual §7.3.

Grounding Assessment: HIGH. ~75% grounded through convergent precedent across financial classification standards, geopolitical risk literature, and supply chain intelligence practice. Novel portion is bounded — the measurement concept is established; only the specific taxonomy design and formal operationalisation are WEO-specific.

Validation Commitment: Review taxonomy at 200 and 400 events. Assess whether any category falls below 3% primary tag usage (indicating potential merger), whether any category consistently strains boundaries (indicating potential split), or whether persistent untaggable events indicate a missing category.

7.2 Primary/Secondary Tag Structure (1 Primary + Max 2 Secondary)

Manual Section: 7.4, 7.7

Category: ADAPTED (MEDIUM-HIGH 60–75%)

Rationale: Each event receives one primary tag (industry whose operational landscape the event's documented provisions most directly address) and up to two secondary tags (industries whose operational landscapes are also identifiably addressed by documented provisions). This structure adapts well-established primary/secondary classification practice from multiple domains.

Evidence Chain:

Precedent	Type	Contribution
ISIC Rev. 4 (UN, 2008)	Direct	Defines "principal activity" alongside secondary activities per economic unit — the canonical primary/secondary classification standard
ICD-10 coding guidelines (adopting UHDDS definitions)	Analogue	One "principal diagnosis" per encounter with secondary diagnoses that "coexist at the time of admission, develop subsequently, or affect the treatment received"
CPC patent classification	Analogue	One "CPC First" (classification most adequately representing the invention as a whole) with additional classifications
GB/T 4754-2017 (China)	Direct (structural)	Principal activity with recognised secondary activities; operational at national scale

Single-Primary Requirement: Universally supported across all precedent systems. The primary tag represents the industry most directly addressed — paralleling ISIC's "principal activity" determination and ICD-10's "principal diagnosis" rule.

Max-2 Secondary Cap: More restrictive than any identified comparable system. ICD-10, CPC, and ISIC all allow unlimited secondaries; GICS allows zero. The 1+2 structure is a parsimony measure preventing tag inflation. Empirical observation from the initial 87-event launch dataset shows median tag assignment of ~2.4 per event, suggesting the cap accommodates typical cases. The constraint may prove too restrictive at scale.

Grounding Assessment: MEDIUM-HIGH. ~72% grounded. The overall architecture (one primary, multiple secondary) is strongly precedented. The specific max-2 cap has no exact precedent — it occupies a unique middle position between systems allowing unlimited secondaries and systems allowing none.

Validation Commitment: After 6 months of coding, analyse the distribution of events where analysts identified 3+ candidate secondary tags but were constrained to 2. If >15% of events strain the cap, revisit whether the ceiling should be raised.

7.3 Deterministic Rule-Based Assignment from Metadata

Manual Section: 7.4

Category: STANDARD PRACTICE

Rationale: Rule Sets A, B, C, and D map existing WEO metadata (domain + event type) to primary industry tags through deterministic rules, with keyword disambiguation for ambiguous combinations. The approach — deterministic rule-based classification from structured metadata — is one of the most established practices in professional classification.

Evidence Chain:

Precedent	Type	Contribution
CMS DRG Grouper (US Medicare, operational since 1982–83)	Direct	Assigns diagnosis-related groups from ICD codes, procedures, age, sex, and discharge status using purely deterministic rules
Merchant Category Codes (ISO 18245)	Direct	Deterministic assignment by payment processors based on business type metadata
EPO patent auto-classification (over 90% accuracy at team level)	Direct	Automated classification from structured patent text
ICD auto-coding literature (JMIR 2021, PMC 2022)	Direct	Confidence-threshold workflows matching WEO's green/amber/red approach

Auto-Assignment Rate: Fully deterministic assignment (metadata-only, no keyword disambiguation) covers approximately 70–75% of events — comparable to documented performance ranges for rule-based classification systems:

System	Auto-Assignment Rate	Source
Measure (2014, BLS) — automated occupational coding	73.2%	<i>Proceedings of the Joint Statistical Meetings 2014</i>
Comparative Agendas supervised ML classification	Precision/F1 surpassing 80% "remained elusive"	Sebők & Kacsuk (2021, <i>Political Analysis</i>)

Including keyword-assisted routing, the initial 87-event launch dataset shows a ~7% human review rate (~93% resolved without manual intervention). Sample size is small; the rate may shift as the database grows. The 70–75% conservative estimate for fully deterministic assignment (excluding keyword-assisted routing) aligns with external benchmarks.

Separately, the Australian Bureau of Statistics achieved ~70% quality relative to human-created occupation code lists using LLM-assisted classification (UNECE HLG-MOS, March 2025), demonstrating feasibility of AI-assisted coding in official statistics contexts.

Key Distinction: The *technique* (deterministic metadata-to-category mapping) is standard practice. Only the *specific rule content* (domain × event type → primary tag mappings) is WEO-specific. This distinction is why the component is categorised as STANDARD PRACTICE rather than WEO SYNTHESIS — the novelty resides in mapping content, not mapping technique.

Grounding Assessment: HIGH. ~82% grounded through convergent precedent across clinical coding, news classification, patent classification, and statistical coding.

Validation Commitment: Track auto-assignment accuracy against human review outcomes for the first 200 events. If override rate exceeds 20% at the 150-event checkpoint, revise rules.

7.4 Mechanism Statement Requirement

Manual Section: 7.7

Category: ADAPTED (MEDIUM-HIGH 60–75%) — with novel elements

Rationale: Each secondary tag requires a one-sentence mechanism statement identifying the specific, document-verifiable pathway from the event's documented provisions to the secondary industry's operations. The statement identifies what the provisions contain and which industry's operations that content addresses — it does not claim what effect the provisions produce.

The *principle* of requiring justification for classification decisions is well-established across multiple domains. The *universal application* — every secondary tag, every time, not only ambiguous cases — and the specific mechanism-statement format go beyond what established systems typically require.

Evidence Chain:

Precedent	Type	Contribution
PRISMA 2020 (Item 5)	Direct	Requires authors to document eligibility criteria with rationale — closest parallel to mandatory justification for every classification decision
ICD coding guidelines / MedDRA	Analogue	Clinical justification linking diagnosis to code; mechanism documentation for adverse event classification
Structured Analytic Techniques (Heuer & Pherson, 2014, 2nd ed., CQ Press/SAGE)	Analogue	Describes techniques as mechanisms by which internal thought processes are externalised in systematic and transparent form (paraphrase)
ISA 230/500 (Audit Standards)	Analogue	Documented reasoning for material judgements
WEO CC evidence standards (Manual Chapter 9)	Internal consistency	Already requires documentary justification for CC classification (CC-V, CC-E, CC-A confidence levels)

Novel Element: PRISMA comes closest to WEO's universal requirement (exclusion reasons for every item). Most classification systems require justification only for borderline cases. The mechanism-specific element — requiring *how* an event's provisions address an industry, not just *that* they do — is the genuine contribution. This ~32% novel portion explains the MEDIUM-HIGH rather than HIGH confidence.

Grounding Assessment: MEDIUM-HIGH. ~70% grounded. The *principle* of classification justification is strongly precedented. The *scope of application* (universal, mechanism-specific) is more demanding than comparable systems.

Validation Commitment: Review mechanism statements at tag review checkpoints (100, 150, 200 events). Assess whether statements add genuine analytical value or become formulaic. Measure inter-analyst agreement on statement quality.

7.5 Cross-Sector Transmission Frequency (CSTF)

Manual Section: 7.9 (quality standards context). Operationally specified in *Towards Coordination Science* §III.3; rationale grounded independently below via E-I Index equivalence

Category: ADAPTED (HIGH 75–90%)

Rationale: When a Coordination Connection links two events with different primary industry tags, that CC represents a cross-sector transmission. CSTF measures the proportion of such cross-sector CCs:

$$\text{CSTF} = \text{Cross-industry CCs} \div \text{Total CCs}$$

This metric is mathematically near-identical to two established network metrics. It is a linear rescaling of the **E-I Index** (Krackhardt & Stern, 1988, *Social Psychology Quarterly*) and the complement of the diagonal trace of **Newman's mixing matrix** (2003, *Physical Review E*, 67, 026126):

Established Metric	Relationship to CSTF
E-I Index = $(E - I) / (E + I)$	$CSTF = (E - I + 1) / 2$
Newman mixing matrix $Tr(\mathbf{e})$	$CSTF = 1 - Tr(\mathbf{e})$

Both the E-I Index and Newman's mixing matrix are implemented in standard network analysis packages (UCINET, igraph). When a metric is a linear transform of a 1988 publication and the complement of a 2003 formalism, it is adapted — not novel.

Evidence Chain:

Precedent	Type	Contribution
Krackhardt & Stern (1988, <i>Social Psychology Quarterly</i>) — E-I Index	Direct (mathematical equivalence)	CSTF is a linear rescaling; both measure external-to-total ratio for categorical network partitions
Newman (2003, <i>Physical Review E</i> , 67, 026126) — mixing matrix	Direct (mathematical equivalence)	$CSTF = 1 - Tr(\mathbf{e})$; modularity Q provides null-model-corrected version
Diebold & Yilmaz (2012) — spillover index	Analogue	Cross-sector transmission ratios using continuous economic flows
Leontief Input-Output framework	Analogue	Inter-industry flow measurement; technical coefficients for cross-sector transmission
Guimerà & Amaral (2005, <i>Nature</i>) — participation coefficient	Analogue	Same concept at node level rather than network level

Novel Portion: ~25% — primarily naming ("CSTF") and domain application (geopolitical AI infrastructure coordination events with typed CCs and industry tags). No published work applies cross-category network metrics to this domain.

Known Gap: CSTF omits the null-model baseline correction that both Newman (2003) and Krackhardt & Stern (1988) emphasise. Without this correction, a high CSTF could result from unequal group sizes (more events tagged "Semiconductors" than "Quantum Computing Infrastructure") rather than genuine cross-sector coordination. The rescaling choice (CSTF's 0–1 range vs. E-I's –1 to +1) is justified for the target audience — institutional investors and policy analysts are more familiar with proportion-based metrics.

Grounding Assessment: HIGH. ~75% grounded through demonstrated mathematical equivalence to established network metrics. The null-model gap is a genuine weakness relative to established E-I Index implementation, preventing a higher confidence level.

Validation Commitment: (1) Report expected CSTF under random tag assignment alongside observed values, enabling users to distinguish genuine cross-sector coordination from artefacts of unequal tag distribution. (2) Cite the E-I Index explicitly as the established precedent. (3) At operational threshold, assess whether CSTF values are sensitive to reasonable tagging alternatives.

7.6 Cross-Sector Transmission Matrix

Manual Section: 7.9 (quality standards context). Operationally specified in *Towards Coordination Science* §III.3; rationale grounded independently below via Leontief/Newman structural equivalence

Category: ADAPTED (HIGH 75–90%)

Rationale: An 11×11 matrix where rows and columns are industry categories and cell values are CC counts between each pair. The matrix visualises how coordination dynamics flow between industries through documented Coordination Connections.

The inter-category flow matrix is one of the most thoroughly precedented analytical structures in quantitative social science. It is structurally equivalent to the Leontief Input-Output transactions table (1936/1941; Nobel Prize 1973), where the Bureau of Economic Analysis publishes standard ~70×70 I-O tables for the US economy.

Evidence Chain:

Precedent	Type	Contribution
Leontief Input-Output transactions table (1936/1941)	Direct (structural equivalence)	N×N matrix where rows and columns represent industries, cells contain flow values
Newman mixing matrix (2003, <i>Physical Review E</i>)	Direct (mathematical equivalence)	e_{ij} fraction of edges connecting node types i and j ; WEO matrix uses raw counts rather than proportions
Scherer (1982, <i>Review of Economics and Statistics</i>) — inter-industry technology flows	Direct (structural analogue)	Patent-based inter-industry technology flow classification between sectors — closest structural match
Diebold & Yilmaz (2012, 2015) — volatility spillover tables	Analogue	N×N matrices of cross-variable forecast error variance shares
UN Comtrade bilateral trade matrices; bibliometric citation matrices; bilateral migration flow matrices	Analogue (structural form)	Established analytical form across multiple disciplines

Novel Portion: ~25% — entirely domain application. The analytical structure is canonical. The data substrate — geopolitical AI coordination event relationships flowing between industry categories through typed CCs — has no existing implementation. No identified platform globally offers cross-sector transmission mapping for geopolitical events as a structured analytical product. The novelty resides in what fills the matrix, not in the matrix form.

Grounding Assessment: HIGH. ~80% grounded. The inter-category flow matrix is an established analytical structure applied to new data, not a novel construction.

Validation Commitment: (1) At operational threshold + 150 events, generate first transmission matrix and assess whether patterns are statistically significant vs. noise. (2) Test whether the matrix is dominated by a single industry category, which would limit analytical value. (3) Consider whether row/column normalisation (as in I-O technical coefficients) or raw counts is more informative.

7.7 Adversarial Quality Test

Manual Section: 7.9

Category: STANDARD PRACTICE

Rationale: The adversarial quality test asks: "If a domain professional filtered by this industry tag and saw this event, would they recognise it as correctly tagged — that this event's provisions genuinely address their industry?" This is an established evaluation methodology under at least three disciplinary names.

Evidence Chain:

Precedent	Type	Contribution
Face validity testing (Mosier, 1947; Nevo, 1985)	Direct (structural equivalence)	"The extent to which a test is subjectively viewed as covering the concept it purports to measure" — structural match is exact
TREC relevance assessment (NIST, since 1992)	Direct	Human assessors judge whether retrieved items are relevant to information needs; Voorhees (2000, 2005) demonstrated relative rankings remain stable across assessors
Information retrieval precision from user perspective	Direct	Of items retrieved (events tagged with industry X), what proportion would a user judge as correctly tagged?
Inter-rater reliability testing (Cohen's kappa, Krippendorff's alpha)	Analogue	Standard methodology for assessing classification agreement

Structural Equivalence: Face validity tests whether a measure "looks right" to the intended user. TREC relevance assessment tests whether retrieved items match user expectations. The WEO adversarial test applies the same logic — testing tag quality against domain-professional judgement. The methodology is standard; only the specific domain context (geopolitical AI infrastructure events) is new.

Terminology Note: The "adversarial" label reflects devil's-advocate testing rather than ML adversarial evaluation (red-teaming). The test is closer to classical face validity assessment than to AI safety adversarial testing.

Grounding Assessment: HIGH. ~83% grounded through demonstrated structural equivalence to face validity methodology and TREC relevance assessment practice.

Validation Commitment: (1) Implement with multiple assessors per tag to enable inter-rater reliability measurement (Cohen's kappa). (2) Apply at tag review checkpoints (100, 150, 200 events). (3) Document pass/fail rates per industry tag. If systematic failure patterns emerge for specific industries, this signals taxonomy or tagging rule revision.

7.8 Workflow Integration (Semi-Automated Classification)

Manual Section: 7.4 (rule-based assignment), 7.9 (quality standards)

Category: STANDARD PRACTICE

Rationale: Industry tagging integrates into the existing AI-assisted event verification workflow. Deterministic rules (Rule Sets A, B, C, and D) are applied first; human review handles secondary tags, overrides, and mechanism statements. This LLM-first, human-review-second pattern is the dominant AI-assisted classification architecture as of 2025.

Evidence Chain:

Precedent	Type	Contribution
Reuters AVISTA / AP automated earnings	Direct	News auto-classification with journalist review
CliniCoCo (<i>Nature, npj Digital Medicine</i> , 2024) / MedCodER (2024)	Direct	AI-human collaborative clinical coding; CliniCoCo reduces coding time by 40%
UNECE HLG-MOS Report (March 2025)	Direct	Statistical organisations using LLMs for classification with human verification; ABS achieving ~70% quality relative to human-created lists
Barberá et al. (2021, <i>Political Analysis</i>) — Comparative Agendas Project	Direct (methodology)	"Classify by machine but verify by humans"
Financial regulatory compliance workflows	Direct	Identification, triage, impact analysis, action assignment with human-in-the-loop governance
MIIT gradient cultivation system (China)	Analogue	Integrated evaluation workflows with automated quantitative screening followed by expert committee review

Novel Portion: ~12%. Only the specific domain context and the integration of mechanism statements into the review step are WEO-specific. The workflow architecture is entirely standard.

Grounding Assessment: HIGH. ~88% grounded — the most firmly grounded component in the industry tagging methodology. The LLM-first, human-review-second classification workflow is extensively documented and operationally validated across clinical coding, news classification, patent classification, and statistical coding.

Validation Commitment: Track analyst review time per event, override rates, and inter-analyst agreement. No special validation required — this is process design, not analytical methodology.

Section 8: Environmental Nexus Tags Rationale

Manual Reference: Chapter 8

8.1 Five-Type ENT Taxonomy Structure

Manual Section: §8.3

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

Rationale: The five Environmental Nexus Tag types (Resource Demand, Environmental Enablement, Environmental Consequence, Environmental Dependency, Environmental Coordination) preserve directional coherence across three analytical axes: AI → Environment (ENT-1 demand, ENT-3 consequence); Environment → AI (ENT-4 dependency); Governance mediation (ENT-2 enablement/intent, ENT-5 operationalisation).

The directional logic maps to Ranganathan's PMEST faceted classification: ENT-1 (Matter — intrinsic property), ENT-2 (Energy — action/justification), ENT-3 (Energy — measured outcome), ENT-4 (Space — contextual/relational condition), ENT-5 (Personality — core governance purpose). The mix of intrinsic and relational properties within a single tag layer is theoretically grounded — Dublin Core's 15-element metadata set accommodates analogous variation between `Type` (intrinsic), `Subject` (topical), and `Relation` (external linkage).

Evidence Chain:

Precedent	Type	Contribution
Ranganathan PMEST model (1967)	Structural	Faceted classification grounding mixed intrinsic/relational properties within a single layer
Dublin Core 15-element metadata set	Structural	Combining intrinsic description, topical classification, and relational linkage in one framework
EU ESRS double materiality (ESRS 1, ESRS 2, 2023)	Analogue	Impact materiality (inside-out: ENT-1/ENT-3) and financial materiality (outside-in: ENT-4) as separate perspectives within a single framework
MITRE ATT&CK technique taxonomy	Analogue	Flat taxonomy covering heterogeneous vectors within a unified classification layer
Preliminary calibration (109 events, 34 qualifying)	Empirical	Five types capture 31% of events without redundancy; ENT-1-dominant distribution reflects empirical reality

Key Design Decisions:

Why five and not four? Merging ENT-2 and ENT-5 would lose the distinction between aspirational environmental language and binding operational constraints — fundamentally different governance phenomena.

Why five and not six? A sixth type capturing physical geographic enablement would trend towards trivial universality (projected 50–67% at scale) and overlap with ENT-1 and ENT-4.

Why is the skewed distribution acceptable? The ENT-1-dominant distribution (~27 of 34, as of April 2026) reflects rapid expansion where resource demand dominates and governance response is a lagging indicator. Quantifying this gap is itself the primary analytical finding.

Grounding Assessment: WEO SYNTHESIS. ~65% grounded. Faceted classification theory and mixed-property architecture draw on established information science (Ranganathan, Dublin Core) and environmental risk frameworks (ESRS double materiality). The ~35% novel element is the specific five-type taxonomy design.

Validation Commitment: Review distribution at 150-event milestone. Reassess if any type falls below 2% of tagged events, ENT-1 exceeds 60%, or ENT-5 remains at zero beyond 2028.

8.2 ENT as Analytical Enrichment Layer (Not Classification)

Manual Section: §8.2

Category: ADAPTED (HIGH 75–90%)

Rationale: ENTs occupy the same architectural position as industry tags: a post-classification enrichment layer that inherits upstream pipeline assurance and produces metadata consumed by downstream analysis. This classify-first-enrich-second pattern adapts the Basel Committee on Banking Supervision's treatment of climate-related financial risks — not as a standalone risk category but as a cross-cutting driver manifesting through existing core categories (BCBS d532 FAQs, December 2022). BCBS 239 Principles 4 (Completeness) and 6 (Adaptability) formalise the pipeline: classify by core dimensions first, then filter by supplementary dimensions without disrupting aggregation logic.

The intelligence community mirrors this with ICD 203's two-tier architecture: five core Analytic Standards applied universally, then nine supplementary Analytic Tradecraft Standards. MSCI's ESG architecture (GICS industry classification first, environmental Key Issues as overlay) further confirms the pattern.

Evidence Chain:

Precedent	Type	Contribution
Basel III climate risk overlay (BCBS d532; FAQs December 2022)	Direct	Classify-first-enrich-second for environmental overlays on existing risk categories
BCBS 239 Principles 4 and 6	Direct	Core classification first, supplementary filtering second; adaptability without disruption
ICD 203 (ODNI, January 2015)	Structural	Two-tier architecture: core standards first, supplementary tradecraft standards second
MSCI ESG architecture	Analogue	GICS industry classification first, environmental Key Issues as overlay

Grounding Assessment: ADAPTED. ~80% grounded. Directly adapted from Basel III climate risk treatment — the strongest institutional precedent for environmental overlays on non-environmental classification systems. The ~20% novel element is application to a geopolitical event database.

Validation Commitment: Confirm at 200-event milestone that ENT assignment has never modified upstream classification and that the ENT layer remains computationally removable. Monitor for architectural drift — if WEO outputs treat ENT as a de facto domain, the independence principle has been violated.

8.3 Binary Qualification with Evidence-Basis Descriptor

Manual Section: §8.2

Category: ADAPTED (HIGH 75-90%)

Rationale: ENT qualification is binary (present/absent) with an evidence-basis descriptor ("Documented" or "Assessed") adapting the IPCC's confidence gating model. The IPCC establishes a hierarchy where confidence is prerequisite to probability estimation; the ENT design adapts this: binary qualification is the primary gate, with the descriptor providing a simplified confidence indicator.

"Documented" means qualifying evidence appears in the event's source documentation or primary regulatory sources (standard for ENT-1, ENT-2, ENT-5). "Assessed" means the analyst applied authoritative third-party data to documented facility characteristics (standard for ENT-4, and ENT-3 where measurement relies on independent monitoring). The two-value system was chosen over three-level confidence to minimise cognitive overhead for a supplementary metadata layer, consistent with ICD 203 permitting differentiated confidence architectures for supplementary assessments.

Why not three-level confidence? ENTs are supplementary metadata. Replicating the full CC confidence architecture (CC-V/CC-E/CC-A) would imply equivalent analytical weight. The evidence-basis descriptor captures the most analytically significant variation without over-engineering the confidence layer.

Evidence Chain:

Precedent	Type	Contribution
IPCC confidence/likelihood gating model (AR5, AR6)	Direct	Binary gate before graded assessment; validates hierarchical approach
ICD 203 (ODNI, January 2015, Section D.6.e.(2))	Structural	Differentiated confidence architectures permitted for supplementary assessments

Grounding Assessment: ADAPTED. ~75% grounded. The ~25% novel element is the specific two-value evidence-basis descriptor applied to environmental intersection tagging.

Validation Commitment: Review at 200-event milestone whether the Documented/Assessed distinction remains analytically productive. If >90% carry the same evidence basis, the descriptor adds no information.

8.4 Architectural Guardrails

Manual Section: §8.2

Category: ADAPTED (HIGH 75–90%)

Rationale: Four guardrails ensure ENTs function as a supplementary enrichment layer without becoming a competing classification system.

Guardrail 1 — Read-only dependency mirrors the Regnology ESG platform pattern, where additional attributes are included without modifying existing data.

Guardrail 2 — Independent removability adapts the computational removability test from modular overlay architectures (LSEG "Plus" layer, ISO Annex SL).

Guardrail 3 — No aggregation by ENT mirrors MSCI's architecture, where environmental Key Issues are assigned within GICS industry classification rather than as an alternative grouping principle.

Guardrail 4 — Explicit documentation of layer role follows ESRS "cross-cutting" terminology, distinguishing supplementary from categorical layers.

Evidence Chain:

Precedent	Type	Contribution
MSCI ESG architecture	Direct	Environmental Key Issues within GICS industry, not as alternative grouping
Regnology ESG platform	Analogue	Additional attributes included without modifying existing data pool
ESRS cross-cutting standards (ESRS 1, ESRS 2)	Structural	"Cross-cutting" terminology distinguishing supplementary from categorical layers

Grounding Assessment: ADAPTED. ~80% grounded. Each guardrail has direct institutional precedent. The ~20% novel element is the specific four-guardrail formulation.

Validation Commitment: Confirm all four guardrails hold at each database doubling milestone.

8.5 ENT-1 Resource Demand — Definition and Threshold

Manual Section: §8.3.1

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: The ≥100MW energy threshold derives from convergent institutional classification of "hyperscale" power demand. The International Energy Agency characterises large hyperscale data centres as those with capacity of 100MW or more (IEA, 2024; IEA, *Energy and AI*, April 2025); the United States Congressional Research Service uses the same characterisation (CRS R48646, August 2025), attributed to industry analysts; industry analysts (Synergy Research Group, Uptime Institute) converge on 100MW as the hyperscale classification boundary.

The threshold captures a qualitative shift: facilities exceeding 100MW typically interconnect directly to high-voltage transmission grid (115–230kV), a structural marker of environmental significance. Two alternative energy qualifiers (documented grid impact assessment, documented power purchase agreement) provide context-sensitive qualification below 100MW.

The ≥100MW energy threshold prevents trivial universality — without it, virtually every physical AI infrastructure event would qualify via the energy vector. The water, land, and minerals vectors do not need numerical thresholds because regulatory and planning systems determine what is environmentally significant enough to require dedicated documentation, providing a self-calibrating gate.

Non-Western validation confirms no jurisdiction uses 100MW as a regulatory boundary — regulatory triggers operate at lower levels (South Korea 10MW, Indonesia ~5MW equivalent, Japan energy-consumption-based). The threshold validates through industry convergence, not regulatory adoption, correctly identifying hyperscale environmental significance regardless of the host jurisdiction's classification approach.

Why 100MW rather than 50MW or 200MW? 50MW risks trivial universality at scale. 200MW would exclude lower-range hyperscale and sovereign compute. 100MW has the strongest convergent institutional anchoring.

Evidence Chain:

Precedent	Type	Contribution
IEA "Data centres and energy" (2024); "Energy and AI" (April 2025)	Direct	100MW characterised as "large hyperscale" capacity baseline
US Congressional Research Service R48646 (August 2025)	Direct	>100MW hyperscale characterisation in US government analytical report, attributed to industry analysts
Industry consensus (Synergy Research Group, Uptime Institute, CBRE)	Direct	100MW+ = market-standard hyperscale power category
DOE ANOPR RM26-4-000 (October 2025)	Structural	>20MW = "large load" — confirms 100MW is well above regulatory floor
EU EED Article 12 (Directive 2023/1791)	Structural	≥500kW IT power = environmental reporting floor — confirms 100MW is appropriately higher for geopolitical significance

Grounding Assessment: EMPIRICALLY GROUNDED. ~80% grounded. The ~20% novel element is application to geopolitical significance context and multi-vector architecture.

Validation Commitment: Review threshold at each database doubling. Check for false negatives at 200 events; check for tag inflation if ENT-1 exceeds 50% of infrastructure events.

8.6 ENT-2 Environmental Enablement — Definition and Linguistic Threshold

Manual Section: §8.3.2

Category: WEO-ORIGINAL

Rationale: ENT-2 captures where environmental intent has been codified in AI infrastructure governance. The qualifying threshold imposes four cumulative criteria: environmental objectives must be (a) explicit, (b) in primary source documentation, (c) in operative provisions (not solely preambles), and (d) directed at the AI infrastructure action. The "operative provisions" test — requiring environmental language in legislative articles, binding conditions, or funding criteria rather than solely in preambles, recitals, or aspirational statements — is the critical discriminator preventing inflation from standardised environmental language. The "documented co-benefit" qualifier ensures that even environmental co-benefits of the AI infrastructure action must be explicitly stated in operative provisions to qualify, preventing unstated environmental side-effects from satisfying the threshold.

The threshold excludes physical substrate enablement (geographic proximity to renewable energy sources) because it trends towards trivial universality at scale (projected 50–67%), overlaps with ENT-1 and ENT-4, and provides analytical value better captured through geographic metadata.

Evidence Chain:

Precedent	Type	Contribution
EU AI Act Annex XI, Section 1, Paragraph 2(e)	Empirical	Qualifying exemplar — binding energy documentation in operative provisions
Canada SCIP (Sovereign Compute Infrastructure Program)	Empirical	Qualifying exemplar — environmental footprint minimisation as documented evaluation criterion in funding
Germany EnEfG (November 2023)	Empirical	Qualifying exemplar — mandatory PUE ≤ 1.2 for new data centres from July 2026, 100% renewable by 2027, waste heat reuse in operative provisions
Singapore DC-CFA2 (December 2025)	Empirical	Qualifying exemplar — PUE ≤ 1.25 , 50% green energy as capacity allocation conditions
Chile PDATA (National Data Centers Plan)	Empirical	Qualifying exemplar — clean production agreements and renewable zone mapping in sovereign planning
Meta Hyperion campus, Louisiana	Empirical	Exclusion exemplar — corporate pledge without sovereign mandate
Hedden (2021) <i>The Accidental Taxonomist</i>	Structural	Metadata-not-tag approach

		for physical geography preserves analytical value without trivially universal tags
GB 40879-2021 (数据中心能效限定值及能效等级), Standardization Administration of China, October 2021. Mandatory national standard (强制性国家标准)	[EB Benchmark]	Binding PUE limits as Chinese mandatory national standard; demonstrates test works for mandatory technical standard form
Inner Mongolia 14th FYP Energy Conservation Plan (内政办发〔2022〕11号), Chapter 4, Section 6: "新建数据中心须达到绿色数据中心建设标准, PUE值不超过1.3"	[EB Benchmark]	Provincial binding implementation using mandatory language (须...不超过); confirms cascade model
Energy Conservation Act (省エネルギー法), Japan, revised April 2023. PUE ≤1.3 binding for new data centres from FY2029	[WB Additional]	Demonstrates test works across Japanese benchmark architecture
AI Basic Act (인공지능 발전과 신뢰 기반 조성 등에 관한 기본법), South Korea, Act No. 20676, effective 22 January 2026. Article 25	[WB Additional]	Correctly-excluded exemplar — sole data centre provision contains zero environmental language; demonstrates discriminating power
Or equivalent binding environmental provisions in operative governance instruments, meeting the same evidentiary verification standard	[Accommodation]	Opens evidence chain to all bloc members following the "or equivalent" pattern (Manual §3.1, §4.6, §4.7)

Grounding Assessment: CATEGORY CREATOR with non-Western validation. ~60–65% grounded. The four-criteria linguistic threshold is WEO-original, but cross-jurisdictional validation across five distinct regulatory architectures (EU statutory, Japanese benchmark, Singapore approval-gating, Chinese cascade, cross-cutting environmental legislation) confirms the operative provisions test produces correct results globally. The ~35–40% novel element remains the specific application to environmental intersection tagging, but the test's discriminating power is empirically demonstrated (correctly qualifying Japan, Singapore, UAE, Indonesia, China; correctly excluding India, South Korea AI Basic Act, Japan Watt-Bit Collaboration).

Validation Commitment: Track ENT-2 population at 200 events. If the EU regulatory stack produces qualifying events at ~5–10% of events at scale, the threshold is functioning.

8.7 ENT-3 Environmental Consequence — Definition and Activation Model

Manual Section: §8.3.3

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

Rationale: ENT-3's two-category activation model ensures WEO relies on verifiable empirical data, consistent with the evidence-only observation standard (Manual §1, Core Principles; §3.9). Acute consequences are evidence-gated because qualifying data (enforcement actions, independent monitoring) becomes available rapidly. Cumulative impacts require a 24-month maturation window because single-cycle reporting data is insufficient — the EU EED first assessment (July 2025, Borderstep Institute) identified significant data quality challenges.

The 24-month window aligns two annual reporting cycles with environmental monitoring practice. The governance instrument scope limitation references instruments producing facility-level environmental operational data (EU EED, EnEfG), not model-level energy documentation (EU AI Act Annex XI).

Evidence Chain:

Precedent	Type	Contribution
xAI Memphis / EPA ruling (January 2026)	Direct	Acute enforcement within ~6–8 months; validates evidence-gated approach
EU EED first assessment (July 2025, Borderstep Institute)	Direct	Single reporting cycle yields data with significant quality challenges
US EPA NPDES permit practice	Analogue	12–24 month operational monitoring before definitive discharge limits
EU EED Delegated Regulation EU/2024/1364	Direct	24 KPIs for annual data centre reporting across three annexes (energy/sustainability, ICT capacity, data traffic); defines data accumulating over 2 cycles

Grounding Assessment: WEO SYNTHESIS. ~65% grounded. Acute evidence-gating follows from verification lag and regulatory practice (~80%). The 24-month cumulative window synthesises EU EED experience, monitoring practice, and reporting cycle mechanics (~60%). The two-category architecture is WEO-original with analogue in NEPA's tiered review hierarchy (~50%).

Validation Commitment: Reassess 24-month window after EU EED third reporting cycle (~mid-2026). Track Building Chips in America Act regulatory gap through CHIPS-funded fab HVM commencement.

ENT-3 Sparsity Decomposition

Category: WEO-ORIGINAL

ENT-3's population at V1.5 publication (n=1) reflects the corpus's maturation stage — the 2024–2025 buildout cohort has not yet crossed the cumulative-lifecycle activation threshold documented in §8.3.3.

8.8 ENT-4 Environmental Dependency — Definition and Evidence Indices

Manual Section: §8.3.4

Category: ADAPTED (HIGH 75–90%)

Rationale: ENT-4 deliberately outsources risk determination to authoritative external indices rather than defining WEO-specific risk thresholds. Three authoritative global indices are designated as default sources: WRI Aqueduct Water Risk Atlas (Version 4.0) for facility-level water stress; ND-GAIN Country Index for country-level climate vulnerability; Munich Re NatCatSERVICE for empirical catastrophe loss data. Together they cover hydrological risk, climate vulnerability, and historical catastrophe exposure.

The key exclusion — generalised global climate risk does not qualify — ensures ENT-4 maintains specificity, aligning with TCFD's emphasis on entity-specific physical risk assessment. Where national environmental data sources provide demonstrably superior resolution, they are designated as complementary sources alongside the global defaults. The global baseline is preserved for cross-jurisdictional comparability.

Evidence Chain:

Precedent	Type	Contribution
TCFD Recommendations (2017; updated 2021)	Direct	Physical climate risk assessment using entity-specific exposure analysis
WRI Aqueduct Water Risk Atlas 4.0	Direct	Authoritative hydrological risk classification; facility-level granularity
ND-GAIN Country Index	Direct	Country-level climate vulnerability; academic and development finance recognition
Munich Re NatCatSERVICE	Direct	Empirical catastrophe loss data; insurance industry standard
J-SHIS (Japan Seismic Hazard Information Station), NIED, 250m mesh resolution	[WB Additional]	Seismic hazard mapping orders of magnitude more granular than Munich Re for Japanese facilities
CGWB NAQUIM (National Aquifer Mapping), Central Ground Water Board, India. 1:50,000 scale	[WB Additional (Non-Aligned)]	Block/aquifer-level groundwater data addressing WRI Aqueduct gaps in fossil groundwater
First National Disaster Risk Census (第一次全国自然灾害综合风险普查), Ministry of Emergency Management (应急管理部), China. County-level multi-hazard	[EB Benchmark]	County-level disaster risk mapping superior to global indices for Chinese facilities
Or equivalent authoritative national environmental data sources providing demonstrably superior facility-level resolution, meeting WEO source quality requirements (Manual §3.1)	[Accommodation]	Preserves global baseline whilst accommodating jurisdiction-specific high-resolution data

Grounding Assessment: ADAPTED. ~80% grounded. Designated indices are authoritative, widely recognised, and freely accessible. Cross-jurisdictional validation confirms global architecture works with complementary national sources. The ~20% novel element is application to a geopolitical event database.

Validation Commitment: Review index designations at each database doubling milestone. If evidence of operational disruptions emerges for ENT-4-tagged events through WEO's standard rolling update process, this would strengthen validation of the index-based identification approach.

The Coupling Argument

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

V1.5 specifies the two ENT-4 water conjuncts as a coupled pair. Conjunct 1 sets a band floor ($\geq 20\%$ BWS). Conjunct 2 requires a documented consumptive stage. Both must be met — neither is sufficient alone.

The coupling exists because neither variable discriminates independently. Basin stress without consumption-character admits facilities whose closed-loop architecture draws negligibly from the stressed basin — the stress is documented but the dependency is not real. Consumption-character without stress context admits consumptive facilities in unstressed basins — the architecture consumes water, but the basin can sustain it. The coupled gate admits only the intersection: facilities whose documented consumptive architecture draws from basins the authoritative index classifies as stressed.

The coupled design is informed by external precedent. The Climate Neutral Data Centre Pact (CNDP, 2024 Revision) couples water stress with water-source type multiplicatively in its WUE_{max} formula, applying tighter compliance ceilings where stress is high and water source is potable. Uptime Institute recommends prioritising water minimisation in water-stressed regions. The AWARE characterisation method (Boulay et al. 2018), recommended under ISO 14046, weights water consumption by location scarcity. WEO applies the coupling as binary gates rather than continuous coefficients, consistent with §8.2's binary design constraint.

Reporting-framework comparison. Mainstream corporate ESG reporting frameworks capture water stress and water consumption as independent variables. SASB TC-SI-130a.2 requires volume disclosure in High/Extremely High BWS areas; CDP accounts for consumption but not by cooling architecture. ENT-4's coupled gate serves a different analytical purpose — identifying facilities whose documented consumptive architecture participates in documented basin stress — which requires the two variables to interact rather than report in parallel.

Evidence Chain:

Precedent	Type	Contribution
CNDCP $WUE_{\max}$ formula (2024 Revision)	Direct	Multiplicative coupling of stress (K_2) and water-source (K_3) coefficients in an operational ceiling
ISO 14046 / AWARE (Boulay et al. 2018)	Direct	Scarcity characterisation as prerequisite for water footprint claims — consumption $\times$ location factor
Uptime Institute (Dietrich 2025)	Normative	Directional recommendation coupling consumption priority to stress severity
SASB TC-SI-130a.2	Structural	Captures volume in High/Extremely High stress — does not classify by architecture
CDP Water Security	Structural	Captures consumption — does not classify by architecture or couple with stress
ENT-1 $\geq 100\text{MW}$ floor pattern (§8.3.1)	Internal	Binary tag with qualifying floor that gates entry without grading magnitude

Grounding Assessment: WEO SYNTHESIS. ~70% grounded. CNDCP's coupled formula, ISO 14046/AWARE's scarcity characterisation, and Uptime Institute's normative direction provide convergent external precedent for coupling stress with consumption-character. The ~30% novel element is the specific implementation as binary gates within an environmental-intersection tagging layer.

Water-Stress Band Floor Rationale

Category: ADAPTED (HIGH 75–90%)

The qualifying band floor sits at Medium-High (20–40%), honouring the WRI Aqueduct index's own band semantics and the defer-to-index design principle documented in §8.3.4.

Index semantics. WRI Aqueduct's 20–40% class is named "Medium-High" — the index places stress onset at 20%, consistent with the European Environment Agency's Water Exploitation Index (WEI+) and the hydrological convention that withdrawal-to-availability $>20\%$ constitutes water stress. Deferring to the index and then overruling its band semantics with a WEO-imposed 40% floor contradicts the design principle that WEO checks whether authoritative data identifies a risk, not makes its own risk determination (§8.3.4).

Citation: WRI Aqueduct 4.0, Baseline Water Stress band definitions (Kuzma et al. 2023, DOI 10.46830/writn.23.00061; reproduced in CDP Technical Note on Water Accounting v3.0, April 2026, p. 8). URL: <https://www.wri.org/aqueduct/help-center/prioritize-basins-thresholds>

CNDCP includes the band. CNDCP assigns $K_2 = 2.5$ to the 20–40% band — intermediate between Low ($K_2 = 5$) and High ($K_2 = 1$). CNDCP treats Medium-High as generating real, if moderate, stress — not as immaterial.

SASB/CDP divergence resolves as different-instrument-different-purpose. SASB TC-SI-130a.2 draws mandatory disclosure at "High (40–80%) or Extremely High (>80%)" (ISSB Exposure Draft SASB-ED-2025-1, Table A6, technical protocol para. 4). CDP's primary threshold sits at 40%. These are disclosure triggers for volume reporting. ENT-4 is a dependency tag within a coupled gate — the 20% floor admits events into a further test (conjunct 2), not into a tag outright. The structural difference between a disclosure trigger and a coupled dependency gate means the two instruments serve different analytical functions despite both using WRI Aqueduct data.

Citations:

- ISSB Exposure Draft SASB-ED-2025-1, "Proposed Targeted Amendments to other SASB Standards," Table A6, TC-SI-130a.2 technical protocol para. 4. URL: <https://www.ifrs.org/content/dam/ifrs/project/enhancing-the-sasb-standards/sasb-ed-2025-1-targeted-amendments.pdf>
- CDP Technical Note on Water Accounting, v3.0, April 2026, p. 8. URL: https://cdn.cdp.net/cdp-production/cms/guidance_docs/pdfs/000/001/561/original/CDP-technical-note-water-accounting-definitions.pdf
- CDP Water Security Reporting Guidance, question W1.2d: "There is no universally accepted methodology for classifying an area as water stressed, nor for identifying whether facilities are located in a water stressed area." URL: <https://guidance.cdp.net/en/guidance?ctype=ExternalRef&idtype=RecordExternalRef&cid=W1.2d&otype=guidance&incchild=1&site=1&gettags=1>

Under the coupled gate, admitting Medium-High cannot reopen trivial universality — a Medium-High tag also requires a documented consumptive stage. The residual population admitted is: consumptive facilities in basins the index classifies as stressed.

Grounding Assessment: ADAPTED. ~80% grounded. WRI's own band semantics, the EEA WEI+ convention, and CNDP's K_2 assignment all place stress onset at 20%. The ~20% novel element is the application within a coupled binary gate — where the 20% floor's permissiveness is structurally offset by conjunct 2's discriminating power.

Consumptive-Process Floor Rationale

Category: WEO-ORIGINAL

No environmental-reporting framework formally gates on cooling-architecture consumption character for risk classification.

Internal consistency. The consumptive-process floor applies the ENT-1 pattern to ENT-4 — a binary tag with a qualifying floor that gates entry without grading magnitude. "Consumptive vs non-consumptive" is a character distinction readable from architecture documentation, exactly as " $\geq 100\text{MW}$ " is readable from capacity documentation. No volume, litres, or WUE figure enters the test at any point; the binary design constraint (§8.2) is respected.

Engineering consensus. The consumptive/non-consumptive distinction is universally recognised in data centre engineering. ASHRAE TC 9.9 classifies cooling by thermal parameters — facility-water supply-temperature classes W1–W5 (original designation) and H1 (5th edition addition) — from which consumption character is derivable. Uptime Institute identifies heat-rejection equipment as having "the greatest influence on water consumption." The EU Code of Conduct for Energy Efficient Data Centres addresses cooling architecture and water sources in dedicated sections. WUE (The Green Grid / ISO/IEC 30134-9:2022) measures site water consumption, approaching zero for dry-cooled and closed-loop facilities. The distinction is an established engineering concept; ENT-4's conjunct 2 formalises it as a gate.

Citations:

- ASHRAE TC 9.9 Thermal Guidelines, 5th ed. (2021), liquid cooling classes W1–W5 / H1.
- ISO/IEC 30134-9:2022, Introduction: "does not specify limits or targets for the KPI." URL: <https://www.iso.org/standard/77692.html>
- Dietrich (2025), Uptime Institute — same citation as coupling argument above.
- EU Code of Conduct Best Practice Guidelines, JRC136986 (2024), Section 5 (Cooling), Section 8.3 (Water Sources). URL: https://e3p.jrc.ec.europa.eu/sites/default/files/2024-04/JRC136986_2024_best_practice_guidelines.pdf

Standards direction. ISO/IEC 30134-9 (WUE) is under formal revision (ISO/IEC AWI 30134-9), with a three-year timeline from 2024. The revision scope explicitly includes whether to incorporate water source type, evaporative cooling type, water stress, and water recycling. The gate aligns with the direction the standards community is actively moving.

Citation: ISO catalogue, ISO/IEC AWI 30134-9 (active revision). Equinix corporate blog, 13 November 2024. Datacenter Jedi, 27 November 2024.

Reporting-framework gap. No mainstream environmental-reporting framework (SASB, CDP, EU Taxonomy, TCFD) formally classifies cooling architectures as consumptive vs non-consumptive for risk purposes. They measure volumes. ENT-4's conjunct 2 formalises a distinction recognised across data centre engineering but not classified in current environmental-reporting standards.

Heat-rejection-terminus rule. "DLC" alone is ambiguous — a DLC server loop can terminate in either evaporative towers or dry coolers. The terminus rule resolves every hybrid case mechanically: liquid-cooled plates with evaporative secondary heat rejection qualify; warm-water DLC with dry coolers does not. The rule prevents both under-inclusion (hybrid consumptive architectures missed) and over-inclusion (closed-loop non-consumptive architectures admitted).

Undocumented-default clause. Where the cooling or process-water architecture is not documented, the conjunct is not met. This prevents inferential tagging — the ENT-4 equivalent of the ENT-1 verifiability standard. A facility's environmental dependency must be evidenced from documented architecture, not inferred from circumstance.

Grounding Assessment: WEO-ORIGINAL. ~55% grounded. The consumptive/non-consumptive distinction has universal engineering recognition (ASHRAE, Uptime Institute, EU CoC, ISO/IEC 30134-9), directional standards movement (ISO/IEC AWI 30134-9 revision), and CNDCP coupling precedent. The ~45% novel element is the formalisation as a binary gate within an environmental-dependency tag — no prior framework uses cooling-architecture consumption character as a formal classification criterion.

8.9 ENT-5 Environmental Coordination — Definition and Medium Scope

Manual Section: §8.3.5

Category: WEO-ORIGINAL for scope selection; ADAPTED (HIGH 75–90%) for Pathway A evidence architecture

Rationale: ENT-5 captures where environmental governance has been operationalised into binding requirements on AI infrastructure (Pathway A) or where dedicated AI infrastructure has been deployed for environmental purposes at sovereign or multilateral scale (Pathway B). The definition resolves a scope spectrum:

- **Narrow scope yields an empty set** in the current database (projected ~3–8 events at 400–600 scale). Binding environmental regulations for AI infrastructure exist (EU EED, EnEfG) but narrow scope would exclude them because they govern the environmental footprint OF AI infrastructure rather than deploying AI FOR environmental purposes.
- **Broad scope creates unacceptable overlap with ENT-2.** Every broad ENT-5 event would also receive ENT-2 (since all events with environmental governance provisions also contain environmental language).
- **Medium scope** produces the analytically productive ~10:1 ENT-1/ENT-5 ratio quantifying the governance gap whilst acknowledging that governance response has begun.

Evidence Chain:

Precedent	Type	Contribution
EU AI Act Article 53/Annex XI	Empirical	Pathway A — binding energy documentation for GPAI providers
EU EED Article 12 (Directive 2023/1791)	Empirical	Pathway A — mandatory annual reporting for data centres $\geq 500\text{kW}$
Germany EnEfG (November 2023)	Empirical	Pathway A — mandatory PUE ≤ 1.2 , 100% renewable by 2027
Singapore DC-CFA2 (December 2025)	Empirical	Pathway A — capacity-gated green requirements
DestinE Phase 3 (February 2026 confirmation)	Empirical	Pathway B — dedicated EuroHPC infrastructure for climate digital twins
France Jean Zay supercomputer	Empirical	Exclusion exemplar — domestic waste-heat qualifies ENT-2 not ENT-5
Keohane coordination definition (1984)	Structural	ENT-5 Pathway B requires sovereign or multilateral scale — aligning with WEO's foundational coordination definition
EDWC programme (发改高技〔2021〕709号) with provincial binding implementation via GB 40879-2021 and approval-process PUE conditions	[EB Benchmark] (Pathway A)	Binding environmental requirements on compute infrastructure at sovereign scale; first EB exemplar; qualification through administrative approval rather than standalone environmental legislation
Supercomputer Fugaku environmental programmes (RIKEN/JAMSTEC): tsunami prediction, earthquake simulation, extreme weather prediction; J-Alert integration	[WB Additional] (Pathway B)	Strongest Pathway B exemplar globally — governance-scale dedicated environmental AI infrastructure
Or equivalent binding environmental operational requirements or dedicated environmental compute infrastructure at sovereign or multilateral scale, meeting the same evidentiary verification standard	[Accommodation]	Covers all bloc members

Key Design Decision:

Why require binding/enforceable constraints for Pathway A? Non-binding pledges are captured by ENT-2 (intent). ENT-5 specifically tracks operationalisation — the point at which intent becomes obligation. Why sovereign/multilateral scale for Pathway B? Corporate AI-for-environment deployments are product offerings on commercial infrastructure, not governance events.

Confidence Assessment: CATEGORY CREATOR for scope selection (~50%); ADAPTED (~80%) for Pathway A evidence architecture. The ~50% novel element is the specific medium scope boundary definition, grounded in empirical population analysis and Keohane's coordination framework. Non-Western validation confirms Pathway A correctly identifies binding requirements across diverse governance architectures (EU statutory, Chinese administrative approval, Japanese benchmark, Singaporean capacity-gating, cross-cutting environmental legislation).

Validation Commitment: If ENT-5 remains at zero beyond 2028 despite documented regulatory maturation, reassess medium scope. Track regulatory pipeline: EU Data Centre Energy Efficiency Package, Japan Energy Saving Act PUE mandate (post-2029), Singapore Digital Infrastructure Act.

Category: STANDARD PRACTICE

The ENT-5 tag name ("Environmental Coordination") invites conflation with the tag's qualifying threshold. The test is operative, not the name: institutional coordination machinery satisfies the tag's name but not its threshold.

8.10 ENT-2/ENT-5 Boundary (Governance Maturity Spectrum)

Manual Section: §8.3.2, §8.3.5

Category: WEO-ORIGINAL

Rationale: ENT-2 marks intent; ENT-5 marks operationalisation. An event may carry both tags when environmental language (ENT-2) generates binding constraints (ENT-5). The dual-tag is informative, not redundant — it measures different positions on the governance maturity spectrum.

Scenario	ENT-2	ENT-5	Rationale
EU AI Act Annex XI — binding energy documentation	Yes	Yes	Environmental mandate in operative provisions + legally binding
Canada SCIP — environmental footprint as evaluation criterion	Yes	Borderline	Environmental language in funding; contractual enforceability arguable
France Jean Zay — waste-heat recovery	Yes	No	Environmental provisions in domestic sovereign compute without binding environmental governance targeting AI infrastructure
GAIIIP \$100B private equity partnership	No	No	Corporate ESG language without sovereign mandate; fails ENT-2's primary source and operative provisions tests

Grounding Assessment: CATEGORY CREATOR. ~45% grounded via governance maturity concepts; ~55% novel calibration. No existing framework applies a comparable intent-versus-operationalisation spectrum to environmental governance in AI infrastructure.

Validation Commitment: At 200 events, assess whether all ENT-5 events also carry ENT-2 (as predicted). If ENT-5 events exist without ENT-2, reassess boundary logic.

8.11 Chinese Cascade Model — Governance Architecture for "Operative Provisions" Test Application

Manual Section: §8.3.2

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: Chinese data centre environmental governance operates through a directive-to-enforcement cascade: central policy documents use directional language (引导 — "guide"; 鼓励 — "encourage") but gain operative force through (a) mandatory national standards (GB 40879-2021, 强制性国家标准) providing legal enforcement baselines, and (b) provincial implementation plans replacing directional language with mandatory language (须 — "must"; 应当 — "shall"; 不低于 — "no less than"; 不得超出 — "must not exceed").

The cascade is confirmed across four provinces spanning two enforcement pathways.

Administrative-hierarchy (Inner Mongolia): DRC links project approval (立项审批) to compliance with energy budget quotas and mandatory PUE standards. **Market-mechanism** (Ningxia): data centres designated as entities subject to renewable energy consumption obligations within provincial electricity trading, with State Grid Ningxia responsible for verification and dispatch. The ENT-2 "operative provisions" test applies correctly to the implementing instrument where binding language appears, consistent with the principle that EU directives require member-state transposition before creating binding obligations.

Evidence Chain:

Precedent	Type	Contribution
Three-Year Action Plan (工信部通信〔2021〕76号, MIIT, 4 July 2021): PUE ≤1.3 for new large data centres (引导)	Empirical	National directional layer — sets targets using guidance language
GB 40879-2021 (Standardization Administration of China, October 2021): mandatory PUE grades (1.5/1.3/1.2)	Empirical	National mandatory standard — legal enforcement baseline
Greater Bay Area hub approval (发改高技〔2022〕66号, NDRC, 7 February 2022): PUE ≤1.25	Empirical	Hub-level binding condition — approval-process enforcement
Inner Mongolia 14th FYP Energy Conservation Plan (内政办发〔2022〕11号): "须...不超过1.3"	Empirical	Provincial mandatory conversion — administrative-hierarchy enforcement
Ningxia 2026 Power Trading Notice (宁发改电力〔2025〕752号): data centre green electricity quota	Empirical	Provincial mandatory conversion — market-mechanism enforcement
EU directive transposition requirement	Structural	Analogous cascade: EU directives require member-state implementation
Chinese 1+N regulatory structure (Methodology Rationale §9.2)	Structural	Precedent for documenting Chinese cascade governance in WEO methodology

Grounding Assessment: EMPIRICALLY GROUNDED. ~80% grounded. The cascade is confirmed across 4 provinces, 2 enforcement pathways, 7 instruments with quoted operative language. EU transposition analogy and established CC Type 6 validation provide structural precedent. The ~20% novel element is application to environmental governance provisions.

Validation Commitment: Confirm cascade in additional EDWC hub provinces as coverage expands. Track whether market-mechanism pathway expands as green electricity trading markets mature.

8.12 Floors and Categorical Tests — Gate Design Principle

Manual Section: §8.3 (cross-cutting)

Category: WEO SYNTHESIS (MEDIUM-HIGH 60–75%)

Rationale: The five ENT gates are heterogeneous in form — quantitative floor, linguistic test, evidence-type test, coupled index-and-process conjunct, legal-character test — because each type's evidence universe is structurally different (§8.4). The uniformity that matters is discriminating power, not gate form.

The design principle governing gate selection: floors belong where a type's qualifying evidence is continuous (MW capacity, stress percentage, consumption character) and categorical tests belong where it is discrete (operative or not, measured or not, binding or not). The five gates instantiate this principle:

- **ENT-1:** $\geq 100\text{MW}$ quantitative floor (energy capacity is a continuous variable; the floor prevents trivial universality)
- **ENT-2:** Four-criteria categorical test (each criterion is a discrete yes/no determination — explicit, primary source, operative provisions, AI-infrastructure-directed)
- **ENT-3:** Measured-outcomes categorical test with temporal window (consequence is either documented and measured or it is not; the window governs when measurement becomes available)
- **ENT-4:** Qualifying-band floor $\times$ consumptive-stage floor (stress severity and consumption character are both continuous variables reduced to binary gates — the coupled architecture documented in §8.8)
- **ENT-5:** Binding-character categorical test with dual pathways (an instrument either creates binding environmental requirements or it does not; a programme either deploys dedicated environmental infrastructure at sovereign scale or it does not)

Adding floors to categorical gates (e.g. grading the "bindingness" of ENT-5 instruments) risks grading what the methodology designs as a binary gate. Adding categorical tests to floor-based gates (e.g. replacing $\geq 100\text{MW}$ with "significant energy demand") replaces verifiable thresholds with subjective characterisations.

Grounding Assessment: WEO SYNTHESIS. ~65% grounded. The floor/categorical distinction is a standard measurement-design principle; its application across a five-type environmental tagging taxonomy is the novel element.

8.13 Warmth Engine Ratio — Positioning and Novelty

Manual Section: §8.5

Category: CATEGORY CREATOR

Rationale: $WER = ENT-1 \text{ count} \div ENT-5 \text{ count}$ — measures the balance between documented environmental resource demand and documented environmental governance within the WEO event corpus. No formal demand-to-governance ratio exists as an established metric in environmental economics, sustainability science, or regulatory analysis.

Novelty. No demand-to-governance ratio appears in the major environmental governance measurement frameworks:

- **OECD Environmental Policy Stringency (EPS) Index and Climate Actions and Policies Measurement Framework (CAPMF):** measure policy stringency and policy adoption. Neither produces a demand-to-governance ratio.
- **Yale Environmental Performance Index (EPI):** measures environmental health and ecosystem vitality outcomes. Covers governance outputs, not the demand-to-governance balance.
- **World Bank Worldwide Governance Indicators (WGI):** measure governance quality across six dimensions. Governance-focused, not demand-facing.
- **UNEP environmental assessments:** monitor environmental state and pressures. Descriptive reporting, not a demand-to-governance ratio.
- **Institutional-fit literature** (Young 2002; Folke et al. 2007; Bergsten, Galafassi & Bodin 2014): examines whether governance institutions match the ecological and environmental systems they are supposed to manage. The field itself acknowledges the gap: "there is a shortage of approaches to systematically and quantitatively examine the level of fit" (Bergsten, Galafassi & Bodin 2014).

Citation: Bergsten, A., D. Galafassi, and Ö. Bodin (2014), "The problem of spatial fit in social-ecological systems: detecting mismatches between ecological connectivity and land management in an urban region," *Ecology and Society* 19(4):6. DOI: 10.5751/ES-06931-190406. URL: <https://doi.org/10.5751/ES-06931-190406>

Structural distinctiveness. The closest adjacent constructs run in inverse orientation or use different denominators. Ekstrom & Young (2009) measure governance-coverage-of-ecology — a Simple Matching Coefficient ($M = \text{ecosystem links covered by law} \div \text{total modelled ecosystem links}$). Carbon-pricing coverage measures governance-coverage-of-demand ($\text{regulated emissions} \div \text{total emissions}$). The Ecological Footprint measures demand $\div$ ecological capacity. WER inverts the orientation: demand in the numerator, governance in the denominator — the inverse of the governance-coverage orientation used in the fit literature.

Citation: Ekstrom, J. A., and O. R. Young (2009), "Evaluating Functional Fit between a Set of Institutions and an Ecosystem," *Ecology and Society* 14(2):16. URL: <https://www.ecologyandsociety.org/vol14/iss2/art16/>

WER as a sector-specific implementation of the fit concept. WER operationalises the problem-of-fit concept (Young 2002; Folke et al. 2007) within a single sector (AI infrastructure) using binary tags with specific qualifying gates — a sector-specific implementation, not a general-purpose governance-gap metric.

Citation: Folke, C., L. Pritchard, F. Berkes, J. Colding, and U. Svedin (2007), "The problem of fit between ecosystems and institutions: ten years later," *Ecology and Society* 12(1):30. DOI: 10.5751/ES-02064-120130. URL: <https://doi.org/10.5751/ES-02064-120130>

Temporal-tracking property. Because WER sits on a living database with a defined event-processing methodology, it can be tracked longitudinally as events enter the corpus — enabling observation of how the ratio evolves. This distinguishes it from the bespoke case-study quantifications in the fit literature, which produce a measurement once without tracking it over time.

Evidence Chain:

Precedent	Type	Contribution
Problem-of-fit literature (Young 2002; Folke et al. 2007)	Conceptual	Governance-ecosystem matching as an established research domain; WER operationalises the concept
Bergsten, Galafassi & Bodin (2014)	Conceptual	Explicit acknowledgement of quantitative gap in governance-fit measurement
Ekstrom & Young (2009)	Structural	Closest methodological analogue (Simple Matching Coefficient); inverse orientation distinguishes WER
Carbon-pricing coverage metrics	Analogue	Governance-coverage-of-demand orientation — WER inverts to demand-over-governance
Ecological Footprint	Analogue	Demand ÷ capacity — WER substitutes governance for ecological capacity

Grounding Assessment: CATEGORY CREATOR. ~40% grounded via the institutional-fit concept, which provides the theoretical frame. ~60% novel: no formal demand-to-governance ratio exists in environmental economics, sustainability science, or regulatory analysis; the specific implementation using binary ENT gates within a living event corpus is without precedent in the literature surveyed.

8.14 WER — Provisionality, n-Disclosure, and Corpus Scope

Manual Section: §8.5

Category: ADAPTED (HIGH 75–90%)

Rationale: Three disclosure provisions — n-disclosure, provisionality status, and denominator-scope transparency — adapt established conventions from adjacent measurement disciplines to WER's context.

n-Disclosure

WER is reported with its component counts: "WER [value] (n: [ENT-1 count]/[ENT-5 count])." This adopts a simplified version of the World Bank Worldwide Governance Indicators (WGI) practice, which publishes, for every governance data point, standard errors, the number of underlying sources, and upper and lower bounds of a 90% confidence interval (Kaufmann & Kraay 2024). WER's n-disclosure is the appropriate simplification for a simple ratio rather than a composite index — the component counts convey the evidential base without the statistical apparatus.

Citation: Kaufmann, D. and Kraay, A. (2024), "The Worldwide Governance Indicators: Methodology and 2024 Update," World Bank Policy Research Working Paper. URL: <https://www.worldbank.org/en/publication/worldwide-governance-indicators/documentation>

Provisionality Threshold

WER is reported as provisional while ENT-5 count < 10. The NCHS Data Presentation Standards for Proportions (Parker et al. 2017) establish suppression thresholds for ratio-based statistics with small denominators, including suppressing counts based on fewer than 10 events. WER's provisionality threshold draws on this convention. The difference: NCHS suppresses the data point entirely; WER publishes it provisionally with n-disclosure — combining transparency (WGI-style) with epistemic honesty about instability (NCHS-inspired threshold). Provisionality is a defined methodological status with stated exit criteria (ENT-5 ≥ 10), not an informal caveat.

Citation: Parker, J.D. et al. (2017), "National Center for Health Statistics Data Presentation Standards for Proportions," *Vital and Health Statistics, Series 2*, No. 175. URL: https://www.cdc.gov/nchs/data/series/sr_02/sr02_175.pdf

Denominator-Scope Transparency

The denominator counts environmental-governance instruments that qualify as WEO events; instruments outside the event corpus are outside the ratio. This applies the epistemic principle underlying PRISMA 2020 (Page et al. 2021) — which requires systematic reviews to report all databases searched, search dates, and limits applied (Items 6, 7), and to document exclusions with reasons (Item 16b) — to corpus-bounded environmental measurement. The principle that absence of evidence is not evidence of absence (Altman & Bland 1995) underpins this disclosure practice: the ratio's denominator measures what the corpus has documented, not what exists in the world. No major environmental policy index (OECD EPS, OECD CAPMF, Yale EPI, World Bank WGI) has adopted PRISMA-style corpus-disclosure norms.

Citations:

- Page, M.J. et al. (2021), "The PRISMA 2020 statement: an updated guideline for reporting systematic reviews," *BMJ*, 372, n71. DOI: 10.1136/bmj.n71. URL: <https://doi.org/10.1136/bmj.n71>
- Altman, D.G. and Bland, J.M. (1995), "Statistics Notes: Absence of evidence is not evidence of absence," *BMJ*, 311, 485.

Corpus-relative scoring as structural precedent. The OECD Climate Actions and Policies Measurement Framework (CAPMF) scores policy stringency relative to its own sample rather than against an external benchmark — a corpus-relative measurement approach consistent with WER's denominator scope.

Citation: OECD, "The Climate Actions and Policies Measurement Framework (CAPMF) — FAQ," 24 April 2025. URL: <https://www.oecd.org/en/data/insights/data-explainers/2025/04/the-climate-actions-and-policies-measurement-framework-capmf.html>

Evidence Chain:

Precedent	Type	Contribution
WGI methodology (Kaufmann & Kraay 2024)	Direct	n-disclosure with uncertainty quantification for governance indicators
NCHS Data Presentation Standards (Parker et al. 2017)	Direct	Suppression threshold ($n < 10$) for ratio-based statistics with small denominators
PRISMA 2020 (Page et al. 2021)	Structural	Corpus-boundary transparency as epistemic best practice
Altman & Bland (1995)	Structural	Absence of evidence $\neq$ evidence of absence — the principle the denominator-scope sentence operationalises
OECD CAPMF (2025)	Structural	Corpus-relative scoring: stringency measured within the sample, not against an external benchmark
WEO epistemic disclosure architecture (CC-V/CC-E/CC-A, evidence basis, scale assessment status)	Internal	Consistent disclosure pattern across platform provisions

Grounding Assessment: ADAPTED. ~75% grounded. WGI n-disclosure, NCHS suppression thresholds, PRISMA corpus-boundary transparency, and OECD CAPMF corpus-relative scoring provide convergent precedent from established measurement disciplines. The ~25% novel element is the specific combination — publishing with n-disclosure rather than suppressing, and applying corpus-boundary transparency to an environmental demand-to-governance ratio — which has no direct precedent in the surveyed environmental governance literature.

Section 9: Coordination Connections Rationale

Manual Reference: Chapter 9

9.1 CC Taxonomy Overview

Manual Section: 9.1, 9.2

Category: WEO-ORIGINAL (Verified Novel IP)

Rationale: The 7-type Coordination Connections taxonomy represents WEO's primary novel intellectual property contribution. Deep research across 25+ platforms (Stratfor/RANE, Eurasia Group, GDELT, ICEWS, ACLED, Stanford HAI, Sayari, Bloomberg, etc.) confirmed no existing product implements typed event-to-event relationship categories comparable to CC.

Key Finding: "No existing product implements typed event-to-event relationship categories comparable to CC."

Synthesis Sources:

- Chinese governance frameworks (1+N policy cascades)
- OFAC sanctions methodology
- ISO/IEC Directives structure
- Academic policy diffusion literature (Shipan & Volden, Gilardi & Wasserfallen)

Novelty Verification: Examination of 25+ platforms across four categories (geopolitical intelligence, AI policy trackers, event databases, entity relationship platforms) found:

- No platform uses typed event-to-event relationship categories
- Entity relationship platforms (Sayari, Maltego) have typed taxonomies but only for entity-entity connections
- CAMEO codes 204 event types but does not code relationships between events

9.1.1 Distinct Actor Requirement

Manual Section: §9.1.6

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: The distinct actor requirement formalises a principle that was implicit in the CC framework's design but not explicitly stated. Coordination, by definition, involves multiple parties — a single actor coordinating with itself is internal decision-making, not inter-actor coordination. The absence of an explicit statement created ambiguity during processing: same-corporation product lineage events (successive model releases) and intra-ministerial policy sequences generated CC candidates that passed the mechanical qualification tests (temporal proximity, thematic similarity) but did not represent coordination between distinct actors.

A corpus-wide investigation across the event database confirmed that same-entity connections, when mechanically qualified, produce analytically misleading results — they inflate the connection count without documenting genuine coordination dynamics. Seven same-entity CC candidates were permanently excluded from the holding pool following this investigation.

The subsidiary clause (consistent with §5.3.2) addresses the boundary case: subsidiaries operating under direct parent-company control are the same actor; subsidiaries with documented strategic autonomy in the relevant domain are distinct actors.

Confidence: 80%. The principle is definitionally grounded (coordination requires multiple parties) and supported by the holding pool investigation. No existing CC in the database is affected — the requirement formalises what was already operationally applied.

Structural co-membership clarification (V1.5).

The V1.4 §9.1.6 text uses exclusively sequential/causal examples (successive policy instruments, sequential product releases, model family lineage, corporate roadmap execution, intra-governmental policy cascades). No example addresses Type 6 Framework Family, which documents structural co-membership (bidirectional, non-causal) rather than sequential coordination. This created an ambiguity: a strict mechanical reading of §9.1.6 would remove valid programme-architecture CCs; a contextual reading preserves them but relies on implicit resolution.

Empirical basis. Corpus investigation confirmed the ambiguity and identified a discriminator: operationally distinct governance. Same-programme CCs where separate national or institutional entities operate under a shared multilateral or regulatory framework — each with its own governance, budget, and operational autonomy — are correctly documented as Coordination Connections. Same-programme CCs where a single corporate operator runs multiple facilities under unified operational control are correctly handled through event metadata. The discriminator is whether the implementing sub-entities have operationally distinct governance, not whether a parent framework document exists.

The corpus contains both patterns: multilateral infrastructure programmes with separate national hosting entities (operationally distinct governance — correctly documented as CCs) and single-operator infrastructure programmes with multiple sites (unified governance — correctly handled through event metadata). A corpus-wide audit confirmed that the operationally-distinct-governance test produces consistent results across all seven CC types.

Resolution. V1.5 adds a three-condition table to §9.1.6 encoding the operationally-distinct-governance discriminator for structural co-membership cases. The table preserves the requirement's universal scope ("applies to all seven CC types") whilst providing explicit guidance for the Framework Family boundary case that the V1.4 examples did not address.

Validation commitment: Monitor for edge cases at the operationally-distinct-governance boundary. If the three-condition test produces inconsistent results for structurally similar programme architectures across jurisdictions, refine the governance-distinctness criterion.

Validation commitment: Monitor for edge cases at the subsidiary boundary. If strategic autonomy assessments for CC purposes diverge from SCP subsidiary clause assessments, reconcile.

The entity-level test covers common-control scenarios: separate legal entities controlled by the same natural person with documented resource sharing are assessed as the same actor, consistent with the subsidiary clause's strategic-autonomy exception.

9.2 Seven Connection Types

Manual Section: 9.2.1

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: Each connection type was validated across 3+ independent systems before inclusion.

Type	Name	Definition	Precedent Inspiration
Type 1	Policy Cascade	Event B cites Event A as legal/policy basis	Chinese 1+N Framework, CSET policy analysis
Type 2	Joint Issuance	Single document with multiple issuing authorities	Chinese multi-ministry governance, OECD, EU regulatory practice
Type 3	Funding Flow	Event A provides funding enabling Event B ($\geq 10\%$ threshold)	OECD Due Diligence Guidance, Government grant standards
Type 4	Regulatory Enablement	Event A establishes framework Event B operates within	ISO/IEC Directives Part 2, Administrative law precedent
Type 5	Response	Event B is documented reaction to Event A	OFAC sanctions methodology, GSDB, CSET
Type 6	Framework Family	Events are part of same standards/institutional series	ISO/IEC standards families, UN framework structures
Type 7	Parallel Policy	Substantive similarity without documented linkage ($\geq 3/4$ criteria)	Shipan & Volden (2008), Gilardi & Wasserfallen (2019)

Academic Literature Foundation:

- Shipan, C. R., & Volden, C. (2008). "The mechanisms of policy diffusion." *American Journal of Political Science*, 52(4), 840-857.
- Gilardi, F., & Wasserfallen, F. (2019). "The politics of policy diffusion." *European Journal of Political Research*, 58(4), 1245-1256.
- Berry, F. S., & Berry, W. D. (1990). "State lottery adoptions as policy innovations." *American Political Science Review*, 84(2), 395-415.

Type 1: Policy Cascade — Authority Hierarchy Test

Category: ADAPTED (HIGH 75–90%) — synthesised from administrative law precedents

Source Frameworks:

- Western administrative law: Constitutional hierarchy principles
- Chinese governance: 1+N framework analysis (CCICED)
- ISO/IEC Directives Part 2: Normative vs informative reference distinction

Authority Hierarchy (Descending):

AL	Authority Level	Examples
AL-1 (Highest)	International treaties, multilateral binding agreements	UN Security Council resolutions, WTO agreements, binding ISO/IEC standards
AL-2	Constitutional provisions, enabling statutes	US Constitution, Basic Law, foundational organic laws
AL-3	Primary legislation	Acts of Congress/Parliament, Chinese NPC laws (法律)
AL-4	Regulations, Executive Orders, Ministerial Decrees	Federal Register rules, State Council regulations (法规), EU Regulations
AL-5	Agency guidance, implementation rules, departmental orders	Ministry notices (通知), agency guidance documents, rules (规章)
AL-6	Awards, grants, individual determinations	Funding awards under frameworks, permits, licences

Decision Rules:

1. **≥2 AL difference** → Policy Cascade
 - Example: Primary legislation (AL-3) → Agency implementation rule (AL-5)
2. **1 AL difference** → Examine language
 - Mandatory language ("pursuant to," "implementing," 根据, 落实) → Policy Cascade
 - Permissive language ("authorised under," "enabled by," 依照) → Regulatory Enablement
3. **Same Level** → Default to Regulatory Enablement unless explicit hierarchical language present
4. **Cross-jurisdictional** → International agreements (AL-1) creating national implementation (AL-3 to AL-5) = Policy Cascade

Chinese Framework Equivalent:

Level	Chinese Term	Authority
Central	中央/国务院	Highest domestic
Ministry	部委	Departmental
Provincial	省级	Regional
Local	地方/市级	Municipal

Document Type Hierarchy:

法律 (Law) > 法规 (Regulation) > 规章 (Rule) > 通知 (Notice) > 意见 (Opinion)

≥2 AL Threshold Rationale:

Clear hierarchy (2+ AL difference) provides unambiguous Policy Cascade classification. Adjacent ALs require linguistic marker analysis because hierarchy alone is insufficient to distinguish mandatory from permissive relationships.

Type 6: Framework Family — Evidence Pathway Grounding

Manual Section: 9.2.1

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: The three Type 6 evidence pathways (common numbering, explicit cross-reference, parent framework document) combine three independently established concepts from standards management and legal instrument classification into a single evidence framework for geopolitical event relationship classification. Each pathway has an identified external basis; the combination and application to WEO's domain is novel.

Research basis: A targeted survey of standards relationship classification systems was conducted across ISO/IEC Directives (Parts 1 and 2), NISO Standards Tag Suite (ANSI/NISO Z39.102), ISO Online Browsing Platform relationship categories, and national standards body concordance systems (BSI, ANSI, DIN, CEN/CENELEC).

Key findings:

1. **ISO/IEC does not formally define "standards family."** The only formally defined structural concept is the multipart standard (ISO/IEC Directives, Part 2, Clause 5.1). ISO's colloquial use of "family" (e.g., "ISO 27000 family") is defined by number range, not committee membership. No surveyed standards body provides a normative definition.
2. **The normative reference is the only relationship type creating binding technical dependency.** ISO/IEC Directives Part 2 defines normative references as documents "referred to in the text in such a way that some or all of their content constitutes requirements of this document." Informative references (bibliography, body text, informative annexes) provide contextual information without imposing requirements.
3. **Committee membership is administrative metadata, not an inter-standard relationship.** The NISO STS schema classifies inter-standard relationships through the `<std-xref>` element — relationship types include "revises," "supersedes," "amends," "adopted-from," and "std-family." Committee attribution and ICS classification appear as metadata but do not constitute formal inter-standard relationships.
4. **Cross-body adoption creates no transitive relationships.** The ISO-IEEE PSDO agreement (2007) defines fast-track adoption and joint development mechanisms. Adoption renumbers a standard into the adopting body's catalogue but creates no normative links to other standards in that catalogue.

Synthesis — pathway mapping to external sources:

WEO Pathway	External Basis	Source	Boundary
Common numbering/naming convention	Informal "family/series" number-range convention used by ISO, BSI, ANSI	ISO Online Browsing Platform; BSI Standards Catalogue	Shared number range required; different numbering under same committee insufficient
Explicit cross-reference	Normative reference as defined by ISO/IEC Directives Part 2, Clause 2	ISO/IEC Directives Part 2 (2021), §6.2.2	Normative Section 2 references only; informative references excluded
Parent framework document	Formal legislative/institutional designation as used in legal instrument classification	Administrative law hierarchy (see §9.2 Type 1 rationale); Vienna Convention on the Law of Treaties	Binding legal instruments and institutional founding documents qualify; administrative committee catalogues do not

Sources:

Institutional/Standards Sources:

- ISO/IEC Directives, Part 2 (2021): Normative vs informative reference distinction (§6.2.2)
- NISO Standards Tag Suite: ANSI/NISO Z39.102-2022, inter-standard relationship taxonomy (element)
- ISO Online Browsing Platform: Standards relationship display and committee attribution metadata
- ISO-IEEE Partner Standards Development Organisation (PSDO) Cooperation Agreement (2007): Fast-track adoption and joint development mechanisms
- BSI Standards Catalogue; ANSI Essential Requirements; DIN 820; CEN/CENELEC Internal Regulations: Surveyed for "standards family" normative definitions (none found)

Confidence: HIGH (85%) — each pathway individually grounded in ISO/IEC Directives Part 2, NISO STS Z39.102, and established legislative instrument classification. Combination into three-pathway structure is WEO synthesis without direct external precedent.

Validation commitment: Review pathway boundaries at 100-event milestone based on classification consistency data.

Pathway (c) parent↔member expansion (V1.5).

Manual Section: §9.2.1 (Type 6 direction convention)

Category: STANDARD PRACTICE (clarification of existing provisions)

Framework Family documents structural co-membership within a programme or institutional framework. Two structurally distinct co-membership patterns exist:

Sibling↔sibling — two events are peer members of a framework, neither of which is the parent framework document. Two standards within a numbered series, two awards under a subsidy programme, two national facilities under a multilateral infrastructure programme. The relationship is bidirectional and symmetric.

Parent→member — one event is the parent framework document and the other is a designated component. A national infrastructure programme designating specific facilities as components; an enabling regulation designating specific initiatives as programme elements. The relationship is directional (parent as source, member as target) and asymmetric — the parent designates the member, not the reverse.

Both patterns satisfy the same pathway (c) evidence standard: a parent framework document that formally identifies events as components of a defined programme or framework. The difference is whether the parent framework document is itself a WEO event (producing a parent→member CC) or an external reference document (serving as evidence for a sibling↔sibling CC between two member events).

Parent→member Framework Family captures programmatic designation — the relationship between a programme-level event and its designated components — where that designation is the substantive inter-event connection. Where the parent→member relationship is already documented through a more specific CC type (Funding Flow, Policy Cascade, Regulatory Enablement), the designation is implicit in the functional relationship and a Framework Family CC is redundant.

External precedent. Parent↔member relationships alongside sibling↔sibling relationships are established in the same source frameworks that ground the V1.4 three-pathway structure. ISO/IEC multipart standards (ISO/IEC Directives, Part 2, Clause 5.1) define a formal parent↔member structure: Part 1 is both a family member and the hierarchical framework part, with subsequent parts as subordinate components. The NISO Standards Tag Suite (ANSI/NISO Z39.102-2022) includes directional relationship types ("adopted-from," "revises," "supersedes") alongside peer types ("std-family") within the same `<std-xref>` taxonomy, confirming that standards-relationship systems accommodate both hierarchical and peer structures. Legislative families operate identically — an enabling statute is both a member of the legislative family and the hierarchical parent from which implementing instruments derive authority (see §9.2 Type 1 rationale, administrative law hierarchy).

Confidence: HIGH. Both co-membership patterns are grounded in the same external precedents (ISO/IEC Directives Part 2, NISO STS, legislative instrument classification). The direction convention formalises the distinction between the two patterns and maps each to a `direction` field value (`bidirectional` for sibling, `source_to_target` for parent→member).

Non-Western pathway validation: A targeted research investigation was conducted to verify whether the three evidence pathways, derived from Western standards body and legislative practice, adequately capture structural dependencies in non-Western AI regulatory and standards architectures. The Chinese ecosystem was assessed as the primary Eastern Bloc standards and regulatory system relevant to the current event population, with comparative assessment of Japanese (METI/MIC), South Korean (AI Basic Act/MSIT), and Indian (MeitY) regulatory architectures.

Non-Western key findings:

1. **Pathway 2 confirmed applicable to Chinese GB/T standards.** SAC/TC260 standards use a 规范性引用文件 (Normative Reference Documents) section with binding language ("构成...必不可少的条款" — "constitutes an indispensable clause") functionally equivalent to ISO/IEC Directives Part 2 normative reference requirements. No pathway modification required.
2. **TC260 AI standards exhibit a decentralised mesh topology** with domain-specific cluster hubs (data annotation around GB/T 42755, large models around GB/T 45288 series, federated learning around GB/T 41867) rather than the hub-and-spoke pattern observed in ISO/IEC SC 42 (centred on ISO/IEC 22989). The topology differs but the pairwise evidence standard is identical.
3. **Chinese 1+N regulatory structures confirmed compatible with Pathway 3.** Provincial and ministerial implementation documents cite the parent State Council directive by specific document number (e.g., "国发〔2017〕35号") in operative provisions. Where "N" documents cite enabling legislation instead, this constitutes a separate connection chain rather than a direct Framework Family link.
4. **No new pathway required.** Mechanisms proposed as requiring methodology expansion (implicit hierarchical delegation, structural convention, experimental frameworks) represent thematic proximity or administrative relationships rather than formal structural dependencies meeting the CC-V evidence standard.

Comparative pathway compatibility:

Regulatory Architecture	Pathway Compatibility	Notes
Chinese GB/T (SAC/TC260)	Pathway 2 — confirmed	规范性引用文件 equivalent to ISO/IEC Section 2
Chinese 1+N (State Council → ministry/provincial)	Pathway 3 — confirmed	"N" documents cite "1" directive by document number in operative text
Japanese (METI/MIC, AI Promotion Act)	Pathways 2 and 3 — confirmed	Statute-driven model; explicit legislative designation
South Korean (AI Basic Act/MSIT)	Pathways 2 and 3 — confirmed	Comprehensive statutory framework; explicit institutional designation
Indian (MeitY, fragmented model)	Inconclusive	Multi-layered, soft-law-reliant architecture; further assessment required at scale

Non-Western sources:

Standards Architecture:

- SAC/TC260 standards documentation: 规范性引用文件 section structure and binding language conventions
- GB/T 42755-2023 (AI Data Annotation for Machine Learning): Normative reference patterns within TC260 ecosystem
- GB/T 45288 series (AI Large Models): Cluster hub cross-reference patterns
- GB/T 41867-2022 (Federated Machine Learning): Domain-specific normative referencing

Regulatory Architecture:

- State Council Next Generation AI Development Plan ("国发〔2017〕35号"): 1+N directive structure
- Interim Measures for the Management of Generative AI Services (CAC, 2023): Legal basis citation patterns (Cybersecurity Law, Data Security Law, Personal Information Protection Law)
- Provincial implementation plans (Gansu, Liaoning, Jilin): Operative text citation of parent directive by document number

Updated confidence: HIGH (85%) — unchanged. Non-Western validation confirms existing pathways are not Western-specific; no expansion required for current event population. Indian regulatory architecture flagged for reassessment at scale.

Type 7: Parallel Policy — Jurisdiction Clarification

Manual Section: 9.2.1

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: The "different jurisdictions" requirement was clarified to explicitly exclude same-body standard pairs and same-programme events. This definition is derived directly from the policy diffusion literature that underpins Type 7:

Source citations:

- Shipan, C. R., & Volden, C. (2008). "The mechanisms of policy diffusion." *American Journal of Political Science*, 52(4), 840-857. Study of policy diffusion across US cities — "jurisdiction" means governmental entities with territorial authority.
- Gilardi, F., & Wasserfallen, F. (2019). "The politics of policy diffusion." *European Journal of Political Research*, 58(4), 1245-1256. Define diffusion units as "country, state, city, etc."
- Graham, E. R., Shipan, C. R., & Volden, C. (2013). "The diffusion of policy diffusion research in political science." *British Journal of Political Science*, 43(3), 673-701. Define policy diffusion as "the process by which policymaking in one government affects policymaking in other governments."
- Simmons, B. A., Dobbin, F., & Garrett, G. (2006). "Introduction: The international diffusion of liberalism." *International Organization*, 60(4), 781-810. "Policy decisions in a given jurisdiction are systematically conditioned by prior policy choices made in other jurisdictions" — jurisdictions are always governmental entities.

Key finding: No source in the policy diffusion literature applies the "jurisdiction" concept to individual standards or individual policy outputs within a single decision-making body. Hooghe, L. & Marks, G. (2003) "Unraveling the Central State, but How? Types of Multi-level Governance" (*American Political Science Review*, 97(2), 233-243) — the closest theoretical analogue for treating a standards body as a jurisdiction-like entity — recognises "task-specific jurisdictions" as functionally specific governance entities, but even these are organisational entities, not individual outputs.

Confidence: HIGH (95%) — directly derived from five independent academic sources with zero scholarly precedent for alternative interpretation.

Type 5: Response — Official Statement Evidence Standard

Manual Section: §9.2.1 (Type 5 evidence requirements: qualifying sources hierarchy, explicit identification standard, non-qualifying sources)

Category: ADAPTED (HIGH 75–90%) — from OFAC sanctions methodology, GSDB attribution conventions, CSET policy analysis evidence standards

Rationale: The Type 5 "Official Statement" Evidence Standard governs what qualifies as evidence that a responding entity explicitly identifies a trigger event. The standard has three components: a qualifying sources hierarchy (five tiers), an explicit identification test (three sufficiency patterns and two insufficiency patterns), and a non-qualifying sources exclusion list.

Why explicit identification, not contextual inference. The explicit identification requirement is the structural boundary between Type 5 (Response) and Type 7 (Parallel Policy). Type 5 documents a relationship the responding entity has officially acknowledged — the response names the trigger. Type 7 documents structural similarity between events where no official inter-event acknowledgement exists. If contextual inference (analyst determination that Event B appears to respond to Event A) satisfied the Type 5 evidence standard, the distinction between documentary and analytical CC types would collapse. The explicit identification requirement preserves the type boundary: where the responding entity names the trigger, the relationship is Type 5; where it does not, the relationship is assessed under Type 7 if structural similarity is present.

Qualifying sources hierarchy. The five qualifying source tiers are ordered by documentary formality and attributional directness:

Tier	Source type	Precedent basis
1	Government press release (.gov domain)	OFAC SDN designation methodology — official government publications are the primary evidence source for sanctions and counter-sanctions attribution
2	Official gazette or register publication	GSDB (Felbermayr et al.) — the Global Sanctions Database uses gazette publications as primary attribution sources for sanctions actions and their stated triggers
3	Regulatory filing preamble or explanatory memorandum	CSET policy analysis methodology — regulatory impact statements and explanatory memoranda document the policy rationale linking a response instrument to its trigger
4	Official spokesperson attribution in verified news sources	Intelligence community sourcing conventions (ICD 203/206) — named, official-capacity statements to media constitute attributable official positions
5	Verified official social media (ministry-level or above)	Contemporary government communication practice — official ministry accounts function as press release channels in jurisdictions where .gov-domain publication is delayed or inconsistent

Tier 1–3 are documentary sources (the evidence is in the document itself). Tiers 4–5 are attributed sources (the evidence is in an official statement captured by a third-party channel). The tier ordering reflects the strength of the attribution chain: self-published government documents carry stronger attribution than third-party-reported statements.

Non-qualifying sources exclusion. Five source categories are excluded because they fail the "official statement" requirement in structurally distinct ways:

Excluded category	Exclusion rationale
Unnamed government sources	Attribution is unverifiable — the responding entity has not officially associated itself with the statement
Leaked documents	Provenance is unverified — leaked documents may be fabricated, selectively presented, or pre-decisional drafts that do not represent the entity's official position
Think tank analysis or academic inference	These are analyst assessments of the trigger-response relationship, not the responding entity's own identification of the trigger — admitting them would collapse the Type 5/Type 7 boundary
News media characterisation without official attribution	A journalist's framing ("widely seen as a response to...") reflects editorial judgement, not the responding entity's official position
Personal social media accounts (even of officials)	Personal accounts do not carry institutional authority — an official's personal commentary is not an official statement of the entity they represent

Explicit identification sufficiency. The three-pattern sufficiency test establishes a specificity threshold: the trigger event must be uniquely identifiable from the statement. Document title/number is maximally specific. Issuing authority + subject + timing narrows to a single event through triangulation. Issuing nation + policy type + specific target achieves the same through alternative dimensions. Generic references ("certain foreign measures") fail because they do not uniquely identify which event is the trigger — multiple candidate triggers may fit the description, and the analyst cannot determine which without applying inference, which is analytical territory (Type 7).

Precedent mapping:

WEO component	External precedent	Application
Official statement requirement	OFAC sanctions administration — enforcement actions cite specific sanctions designations by OFAC programme and SDN entry number	Adapted: WEO requires the responding entity's statement to identify the trigger event with comparable specificity
Attribution hierarchy	ICD 203 (Analytic Standards) / ICD 206 (Sourcing Requirements) — graduated source reliability and attribution standards	Adapted: WEO's five-tier qualifying sources hierarchy maps institutional attribution strength
Gazette/register evidence	GSDB (Global Sanctions Database) — sanctions events indexed by official gazette publication with explicit trigger attribution	Adapted: WEO accepts gazette publications as qualifying evidence for Response CC trigger identification
Explicit naming test	CSET policy analysis — tracks policy instruments and their stated antecedents through document-level citation analysis	Adapted: WEO's sufficiency table operationalises the CSET analytical principle as a deterministic classification gate

Confidence: HIGH (85%). Each component is individually grounded in established evidence methodology (OFAC, GSDB, CSET, ICD 203/206). The combination into a single evidence standard with sufficiency markers and exclusion list is WEO synthesis applied to an established analytical principle. The standard is consistent with the single Type 5 CC in the corpus (CC-126-217-5: head-of-state statement at an international summit, qualifying via Tier 4, with explicit trigger identification via the "Issuing nation + policy type + specific target" pattern).

Validation commitment: Monitor Type 5 CC candidate assessments for evidence-standard boundary cases. If legitimate Response CC candidates are systematically excluded by the explicit identification requirement due to euphemistic official language conventions in specific jurisdictions (e.g. Chinese diplomatic register, ASEAN communiqué conventions), assess whether a jurisdiction-specific accommodation is warranted, following the precedent established by the Chinese regulatory temporal accommodation (§9.4.1).

9.3 CC Confidence Levels

Manual Section: 9.3

Category: ADAPTED (HIGH 75–90%) — from ICD 203/206, ISO/IEC frameworks

Rationale: The three-tier CC confidence system (CC-V/CC-E/CC-A) provides graduated evidentiary standards for inter-event relationship claims.

Level	Definition	Derivation Source
CC-V (Verified)	Documentary proof of connection	ISO/IEC normative reference requirements; legal citation standards
CC-E (Established)	Authoritative statement documents connection	ISO/IEC informative reference standard; ICD 206 sourcing requirements
CC-A (Analytical)	Analyst assessment based on observable evidence	Policy diffusion research methodology (Shipan & Volden); 4-criteria assessment framework

Design Principle: CC-A is the only confidence level permitting analyst assessment rather than documentary proof. Type 7 (Parallel Policy) is the sole CC type assessed at CC-A, a structural entailment of its definition: independent convergence is evidenced by observable similarity between instruments, not by documentation establishing a coordination relationship. CC-A is both the minimum and maximum confidence level for Type 7 — the analytical determination that constitutes the connection cannot, by construction, produce documentary-grade evidence of the connection itself (see CC-A Confidence Ceiling below).

9.3.1 CC-A Evidence Model for Parallel Policy

Category: CATEGORY CREATOR

Type 7 (Parallel Policy) connections capture structural similarity between independently-developed policies. The "connection" is an analytical determination, not a documentary relationship — accordingly, the evidence model differs from CC-V and CC-E.

CC-A Evidence Requirements:

Component	Requirement
Event A documentation	≥1 primary source establishing event exists with documented characteristics
Event B documentation	≥1 primary source establishing event exists with documented characteristics
Criteria assessment	All 4 similarity criteria evaluated (mechanism, industry, scope, timing)
Criteria rationale	Explicit evidence cited for each criterion determination, including sub-category characteristic(s) for Criterion 1 and shared industry tag(s) for Criterion 2
Threshold	Criterion 1 (Mechanism Similarity) met as mandatory gate; ≥3/4 criteria met total

Rationale: Rigour for Parallel Policy derives from systematic criteria assessment with explicit threshold requirements, not source multiplication. The 4-criteria framework operationalises the policy diffusion literature (Shipan & Volden, 2008; Gilardi & Wasserfallen, 2019) into reproducible analytical determination. Criterion 1 (Mechanism Similarity) is a mandatory gate because mechanism convergence is definitional to the type — without it, the remaining criteria document thematic, geographic, and temporal coincidence rather than parallel instrument adoption (§9.5.2). Criterion 2 (Industry Similarity) is assessed deterministically through the industry tagging system (Chapter 7), eliminating subjective "target" judgments (§9.5.3).

9.3.2 CC-A Confidence Ceiling and Coordination Evidence Routing

Manual Sections: §9.3.2, §9.4.3

Category: CATEGORY CREATOR (HIGH 75–90%) — derived from structural analysis of Type 7 definition

Type 7 Parallel Policy confidence is CC-A (floor = ceiling). The CC-A designation is not a minimum awaiting documentary elevation — it is the logically entailed confidence level for a connection type whose defining characteristic is the absence of formal coordination. CC-V and CC-E are documentary confidence levels: CC-V asserts that a document directly links the two events; CC-E asserts that an authoritative statement documents the connection. A Type 7 connection is, by definition, not a documentary connection — it is an analytical determination of structural similarity between independently developed instruments. If documentary proof of the connection itself existed, the relationship would satisfy a Type 1–6 definition (which requires such documentation) and would not be classified as Type 7.

Coordination evidence routing. When documentary coordination evidence exists between the connected events' jurisdictions (bilateral agreements, joint statements, strategic partnerships, ministerial joint task forces), the analyst assesses Types 1–6 first. If a documentary type is satisfied, the CC is classified under that type at the appropriate documentary confidence level. If no documentary type fits — because the instruments were independently legislated through separate domestic processes despite the bilateral coordination environment — the CC remains Type 7 at CC-A. Domain-level coordination evidence is recorded in the `evidence_detail` field as analytical context but does not elevate the confidence level, because the CC determination itself remains analytical.

Empirical basis. The corpus audit (June 2026, 154 CCs across 181 events) identified two Type 7 CCs carrying confidence levels above CC-A: CC-06-05-7 at CC-E and CC-188-190-7 at CC-V. Independent investigation (independent verification, V1.4 Manual, full corpus) found:

- Neither CC satisfies any Type 1–6 definition. Both connect independently legislated sovereign instruments enacted through separate domestic processes with no cross-citation, no shared parent framework, no funding relationship, and no explicit identification of one instrument by the other.
- The coordination evidence in both cases pertains to the *domain-level relationship between two countries* — not to the *design relationship between the two specific instruments*. CC-06-05-7's bilateral evidence (JUCIP Basic Principles, Biden-Kishida Joint Statement) documents US-Japan semiconductor policy coordination. CC-188-190-7's bilateral evidence (EU-Australia MoU on Critical Minerals) documents strategic coordination on critical minerals policy. Neither bilateral instrument links the two specific legislative events to each other.
- The Biden-Kishida Joint Statement (April 2024) — the primary source cited for CC-06-05-7's elevation — postdates both connected events by nearly two years. It is retrospective domain-level coordination evidence, not event-pair connection evidence.
- The EU-Australia MoU was signed five days after the CRMA's entry into force and nine months before the CMPTI's enactment. It documents a strategic partnership, but does not designate the two legislative instruments as its implementation components. Both instruments cite their respective domestic frameworks as their operative basis.

The domain-level / event-pair distinction. A CC's confidence level reflects the evidentiary basis for the specific inter-event claim, not the strongest available evidence about the broader relationship between the actors involved. Bilateral MoUs and summit statements are strong evidence about the actor-pair relationship — they demonstrate that the policy environment is one of active coordination rather than genuinely independent parallel adoption. This is analytically valuable context. But the CC itself (the claim that these two specific instruments are structurally parallel) is established by the similarity rubric, not by the bilateral evidence. The bilateral evidence explains *why* the instruments converge (shared strategic interests, aligned threat assessments); the similarity assessment establishes *that* they converge (structural match across the four criteria). The former is context; the latter is the connection.

Practical outcome. Both CCs are reclassified to CC-A under V1.5. Bilateral coordination evidence was retained in `evidence_detail` as analytical context, providing richer characterisation of the policy environment without misrepresenting the evidentiary basis of the connection determination. A Parallel Policy arising within a documented bilateral coordination relationship is analytically distinct from one arising through genuinely independent convergence — the CC-A confidence level is honest about how the determination is made; the context field adds the granularity.

Precedent. Types 2, 3, and 6 each have a single permissible confidence level (CC-V). Type 1 permits CC-V and CC-E, reflecting the tiered documentary evidence standard introduced at V1.5 (see §9.4 Rationale). All non-Type-7 confidence levels are documentary in character — CC-V and CC-E both assert that documents establish the inter-event relationship, differing only in the directness of the documentary link. Type 7 is structurally distinct: it is the sole analytical CC type, and its CC-A confidence level reflects that the determination rests on structural similarity assessment, not documentary proof. The floor = ceiling principle applies to Type 7 because the nature of the evidence is inherently different in kind, not merely different in strength.

Confidence Assessment: HIGH. The ceiling follows deductively from the type definition and the confidence level definitions. The routing rule operationalises the deduction procedurally.

Validation Commitment: Monitor for cases where the routing rule is invoked (coordination evidence exists but no Type 1–6 fits). If a pattern emerges where coordination evidence consistently fails to satisfy any documentary type despite being event-pair-specific rather than domain-level, reassess whether the Type 1–6 definitions have a coverage gap.

9.3.3 Type 1 Evidence Route Clarification

Manual Sections: §9.2.1 (Type 1 evidence requirements), §9.3.2 (minimum confidence table)

Category: STANDARD PRACTICE (clarification of existing provisions)

The V1.4 Manual §9.2.1 lists three evidence routes for Policy Cascade connected by implicit disjunction: (a) child event explicitly cites parent event as legal/policy basis; (b) parent event issues directive requiring child event's actions; (c) implementation relationship documented in official sources. In practice, route (a) — child-cites-parent — dominated application, with routes (b) and (c) underutilised. V1.5 makes the disjunction explicit: any single route independently establishes the cascade relationship.

Empirical basis. Examination of 12–14 established frameworks across policy diffusion, process tracing, trade policy, and regulatory analysis found unanimous agreement: no examined framework imposes a citation-only evidence standard for cross-jurisdictional policy influence (Dolowitz & Marsh 2000, DOI: 10.1111/0952-1895.00121; Elkins & Simmons 2005, DOI: 10.1177/0002716204272516; Shipan & Volden 2008, DOI: 10.1111/j.1540-5907.2008.00346.x). Every framework accepts adjacent documentary evidence — parliamentary records, regulatory impact assessments, diplomatic records — under disciplined conditions.

Targeted evidence assessment of four cross-jurisdictional cascade candidates in the WEO corpus confirmed the V1.4 evidence standard's existing scope as adequate:

- *Trilateral semiconductor coordination → national implementing instruments:* Negative finding. The coordinating event has no formal parent instrument (no treaty, no MOU, no published text), so all three evidence routes are structurally unavailable. The relationship is correctly captured at the event level rather than through a CC — an example of a documented coordination boundary rather than an evidence-standard gap.
- *Cross-jurisdictional semiconductor incentive cascades:* Government sources in the adopting jurisdiction benchmarked against the originating instrument's provisions in pre-enactment documentation. However, the evidence establishes competitive policy benchmarking — designing a comparable instrument — not a hierarchical cascade. These relationships are correctly classified as Type 7 Parallel Policy.
- *Chinese intra-hierarchy cascades:* Provincial government documents cite central framework programmes by instrument number using explicit 根据 citations, satisfying the existing citation standard. The evidence exists at the correct administrative level; no evidence-standard expansion is required.

Resolution. V1.5 makes explicit that the three evidence routes are operationally equal — any single route independently establishes the cascade relationship. The confidence level is determined by the evidence's documentary character: direct citation (CC-V) or evidence requiring interpretation of official sources (CC-E). CC-A is not available for Type 1; where no documentary evidence establishes the inter-event connection through any route, the relationship is assessed under Type 7 if structural similarity is present.

Scope of change. This is a clarification of existing Manual provisions, not an expansion of evidence admissibility. The three routes were present in V1.4. The V1.5 contribution is: (a) explicit statement that they are operationally equal, and (b) confidence-level mapping by evidence character.

Confidence Assessment: HIGH. The clarification restates what the Manual already provides; the investigation confirms the existing scope is adequate.

9.3.4 Independent Adoption Check

Manual Section: §9.1.7

Category: STANDARD PRACTICE (codification of implicit principle)

Before classifying any relationship at Types 1–6, the analyst considers whether both events may have arisen independently in response to common conditions. This check is the procedural complement to the Type 7 routing rule (§9.4.3): where independent adoption cannot be ruled out and no documentary evidence establishes the inter-event connection, the relationship routes to Type 7 rather than a documentary type.

The principle was implicit in the V1.4 framework — the documentary evidence requirement for Types 1–6 and the Type 7 definition ("without formal coordination") together entail that undocumented convergence routes to Type 7. The V1.5 codification makes this explicit as a named check, ensuring it is applied systematically rather than being encountered only through the Type 7 routing rule.

Confidence Assessment: HIGH. Follows from the existing type definitions and the Type 7 routing rule.

9.4 Evidence Requirements

Manual Section: 9.4

Category: STANDARD PRACTICE

Rationale: Evidence requirements per connection type follow established verification principles from Chapter 3, adapted for inter-event relationship claims.

9.4.1 Temporal Windows

Category: EMPIRICALLY GROUNDED (MEDIUM-HIGH 60–75%)

Rationale: Temporal windows reflect domain-specific implementation cycles and response dynamics.

Response CC Derivation — Response Bands

Category: EMPIRICALLY GROUNDED (HIGH 75–90%)

Rationale: Response CC temporal windows are calibrated to the response evidence type — the category of official output that records the response. Empirical investigation across 46 dated trigger-response cases spanning three domains (Technology, Security, Trade) and two entity types (sovereign, corporate) demonstrates that the response evidence type is the strongest predictor of measured lag.

The V1.4 calibration used domain-specific windows derived from domain-characteristic sources (OFAC enforcement timing for Security, WTO DSU consultation periods for Trade, product development cycles for Technology, infrastructure project timelines for Energy/Infrastructure). Operational application revealed a systematic exclusion: well-evidenced Technology-domain responses appearing in annual reports (9–15 months) and IPO prospectuses (18–38 months) were excluded by the 180-day window, even where the response document explicitly named the trigger and documented a strategic pivot.

Empirical basis: Three independent deep research inputs (investigating the Technology domain) and two further inputs (investigating Security and Trade domains, including both sovereign and corporate responses) produced a consolidated dataset of 46 qualifying trigger-response cases. Qualifying cases required explicit naming of the trigger in the response document, a resolved gap with exact dates, and post-dating.

Finding: Response evidence type predicts response lag. Two independent discriminator analyses sorted the dataset three ways — by domain, by responding entity type, and by response evidence type — and computed within-group coefficient of variation (CoV) for each grouping. Results:

Criterion	Domain	Entity type	Response evidence type
Lowest within-group CoV	1.07	1.13	0.31
Groups with CoV < 0.75 (N ≥ 3)	0 of 3	0 of 2	4 of 7

Response evidence type is the only axis producing tight clusters. Four response-evidence-type groups achieve CoV below 0.75: national programme announcements (0.31), IPO prospectuses (0.33), annual/periodic corporate reports (0.52), and WTO notifications (0.73).

Three disclosure mechanisms correspond to the three Bands:

Band A evidence types (press statements, executive decrees, compliance filings, regulatory acts) are produced at or near the time of the response decision. The response and the document are the same act — a presidential decree enacting counter-sanctions, a ministry announcement imposing retaliatory tariffs, a compliance filing disclosing a material event. The gap reflects institutional process speed: how long it takes to draft, approve, and publish the instrument.

Band B evidence types (annual reports, quarterly filings, national legislation, national programme announcements) surface at the next available disclosure cycle. The responding entity decides promptly, but the documentary evidence appears only when the reporting or legislative cycle produces a vehicle capable of recording it. The gap is a mechanical artefact of where the trigger falls in the fiscal year or legislative calendar.

Band C evidence types (IPO prospectuses, major restructuring disclosures) require the completion of a multi-year process before the document can exist. The responding entity cannot produce an IPO prospectus documenting a strategic pivot until the pivot — chip redesign, foundry migration, capital raise — is substantially complete.

This three-mechanism structure is domain-independent. Security-domain executive decrees and Trade-domain ministry announcements both cluster at 0–9 days. Technology-domain and Trade-domain annual reports both cluster at 106–342 days. Technology-domain IPO prospectuses cluster at 608–1,155 days. The lag tracks the disclosure vehicle.

Deliberate-pause cases represent a distinct fourth mechanism: the responding entity prepared a response but deliberately withheld it pending negotiation or strategic calculation. Two cases in the dataset exhibit this pattern (one Security-domain strategic commitment reduction at 365 days; one Trade-domain retaliatory tariff at 456 days with a documented 12-month deferral between notification and enactment). These fall within Band B but their gap reflects a policy choice.

Band boundaries:

Boundary	Empirical basis
A/B at 90 days	Natural gap in the dataset between 76 days (highest Band A case) and 106 days (lowest Band B case). 29 of 46 cases fall in Band A; 14 in Band B
B/C at 18 months (548 days)	Natural gap between 456 days (highest Band B case) and 608 days (lowest Band C case). All three Band C cases are IPO prospectuses from two independent entities
C ceiling at 36 months (1,095 days)	Captures all qualifying cases. The longest measured gap (1,155 days) exceeds 36 months, but the same response document measured from a more proximate trigger gives 791 days (within Band C). The ceiling selects for the most proximate trigger

Why Response Bands use a different axis from other CC types: For Policy Cascade, Funding Flow, Regulatory Enablement, and Parallel Policy, the temporal window measures how long the coordination process takes to play out in a given domain — Energy/Infrastructure cascades are slow, Security responses are fast. The lag is domain-driven because the events themselves occur on domain-specific timescales. For Type 5 Response CCs, the temporal window measures how long until documentary evidence of the response appears. The response decision is typically rapid regardless of domain; what varies is the disclosure vehicle. Domain-specific windows for other CC types and response-evidence-type-specific Bands for Type 5 both follow the same calibration principle: match the window to the mechanism that generates the lag.

Sources:

- Crosignani, M., Han, L., Macchiavelli, M., & Silva, A. F. (2024). "Securing Technological Leadership? The Cost of Export Controls on Firms." Federal Reserve Bank of New York Staff Reports, no. 1096. DOI: 10.59576/sr.1096. Forthcoming in *Journal of Financial Economics*, vol. 175 (2026). Finding: immediate equity market reaction (~US\$130 billion market-capitalisation decline) with no evidence of reshoring or friend-shoring within three years.
- Center for Strategic and International Studies (CSIS). "Delays and Uncertainty in the Export Licensing Process: Findings from a CSIS Survey of U.S. Technology Exporters." Finding: median export-licence review time of 210 days for firms listing China as a primary export market, against a 90-day statutory benchmark.
- Winston & Strawn LLP. "China Places Export Controls on Raw Materials Used in Semiconductor Manufacturing."
- Berry, F. S., & Berry, W. D. (1990). "State lottery adoptions as policy innovations." *American Political Science Review*, 84(2), 395–415. DOI: 10.2307/1963526.
- Shipan, C. R., & Volden, C. (2008). "The mechanisms of policy diffusion." *American Journal of Political Science*, 52(4), 840–857. DOI: 10.1111/j.1540-5907.2008.00346.x.
- Mallinson, D. J. (2021). "Policy innovation adoption across the diffusion life course." *Policy Studies Journal*, 49(2), 335–358. DOI: 10.1111/psj.12406.

Confidence: HIGH. The response-evidence-type-predicts-lag finding converges across five independent research inputs, two independent discriminator analyses, and 46 qualifying cases spanning three domains and both entity types. The finding is consistent with the sole Type 5 CC in the corpus (CC-126-217-5, Band A, 21 days, press statement evidence type).

Validation commitment: Monitor Band boundary robustness as new Type 5 Response CCs enter the corpus. If quarterly-filing cases systematically cluster above 90 days, reassess the A/B boundary. If new response evidence types emerge that do not fit the three-Band structure, assess whether a fourth Band is warranted or whether the existing Bands absorb them.

Parallel Policy Derivation Sources:

Parallel Policy temporal windows are domain-specific, independently calibrated to policy diffusion timescales (Shipan & Volden, 2008; Berry & Berry, 1990).

Domain	Parallel Policy CC	Derivation Source
Energy/Infrastructure	24 months	Multi-year project timelines
Security	6 months	OFAC enforcement timing patterns
Technology	12 months	Product development cycles
Trade	12 months	WTO DSU Article 4.7 consultation period

Academic Literature Foundation:

- Berry, F. S., & Berry, W. D. (1990). "State lottery adoptions as policy innovations." *American Political Science Review*, 84(2), 395–415. DOI: 10.2307/1963526.
- Shipan, C. R., & Volden, C. (2008). "The mechanisms of policy diffusion." *American Journal of Political Science*, 52(4), 840–857. DOI: 10.1111/j.1540-5907.2008.00346.x.
- Mallinson, D. J. (2021). "Policy innovation adoption across the diffusion life course." *Policy Studies Journal*, 49(2), 335–358. DOI: 10.1111/psj.12406.

Status: Response CC calibration empirically validated — see Response Band Calibration above. Other CC type temporal windows: provisional calibration pending further empirical validation.

Chinese Regulatory Temporal Accommodation

Manual Section: §9.4.1 (Temporal Windows)

Category: ADAPTED (HIGH 75–90%)

Rationale: The 根据 clause in Article 1 of a Chinese departmental regulation is a mandatory drafting feature established by the *Regulations on Procedures for the Formulation of Rules* (《规章制定程序条例》, State Council Order No. 322, revised 2017). The NPCSC *Legislative Technical Standards (Trial)* (No. 2) (法工委发[2011] 5号, §13) draws an explicit distinction: 根据 is for citing the Constitution or a parent statute as a legislative basis in Article 1, whilst 依据 (yījù) is for citing specific clauses in operative provisions. This confirms that 根据 establishes an authority dependency — analogous to US CFR authority citations (1 CFR §21.40–.43) or EU "Having regard to" preambles — rather than a content dependency analogous to ISO/IEC Clause 2 normative references.

Chinese regulatory families constructed through 根据 chains exhibit 24–42 month spans between parent and child instruments. The CAC AI regulatory family demonstrates this pattern: Algorithm Recommendation Provisions (March 2022) → Deep Synthesis Provisions (January 2023, ~10 months) → Generative AI Measures (August 2023, ~17 months) → AI Content Labelling Measures (September 2025, ~42 months from parent). Comparative data confirms these timescales are normal: EU Level-1 financial-services acts average 25 months from proposal to publication with a range extending to 51 months (Clifford Chance, February 2025); US rulemaking under enabling statutes commonly takes 3–5 years.

The accommodation resolves a routing asymmetry identified through multi-source research (unanimous convergence across independent inputs). The same functional relationship — a parent authority instrument and a child implementing regulation separated by a multi-year gap — is classified as different CC types depending on jurisdiction. Western authority chains (e.g., an enabling statute and the agency rule issued under it) typically qualify as Type 6 (Framework Family) through Pathway 3, which has no temporal window, or as Type 1 (Policy Cascade), which has longer windows (up to 60 months for Energy/Infrastructure). Chinese 根据 chains are classified as Type 4 (Regulatory Enablement) — the child regulation operates within the framework the parent created — where the Technology domain window is 24 months. The result is that functionally equivalent authority chains pass or fail temporal qualification based on which CC type they are routed through, not on the strength of the documented relationship.

The three-constraint design (explicit full-title citation, CC-V grade, 48-month bound) ensures the accommodation is narrow: it applies only to formally documented authority chains, not to thematic proximity or informal policy influence.

The FATF Mutual Evaluation Methodology provides the strongest precedent for legal-tradition accommodation, requiring assessors to account for "the nature of the legal system (civil law or common law) and institutional framework." GDPR adequacy assessments apply functional equivalence rather than formal identity (*Schrems II*, Case C-311/18, ¶105).

Sources: *Regulations on Procedures for the Formulation of Rules* (State Council Order No. 322, 2017); NPCSC *Legislative Technical Standards (Trial) No. 2* (法工委发 [2011] 5号); Clifford Chance, *How Long is the EU Legislative Process?* (February 2025); FATF *Procedures for Fourth Round Mutual Evaluations*; CJEU, *Schrems II* (Case C-311/18).

Confidence: 78–85%. Authority-dependency characterisation well-sourced through NPCSC drafting standards (with stronger sourcing from two of three independent research inputs). Temporal normality conclusively established with comparative data. Absence of dedicated peer-reviewed scholarship on the 根据 clause as a legal institution prevents confidence above 85%.

Validation Commitment: Monitor for published Chinese administrative law scholarship analysing the 根据 clause. If empirical evidence emerges that 根据 functions as content dependency rather than authority dependency, reassess whether the accommodation belongs in §9.4.1 or should migrate to Type 6 Pathway 2.

9.4.2 Funding Attribution Threshold (10%)

Category: ADAPTED (HIGH 75–90%) — multi-source precedent

Threshold:

- **Primary:** ≥50% = Strong funding dependency
- **Significant:** 10–49% = Meaningful funding connection
- **Minor:** <10% = Not documented as Funding Flow

Primary Precedent Sources:

Authority	Threshold	Application
Basel III Large Exposure	≥10% Tier 1 Capital	Large exposure definition threshold; 25% single-counterparty limit applies
IMF BPM6, paragraph 6.12	≥10% voting power	Direct investment classification — ownership of equity entitling investor to 10% or more of voting power
US GAAP ASC 280-10-50-12	≥10%	Segment reporting materiality
IFRS 8 paragraph 13	≥10%	International accounting materiality
Basel III Significant Investment	>10% ownership	Investment significance threshold
CFIUS Passive Investment	<10% safe harbour	31 CFR §800 — below 10% = passive
WTO SCM Agreement Article 11.9	1% de minimis	WEO's 10% is 10× the international minimum
SEC SAB 99	5% preliminary screen	Makes 10% (2×) clearly material

WEO Application:

The 10% threshold prevents trivial linkages whilst capturing meaningful dependencies. Aligned with international financial reporting standards for "significant" contribution versus "causing" relationships.

Note: OECD Due Diligence Guidance provides conceptual framework ("contributes to" vs "causes" vs "directly linked") but no specific percentage. 10% threshold derived from convergent international standards listed above.

WTO Threshold Alignment: The 10% threshold is independently validated by convergent regulatory practice (SEC ASC 280, Basel III, IMF BPM6). WEO's alignment with this cross-regulatory consensus — at 10× the WTO SCM de minimis (1%) and 2× the SEC SAB 99 preliminary screen (5%) — ensures funding attribution captures relationships that are unambiguously material across financial, trade, and investment frameworks.

Denominator clarification (V1.4): The denominator for funding attribution is the direct funding instrument value, not total project cost. This follows standard grant-accounting practice: when a government award of US\$7.86B enables a US\$100B total project (including private investment and debt), the attribution question is "what percentage of the award traces to the enabling policy?" — not "what percentage of the total project does the award represent?" The former correctly captures the funding dependency; the latter dilutes it with capital the policy did not provide. This clarification resolves the CHIPS Act cluster inconsistency where total-project framing produced sub-threshold attribution (7%) for awards that are 100% traceable to the enabling legislation.

9.4.3 Linguistic Marker Reference

Category: EMPIRICALLY GROUNDED (HIGH 75–90%) — from policy citation research

Source: Zhang & Guan (2022); Chinese policy citation literature

High-Confidence Markers (validated across multiple sources):

Chinese	Pinyin	Meaning	CC Type
根据	gēnjù	According to / Based on	Policy Cascade
落实	luòshí	Implement / Put into effect	Policy Cascade
贯彻	guàncè	Carry out / Execute	Policy Cascade
依照	yīzhào	In accordance with	Policy Cascade / Regulatory Enablement
联合	liánhé	Joint / Coordinated	Joint Issuance

Medium-Confidence Markers (require additional verification):

Chinese	Pinyin	Meaning	Notes
配套	pèitào	Supporting / Complementary	May indicate Framework Family
配套政策	pèitào zhèngcè	Supporting policies	Framework Family indicator
针对	zhēnduì	In response to / Targeting	Requires verification of intent
卡脖子	qiǎbózi	Bottleneck / Chokepoint	Security/technology context
拨款	bōkuǎn	Appropriation / Funding	Funding Flow indicator
资助	zīzhù	Subsidise / Fund	Funding Flow indicator
授权	shòuquán	Authorise	Regulatory Enablement indicator
系列	xìliè	Series	Framework Family indicator
回应	huíyìng	Response	Response CC indicator

English Marker Equivalents:

CC Type	English Markers
Policy Cascade	"pursuant to," "in accordance with," "implementing," "as required by"
Regulatory Enablement	"authorised under," "enabled by"
Response	"in response to," "following," "retaliation for"
Framework Family	Same numbering series, explicit family membership

Evidence Quality by Confidence Level:

Level	Label	Evidence Standard
CC-V	Verified	Official document explicitly establishes connection (ISO/IEC normative reference standard)
CC-E	Established	1 Primary + 2 Corroborating sources (ICD 206 corroboration standard)
CC-A	Analytical	Analyst assessment based on observable evidence (Parallel Policy only)

Chinese Policy Citation Research:

Ba et al. (2022) establish that policy citations "perform the legal function of delimiting the policy rights or legislative reference" by identifying and formally linking back to earlier instruments in the Chinese policy hierarchy.

Zhang & Guan (2022) formalise Chinese policy bibliometrics by classifying policy citation types and constructing citation and co-citation networks to trace how focal policies shape subsequent policymaking. Their analysis demonstrates that regional policymakers tend to cite national policies in their local policy documents, with citation patterns reflecting hierarchical governance structure.

Citations:

- Ba, Z., Ma, Y., Cai, J., & Li, G. (2022). "Spatio-temporal dynamics and determinants of new energy vehicle diffusion in China." *Journal of Cleaner Production*.
- Zhang, C., & Guan, J. (2022). "How policies emerge and interact with each other? A bibliometric analysis of policies in China." *Science and Public Policy*, 49(3), 441-459.

9.5 Parallel Policy Similarity Criteria

Manual Section: 9.4.3

Category: CATEGORY CREATOR (derived from academic framework)

Four Similarity Criteria:

Criterion	Assessment
Mechanism Similarity	Same policy tool type with documented sub-category convergence (mandatory gate)
Industry Similarity	Same industry per industry tagging system (Chapter 7)
Scope Similarity	Comparable geographic/sectoral reach
Timing Overlap	Events within domain-specific temporal window (see §9.4.1)

Threshold: Criterion 1 (Mechanism Similarity) is a mandatory gate — if not met, no further assessment is conducted. Of the remaining three criteria, at least two must be met, producing a minimum total of 3 of 4 criteria for Parallel Policy classification.

Academic Foundation:

- Shipan & Volden (2008): Established policy diffusion mechanisms (Learning, Competition, Imitation, Coercion)
- Gilardi & Wasserfallen (2019): Refined criteria for identifying parallel policy adoption

WEO Threshold ($\geq 3/4$) — Category Creator Calibration:

The 3-of-4 threshold is WEO-original calibration. Academic literature establishes the conceptual framework for similarity assessment but does not specify quantitative thresholds:

Threshold	Classification	Rationale
4/4	Very strong similarity	Rare; may indicate explicit coordination
3/4	Strong enough to document	Parallel Policy threshold
2/4	Insufficient	Could be coincidental
1/4 or 0/4	No connection	No documented relationship

Threshold Selection Rationale: Selected to balance:

1. Sufficient stringency (avoids spurious connections)
2. Practical applicability (4/4 too restrictive)
3. Definitional integrity (Criterion 1 mandatory gate — mechanism convergence is the defining characteristic of the Parallel Policy type; its absence precludes classification regardless of the remaining criteria score)

Threshold Derivation — Empirical Basis

Academic Foundation:

The policy diffusion literature establishes that parallel policy adoption occurs through four mechanisms (Shipan & Volden, 2008):

Mechanism	Description	Observable Indicators
Learning	Jurisdictions observe and adopt successful policies from others	Similar policy design, temporal sequence
Competition	Jurisdictions adopt policies to maintain competitive position	Similar targets, similar timing
Imitation	Jurisdictions copy policies for legitimacy without deep analysis	High mechanism similarity, rapid adoption
Coercion	External pressure drives adoption	Explicit conditionality, power asymmetry

All four mechanisms produce observable similarity across the same dimensions: mechanism, target, scope, and timing. The academic literature does not specify quantitative thresholds for "sufficient similarity" — this is the gap WEO fills.

Threshold Calibration Analysis:

Research on policy diffusion patterns (Berry & Berry, 1990; Gilardi & Wasserfallen, 2019) suggests:

Similarity Level	Observed Pattern	Interpretation
4/4 criteria	Rare (<5% of policy pairs)	Strong indicator of explicit coordination; may warrant Joint Issuance investigation
3/4 criteria	Uncommon (10-20% of policy pairs)	Consistent with diffusion mechanisms; sufficient for Parallel Policy classification
2/4 criteria	Common (30-40% of policy pairs)	Likely coincidental; policies in same domain often share some characteristics
1/4 or 0/4 criteria	Very common (>50% of policy pairs)	No meaningful similarity

WEO Calibration Decision:

The 3-of-4 threshold was selected based on:

1. **Distinctiveness:** 3/4 similarity is uncommon enough (~10-20% of policy pairs) to indicate meaningful connection rather than coincidence
2. **Observability:** All four criteria are objectively assessable from policy documentation
3. **Conservatism:** Setting threshold at 3/4 rather than 2/4 reduces false positives whilst accepting some false negatives (missed connections)
4. **Analytical designation:** Parallel Policy is the only CC type permitting CC-A (Analytical) confidence, explicitly acknowledging the inferential nature of similarity-based connections

Sensitivity Analysis:

If threshold were...	Effect
2/4	~2-3× more Parallel Policy connections; higher false positive rate; diluted analytical signal
4/4	~60-70% fewer Parallel Policy connections; captures only near-identical policies; misses genuine diffusion
3/4 without mandatory gate	Admits connections where Criterion 1 (Mechanism Similarity) fails — i.e., events using different policy tool types that share industry, scope, and timing. These document thematic and temporal coincidence, not parallel instrument adoption. The corpus audit identified 2 cross-category CCs that would have survived 3/4 without the mandatory gate; both were removed (§9.5.2)
3/4 with mandatory gate but without specificity sub-test	Admits all same-category matches at the binary level. In the Investment/Buildout category, this captures "both deploy capital for AI infrastructure" — a determination so broad it carries minimal analytical signal for 8 of 50 Infrastructure-dominant PPs (§9.5.2). The sub-test preserves the 42 PPs whose mechanism match reveals documented supply-chain interdependencies or institutional parallelism

Citations:

- Berry, F. S., & Berry, W. D. (1990). "State lottery adoptions as policy innovations." *American Political Science Review*, 84(2), 395-415.
- Gilardi, F., & Wasserfallen, F. (2019). "The politics of policy diffusion." *European Journal of Political Research*, 58(4), 1245-1256.
- Shipan, C. R., & Volden, C. (2008). "The mechanisms of policy diffusion." *American Journal of Political Science*, 52(4), 840-857.
- Mallinson, D. J. (2021). "Policy innovation adoption across the diffusion life course." *Policy Studies Journal*, 49(2), 335-358.

Validation Commitment: The 100-event milestone commitment is fulfilled by the corpus audit (June 2026, 154 CCs across 181 events). The audit produced the first structured dataset of PP failure patterns, specificity gradients, and structural boundary cases across the complete Type 7 corpus (74 PPs ranked, 10 removed, 2 confidence reclassifications). Findings are documented in §9.5.2 (Criterion 1 calibration), §9.5.3 (Criterion 2 validation), and the CC-A Confidence Ceiling subsection of §9.3 (confidence reclassifications). **Next milestone:** Review at database doubling (~300 events / ~300 CCs). Assess whether the specificity distribution has shifted, whether the indicative sub-category table (Manual §9.4.3) needs expansion, and whether the industry-tag binary test remains adequate or requires refinement.

9.5.1 Type 7 — Corporate Product Release Exclusion

Manual Section: §9.1.8 (Corporate Product Release Scope Limitation), §9.4.3 (Criterion 1: Mechanism Similarity)

Category: STANDARD PRACTICE

Rationale: The five mechanism categories in Criterion 1 (Subsidy/Incentive, Restriction/Control, Standard/Mandate, Framework/Governance, Investment/Buildout) are exclusively policy instruments. Corporate product releases are market actions, not policy tools, and do not fit any enumerated category. Follows from strict application of existing criteria.

9.5.2 Criterion 1 — Mandatory Gate and Mechanism Specificity Sub-Test

Manual Section: §9.4.3 (Criterion 1: Mechanism Similarity)

Category: CATEGORY CREATOR (HIGH 75–90%) for the mandatory gate + WEO SYNTHESIS (MEDIUM-HIGH 60–75%) for the specificity sub-test

Mandatory Gate

Criterion 1 is elevated to a mandatory qualifying gate: if Mechanism Similarity is not met, no further assessment is conducted and the connection does not qualify, regardless of how many remaining criteria would be satisfied.

Mechanism convergence is definitional to the Parallel Policy type. "Parallel Policy" means structurally parallel adoption of similar policy instruments — the type captures independent convergence on *similar instruments*, not independent action in *similar domains*. Without mechanism similarity, the remaining criteria (industry, scope, timing) document thematic, geographic, and temporal coincidence between events that may employ entirely different policy tools. A semiconductor export control and a semiconductor investment subsidy share industry, may share scope, and may share timing — but they are not parallel instruments. They are different responses to the same policy challenge.

The mandatory pre-filter pattern is established within WEO methodology. The Distinct Actor Requirement (§9.1.6) is a mandatory pre-filter before any CC assessment — same-actor events cannot constitute a CC regardless of other criteria. The jurisdictional pre-filter for Type 7 is mandatory before the similarity rubric. The F1+F2 double-fail exclusion (§4.9) is a mandatory qualifying gate within significance assessment. The structure is consistent: definitional characteristics must be present before detailed assessment begins.

Mechanism Specificity Sub-Test

Category-level matching (both events falling within the same row of the five mechanism categories) is necessary but not sufficient. The analyst must additionally identify at least one documented sub-category characteristic — a specification, instrument sub-type, or structural feature finer-grained than the category label — shared by both events. Where no sub-category characteristic can be identified from verified source documentation, Criterion 1 is not met.

Porosity problem. The five mechanism categories (Subsidy/Incentive, Restriction/Control, Standard/Mandate, Framework/Governance, Investment/Buildout) were designed with policy instruments as the primary use case. For Policy events, category-level matching is naturally discriminating — a risk-tiered regulation and a liability amendment are recognisably different governance instruments, and their similarity at the Framework/Governance category level carries genuine analytical signal. For Infrastructure events, the Investment/Buildout category is porous: all commercial AI compute deployments, sovereign GPU procurement programmes, investment fund vehicles, and facility construction projects fall within the same category. The category-level match "both deploy capital for AI infrastructure" is trivially universal for Infrastructure events and carries minimal analytical signal.

The sub-test addresses this porosity by requiring a documented shared characteristic below the category level. This preserves analytically informative connections — two events sharing the same GPU architecture reveal supply-chain interdependence; two sovereign frontier supercomputers benchmarked on the same performance standard reveal institutional parallelism — whilst filtering connections whose mechanism match amounts to "both are data centres" or "both spend money on AI."

Calibration evidence (June 2026 corpus audit). The corpus audit ranked all 74 Type 7 PPs by Criterion 1 mechanism specificity — 50 Infrastructure-dominant, 15 Policy/Governance-dominant, and 9 previously unranked PPs assessed separately:

Specificity	Infrastructure	Policy/Governance	Previously unranked	Total	%
HIGH	22	8	6	36	48.6%
MEDIUM	20	5	3	28	37.8%
LOW	8	2	0	10	13.5%
Total	50	15	9	74	

The distribution exhibits a natural break at the MEDIUM/LOW boundary. All 36 HIGH-specificity PPs identify documented supply-chain interdependencies (shared hardware architectures, identical instrument types, matched institutional forms) or near-identical governance instruments (both binding AI-primary statutes, both advisory AI safety institutions). All 28 MEDIUM-specificity PPs identify genuine institutional or structural parallelism with documented variation in one dimension (different instrument sub-type within the same category, enforcement vs advisory character, sovereign vs corporate actor type). The 10 LOW-specificity PPs operate at the category boundary — their mechanism match holds at the category level but their internal structures diverge significantly in instrument type, actor class, or target industry.

The sub-test cuts at this empirically identified noise floor. All 10 LOW-specificity PPs fail the sub-test (no documented sub-category characteristic can be identified from verified sources that both events share). All 28 MEDIUM and 36 HIGH PPs pass. The retroactive impact is 8 removals through the sub-test plus 2 removals through the mandatory gate (cross-category matches), totalling 10 removals (13.5% of the 74-PP corpus). Zero MEDIUM or HIGH PPs are affected.

Event-type asymmetry confirmed. Policy/Governance PPs are more naturally discriminating than Infrastructure PPs: 53.3% HIGH in the Policy/Governance cohort versus 44.0% in the Infrastructure cohort. Policy instruments have inherently varied mechanisms, so "same mechanism type" carries more analytical signal. The LOW cohort is correspondingly smaller for Policy PPs (13.3% vs 16.0%). The sub-test is most needed for Infrastructure PPs but applies uniformly for methodological consistency.

Extension Mechanism

The indicative sub-category table (Manual §9.4.3) provides examples for each mechanism category but is not exhaustive. Each row includes "or comparable documented characteristic" as the extension mechanism, ensuring future event types not yet encountered can be accommodated without taxonomy maintenance. The `specification_overlap` field makes every determination auditable — the analyst records the specific characteristic identified, whether from the indicative table or from the extension mechanism.

Vendor-Dominance Clause

For Investment/Buildout, shared vendor alone does not satisfy the sub-test where the vendor holds dominant market share in the relevant technology segment. Without this clause, "both use [dominant vendor]" would satisfy the sub-test for any two events deploying the dominant vendor's products — a match driven by market structure rather than analytically informative structural convergence. The clause requires at least one specification beyond the vendor name (architecture generation, networking standard, or deployment model) to be documented for both events, establishing supply-chain interdependence at a level finer than market ubiquity.

The clause is anonymised and evergreen. No vendor is named; the test is whether the vendor holds dominant market share in the relevant segment, assessed at the time of the CC determination. This ensures the clause survives vendor market-share changes without requiring methodology amendment.

Evidence Standard

The sub-test requires that the shared sub-category characteristic be documented in "verified sources" — primary or corroborating sources within each event's verified source package. This aligns with the event verification framework (§3.1), which imposes source quality requirements at the event level rather than per-fact. Individual facts can come from any source in the verified package. Requiring "primary documentation" specifically for the sub-category characteristic would be stricter than the event-level standard and would create an asymmetry: a hardware specification documented in a corroborating trade press article would be a verified fact about the event but could not be used for the sub-test. The "verified sources" standard avoids this inconsistency.

Confidence Assessment: HIGH (mandatory gate — deductive from type definition). MEDIUM-HIGH (specificity sub-test — 73–80%; empirically grounded in the June 2026 corpus audit data with natural-break confirmation; extension mechanism introduces controlled analyst judgment).

Validation Commitment: Review specificity distribution at database doubling milestone (~300 CCs). Assess whether the indicative sub-category table needs expansion for new event types. Monitor for edge cases where the extension mechanism produces inconsistent determinations across analysts. Review vendor-dominance clause if dominant market shares shift significantly within tracked technology segments.

9.5.3 Criterion 2 — Industry Similarity

Manual Section: §9.4.3 (Criterion 2: Industry Similarity)

Category: ADAPTED (HIGH 75–90%) — industry tagging formalisation of existing criterion

Rename: "Target Similarity" → "Industry Similarity"

The V1.4 Criterion 2 ("Target Similarity") used the operative word "target" — a word denoting strategic intent. The industry tagging system (Chapter 7) uses the governing verb "address" — what an event's documented provisions actually do. These are different concepts. A semiconductor manufacturing subsidy may strategically *target* AI supremacy, but its documented provisions *address* semiconductor manufacturing. The old Criterion 2 permitted an analyst to assess "both target AI infrastructure" and score MET on the basis of inferred strategic intent rather than documented provisions, without a defined vocabulary or deterministic test.

"Industry Similarity" replaces "Target Similarity" to align the criterion with its operative assessment mechanism and to eliminate the intent/provisions ambiguity.

Industry-Tag Binary Test

Criterion 2 is assessed through the industry tagging system: shared tag (primary or secondary) = MET; no shared tag = NOT MET. The four-level alignment table in V1.4 (Exact/Category/Domain/No Match) is retired. The binary test eliminates a judgment step between "Exact" and "Category" that did not change the pass/fail outcome — both produced MET under V1.4 practice. The binary formulation removes dead-weight analyst judgment without changing the operative threshold.

The processing dependency is codified: industry tags (Chapter 7) must be assigned to both events before Criterion 2 can be assessed. This was implicit in V1.4 (Criterion 2 could not be assessed without knowing the events' industry characteristics) and is made explicit in V1.5 to formalise the processing sequence.

Corpus Validation

72 of 74 existing Type 7 PPs share at least one industry tag:

Tag overlap status	Count	%
Same primary tag	63	85.1%
Different primary, bridged by secondary	9	12.2%
Zero overlap	2	2.7%

The formalisation validates the existing corpus: 97.3% of PPs already satisfy the industry-tag test. The 2 PPs without shared tags: one drops from 4/4 to 3/4 (survives — Criterion 2 is the single unmet criterion and the $\geq 3/4$ threshold is still satisfied); the other was already assessed as Criterion 2 NOT MET under V1.4 and is unaffected. The industry-tag formalisation produces zero additional removals beyond those already attributable to the Criterion 1 changes.

Confidence Assessment: HIGH. The rename resolves a documented ambiguity. The binary test is deterministic. The corpus validation demonstrates that the formalisation captures existing practice rather than disrupting it.

Validation Commitment: Review industry-tag test adequacy at database doubling milestone. If the industry taxonomy expands (new categories added per Chapter 7 review), assess whether the binary test's discriminating power changes. Monitor for false negatives — event pairs with genuine industry convergence that fail the tag test due to tagging idiosyncrasies rather than substantive divergence.

9.5.4 Structural Correspondence — PP Criteria and WEO Classification Axes

Category: STANDARD PRACTICE

The four PP criteria operate on the same analytical axes as WEO's classification systems, at one level of granularity finer:

WEO classification axis	Categories	PP criterion	Categories
Event Type (§3.3)	4 types	Criterion 1 (Mechanism)	5 mechanism categories
Domain → Industry Tags (§6, §7)	4 domains → 11 industry tags	Criterion 2 (Industry)	11 industry tags
Geographic classification	—	Criterion 3 (Scope)	Geographic match + scale
Temporal verification	—	Criterion 4 (Timing)	Domain-specific windows

The PP rubric assesses whether two events are similar along the dimensions WEO considers analytically meaningful for classifying events. Criterion 1's mechanism categories subdivide Event Type. Criterion 2's industry tags are the industry tagging vocabulary from Chapter 7. Criterion 3's geographic scope is the same jurisdictional-level analysis used in bloc classification. Criterion 4's temporal windows are domain-specific, drawing on the same domain structure as Chapter 6.

Each criterion tests similarity along a dimension the methodology uses for event classification.

9.5.5 Type 7 Analytical Character and Closure

Manual Sections: §9.1.2, §9.4.3

Category: CATEGORY CREATOR (structural characterisation of CC type space)

Type 7 is the sole analytical CC type. This characterisation is a structural necessity, not an arbitrary designation. Independent convergence — where actors adopt similar instruments without formal coordination — is evidenced by the structural match between the instruments themselves. The evidence demonstrates parallelism, not coordination intent. The CC determination is an analytical assessment of observable similarity, conducted through a defined rubric with explicit criteria and threshold. Documentary evidence of the connection itself is definitionally unavailable: if such evidence existed, the relationship would satisfy a Type 1–6 definition (all of which require documentary proof of the inter-event link) and would not be classified under Type 7.

The CC-A confidence level is the logical entailment. CC-V and CC-E assert the existence of documents that establish the connection — a primary source directly linking the events, or an authoritative statement documenting the connection. A connection whose defining characteristic is structural similarity *without* formal coordination cannot produce such documents without contradicting its own classification. CC-A (analyst assessment based on observable evidence) is the only confidence level consistent with the type definition.

Closure. The word "sole" is deliberate and carries a closure function. A future proposal for an additional analytical CC type would need to demonstrate the same structural impossibility: a relationship class where documentary evidence of the connection is *definitionally* unavailable, not merely difficult to obtain. "Difficult to obtain" is CC-E territory — authoritative statements may be hard to locate but are not structurally impossible. "Definitionally unavailable" is what makes the analytical exception warranted and bounded. The current CC type space is closed under this test: Types 1–6 cover documentary relationships; Type 7 covers the single identified class of non-documentary relationship (structural similarity through independent convergence). Reopening the analytical exception would require identifying a second structural impossibility — a relationship class that cannot, by construction, produce documentary evidence of the connection.

Confidence Assessment: HIGH. Follows deductively from the CC type definitions and the confidence level definitions.

9.5.6 Criterion 3 — Scope Comparator Hierarchy

Manual Section: §9.4.3 (Criterion 3: Scope Similarity)

Category: ADAPTED (HIGH 75–90%) for instrument-specific comparators, 3× threshold, and binary discipline + WEO SYNTHESIS (MEDIUM-HIGH 60–75%) for hierarchical ordering and the inapplicable/indeterminate distinction

Rationale: V1.3 specified scope similarity through geographic matching and a financial scale ratio test ("Within 3× of each other") without codifying metric selection across event types where financial data may be unavailable or inapplicable, and without distinguishing between events where financial comparison is meaningless by nature and events where data would be meaningful but is undisclosed. V1.4 introduces a two-level metric selection hierarchy, the inapplicable/indeterminate distinction, and codification of range-straddling and structural comparability provisions.

Metric Selection Hierarchy (WEO SYNTHESIS)

The practice of matching comparability metrics to instrument type is established across apex institutions: IEA/IRENA use MW/GW for energy programmes (IRENA, *Renewable Capacity Statistics 2026*); IMF uses GDP-share for fiscal policy (*Fiscal Monitor April 2026*); IEG uses segmented methodology by project type (*Project Performance Ratings Codebook*, September 2015). No institution codifies a metric-selection hierarchy — the ordering is a WEO contribution consistent with the OMB Circular A-4 (November 2023) monetise → quantify → qualify cascade, which governs assessment depth rather than cross-programme comparator selection.

The two-level ordering (financial → physical capacity) reflects a precision hierarchy. When neither metric is available, scope is assessed on geographic matching alone. Across 75 Parallel Policy CCs at V1.5 publication, all assessments lacking quantitative data returned Met, consistent with the expectation that events passing mechanism and industry similarity at the same jurisdictional level are inherently scope-comparable.

3× Ratio Threshold (ADAPTED)

The geometric ratio of 0.33× to 3.0× is the established institutional standard for defining comparable organisational scale in executive compensation benchmarking, universally applied by tier-one consulting firms: FW Cook ("1/3× to 3× revenue," Lum and Jordan, 14 October 2021), Meridian Compensation Partners (Eaton Corporation plc DEF 14A, 2023/2025), Pearl Meyer (0.33×–3.0× revenue), and Compensia (0.33×–3.0× revenue). ISS applies a tighter 0.4×–2.5× band. The 3× upper bound reflects logarithmic parity — a half order of magnitude ($10^{0.5} \approx 3.16$) — representing the boundary beyond which entities are no longer structurally comparable in scale.

Internal calibration across numerically-assessed Parallel Policy CCs reveals a natural discontinuity:

Ratio Range	Qualifying (Met)	Non-Qualifying (Not Met)
1.0×–2.8×	20	0
2.8×–3.5×	0	2
3.5×–5.0×	0	3
>5.0×	0	10

Note: The ratio-range distribution above reflects the V1.4 count of 35 quantitative Parallel Policy CCs. At V1.5 publication, the corpus comprises 75 Type 7 CCs, of which 46 are quantitative (§9.4.3, Scope Comparator Hierarchy). The distribution above is the V1.4 derivation basis.

The 3× threshold falls within the observed gap between qualifying ratios ($\leq 2.77\times$) and non-qualifying ratios ($\geq 3.50\times$). No alternative ratio has stronger grounding in either external precedent or observed data.

Binary Discipline, Range-Straddling, and Structural Comparability

The 3× threshold is applied as strict binary pass/fail, consistent with ISO/IEC 17025 and the ILAC G8 simple acceptance rule. Graduated alternatives exist (Basel III CCB, EU Merger 2023 Notice, FATF) — the binary approach is a deliberate design choice prioritising replicability over case-by-case flexibility.

Range-straddling: where figures are reported as ranges crossing the threshold, the conservative bound governs, consistent with ISO/IEC 17025 conformity assessment practice.

Structural comparability: scope encompasses mechanism-structure comparability beyond numerical proximity. The analyst documents the rationale for any override in the evidence record.

Data-Availability Asymmetry (WEO SYNTHESIS)

Events assessed as indeterminate pass scope on geographic matching alone — the same outcome as events assessed as inapplicable. This means less-documented events face a less stringent assessment than well-documented events. This asymmetry is inherent to evidence-based frameworks: ISS cannot screen companies that do not report revenue; FATF cannot score criteria for which data is absent. Three considerations support the design: penalising regulatory events for lacking budgets would be a category error; penalising events for jurisdictional opacity would create structural visibility asymmetry; and the $\geq 3/4$ threshold ensures scope is one of four gates, not a standalone qualifier. The inapplicable/indeterminate distinction creates an audit trail for milestone review. At V1.5 publication, 46 CCs are quantitative, 14 inapplicable, and 15 indeterminate.

Sources:

Institutional: IRENA (2026); IMF *Fiscal Monitor* (2026); IEG *Codebook* (2015); OMB Circular A-4 (2023); ISO/IEC 17025:2017; ILAC G8:09/2019.

Corporate Governance: FW Cook (Lum/Jordan, 2021); Eaton DEF 14A (2023/2025, Meridian); Pearl Meyer; Compensia; ISS *U.S. Peer Group FAQ* (v.2025.12); CAP S&P 1500 insurer study (0.36×–4.55× observed dispersion).

Academic: Albuquerque, De Franco & Verdi (2013) *JFE*; Bizjak, Lemmon & Nguyen (2011) *JFE*; Faulkender & Yang (2010) *JFE*.

Confidence: MEDIUM-HIGH (72–80%). Instrument-specific practice ~85%. 3× threshold grounded in corporate governance practice, consistent with observed CC data ~78%. Binary discipline ~75%. Hierarchical ordering ~68%. Inapplicable/indeterminate distinction ~70%.

Validation Commitment: Review ratio distribution at database doubling milestones. Monitor for X-fold ratio tests in comparative policy literature. Review structural comparability overrides at 200-event milestone. Review indeterminate-status CCs at 200-event milestone; if >15% of Parallel Policy CCs, assess whether mitigation is required.

9.6 Derivation Documentation

9.6.1 Connection Type Derivation

High-Consensus Types (Multi-Source Validation)

The following connection types appear across 3+ independent systems:

Type	Sources	Consensus
Policy Cascade	Chinese 1+N Framework, CSET, OECD	HIGH
Joint Issuance	Chinese Governance, OECD, OFAC/EU	HIGH
Funding Flow	OECD Due Diligence, Government Grant Standards, Federal Budget Practice	HIGH
Regulatory Enablement	ISO/IEC Directives, OECD, Administrative Law	HIGH
Response	OFAC Sanctions, GSDB, CSET	HIGH
Framework Family	ISO/IEC Standards, UN Framework Families, National Standards Bodies	HIGH
Parallel Policy	Policy Diffusion Literature (Shipan & Volden), Gilardi & Wasserfallen (2019), Berry & Berry (1990)	MEDIUM

Research Sources:

Primary Institutional Sources:

- ISO/IEC Directives, Part 2: Normative vs. informative reference distinction
- OECD Due Diligence Guidance: "Causes," "contributes to," "directly linked" relationships
- OFAC Sanctions Programs: Escalation sequences, response patterns
- Chinese State Council Gazette: 1+N framework documentation practices

Academic Literature:

- Shipan & Volden (2008): Policy diffusion mechanisms (Learning, Competition, Coercion, Emulation)
- Gilardi & Wasserfallen (2019): Politics of policy diffusion
- Berry & Berry (1990): Event history analysis of policy adoption

Policy Analysis:

- CSET AI Policy Research: Sanction cascade, regulatory mimicry, forum shopping response
- CCICED: China 1+N framework analysis
- Carbon Brief: Chinese policy documentation analysis

9.6.2 Threshold Derivation

Temporal Windows:

Domain	Source	Derivation
Energy/Infrastructure (365-day Response)	Policy implementation research	Long-horizon commitment cycles
Security (30-day Response)	OFAC enforcement patterns	Sanctions response typically within 1 month
Technology (180-day Response)	Product development cycles	Minimum iteration period
Trade (60-day Response)	WTO DSU Article 4.7	Consultation period before panel request

Note: The V1.4 methodology used domain-specific response windows. V1.5 recalibrated to Response Bands by response evidence type (§9.4.1). The domain-specific values above are the V1.4 derivation basis.

Citation Standards:

Standard	Source	Application
Normative reference	ISO/IEC Directives Part 2	Mandatory compliance relationship
Informative reference	ISO/IEC Directives Part 2	Guidance/background relationship
1+N qualification	CCICED analysis	Chinese policy cascade identification

Funding Attribution:

Threshold	Source	Rationale
10% minimum	OECD "contributes to" standard	Meaningful contribution vs. trivial link
50% primary	Government grant reporting	Primary funding source designation

9.6.3 Gap Analysis

Deferred Connection Types:

Type	Gap Reason	Research Needed
Escalation Sequence	Requires serial temporal analysis	Conflict escalation literature
Cross-Agency Referral	Low public documentation visibility	Internal government process research
Genealogical Traceability	Requires supply chain mapping	ISO 9001 traceability implementation

Known Limitations:

Limitation	Impact	Mitigation
GDEL/ICEWS no explicit relationship typology	Cannot benchmark against major event databases	Rely on regulatory/legal framework sources
Chinese informal coordination underdocumented	May miss 部际联席会议 (bùjì liánxí huìyì, inter-ministerial conference) patterns	Enhanced Chinese-language research
Policy diffusion mechanisms difficult to operationalise	"Competition" and "emulation" hard to distinguish from documentation	Conservative classification; defer ambiguous cases

9.7 Novelty Verification

Status: Verified novel via systematic research

Research Scope: 25+ platforms across geopolitical intelligence, AI policy tracking, event databases, and entity relationship platforms

Key Platforms Examined:

- Geopolitical Intelligence: Stratfor/RANE, Eurasia Group, Oxford Analytica, Control Risks
- AI/Tech Policy Trackers: OECD AI Policy Observatory, Stanford HAI, Carnegie
- Event Databases: GDEL, ICEWS, UCDP, ACLED
- Entity Relationship Platforms: Bloomberg, Sayari, Maltego

Conclusion: No comparable typed event-to-event relationship taxonomy found. Coordination Connections (CC) represents genuine novel contribution. The 7-type taxonomy qualifies as copyrightable "creative selection and arrangement."

Section 10: Infrastructure Threads Rationale

Manual Reference: Chapter 10

Manual Section: §10.1–10.5

Category: WEO SYNTHESIS (HIGH 75–90%)

Rationale: Infrastructure Threads record documented material facts — that the asset named in one event is the asset whose origin another records — requiring no interpretation. The layer carries no assessment apparatus: no connection type, no confidence grade, no criteria rubric. The four predicates (`fabricated_at` , `deploys` , `trains_on` , `powered_by`) cover the material relationships the corpus documents between AI infrastructure assets and events — fabrication, deployment, training, and power. The layer sits beside the Coordination Connection taxonomy because it documents a different object class: a material fact rather than a relationship between verified events.

The two-layer pattern — a binary, ungraded factual link layer alongside a graded, assessed relational taxonomy — draws on the factual-versus-rhetorical distinction in the Citation Typing Ontology (CiTO; Shotton 2010, SPAR Ontologies) and the observation-versus-inference split in CIDOC-CRM's CRMinf argumentation model. Structural parallels appear in knowledge-graph co-occurrence practice and the binary `ObjectEvent` of GS1 EPCIS. The four link types correspond to the AI-compute supply-chain decomposition documented in Sastry et al. (2024) and the OECD's semiconductor value-chain mapping (2025).

Key Design Decisions:

Why ungraded? A Coordination Connection's confidence grade reflects the strength of evidence for a relationship that must be interpreted. A thread requires no interpretation — the material fact is documented or it is not. Grading a thread would attach an assessment apparatus to a record that makes no assessed claim.

Why a separate layer rather than an eighth CC type? A thread documents a material fact, not a relationship between events involving distinct actors. It is actor-agnostic where §9.1.6 requires distinct actors. Folding it into the CC taxonomy would attach relational apparatus — typing, grading, criteria assessment — to a record that makes no relational claim.

Why four predicates? The set matches the material-relationship classes the corpus documents. Additional predicates enter through the Validation Commitment if the corpus documents asset relationships outside these four.

Evidence Chain:

Precedent	Type	Contribution
CiTO factual-vs-rhetorical split (Shotton 2010, SPAR Ontologies)	Structural	Binary ungraded factual link layer coexisting with assessed relational layer over the same objects
CIDOC-CRM/CRMinf observation-vs-inference split	Structural	Graded belief values attach only to inferences; observations are factual and ungraded
Knowledge-graph co-occurrence vs semantic links	Structural	Undirected factual co-presence links alongside directed assessed predicates
GS1 EPCIS <code>ObjectEvent</code>	Structural	Binary asset-present-at-location record; evidence inherited from captured event data
Sastry et al. 2024 (<i>Computing Power and the Governance of AI</i>)	Structural	Fabrication → chip → deployment → power decomposition grounding the four link types
OECD <i>Mapping the Semiconductor Value Chain</i> (2025)	Structural	Design-fabrication-ATP decomposition grounding <code>fabricated_at</code>

Grounding Assessment: WEO SYNTHESIS. ~80% grounded. The epistemic-layering pattern draws on four established precedents across citation ontology, cultural-heritage argumentation modelling, knowledge-graph practice, and supply-chain event capture. The AI-compute supply-chain decomposition is externally documented in governance and industry-mapping literature. The ~20% novel element is the fusion of these into a four-predicate assessment-free factual layer built to sit beside the Coordination Connection taxonomy.

Validation Commitment: Review link-type coverage at database doubling; monitor for asset types requiring additional predicates.

Section 11: Category Creator Documentation

11.1 Purpose

This section consolidates the validation commitments attached to WEO's calibration choices — category creator components and the adapted or synthesised components that carry review commitments.

11.2 Category Creator Components

Component	Section	Type	Validation Commitment
Level 5 operational evidence (four-pathway gate)	3.5.4	Synthesis	Review as SC 42 database and EU AI Act harmonisation pipeline mature; monitor FATF 5th Round methodology
PIET 3-of-4 gate architecture	3.7.1	Calibration	Review against cumulative PIET event data
PIET trade flow and price thresholds	3.7.2	Calibration	Review against each qualifying PIET event; thin-market sensitivity analysis
OEP 10% hardware deployment threshold	3.8.3	Provisional	Review at 10-event OEP Pathway C milestone
Conflict-era six-month review interval	3.10.1	Calibration	Review interval at first conflict-era determination; assess whether 6 months is operationally appropriate
Third Bloc Formation transitional classification	3.12.1.3	Synthesis	Reassess when third PAA candidate approaches qualification
J-Tier D AI Hub Exception	4.6	Designation	Annual designation review
F3 AI Centrality (mandatory gating factor)	4.6	Threshold	100-event milestone fulfilled (corpus audit, June 2026); review at 200-event milestone
Buildout vs Restriction typology	5.2	Classification	Classification consistency monitoring
Principal AI actor framework (7-dimension, ≥ 3 threshold)	5.4	Classification	Reassess at methodology version milestones; threshold sensitivity review as additional nations develop AI infrastructure
Per-dimension qualification principles (7 principles)	5.4.1	Calibration	Review against cumulative actor assessment data; edge-case stress test refinement at each methodology version milestone

Dimension 2/3 process-node threshold ($\leq 7\text{nm}$)	5.4.1	Calibration	Review against fabrication frontier at each methodology version milestone; recalibrate downward when ≥ 4 actors manufacture at threshold node independently
Dimension 5 benchmark saturation dependency	5.4.1	Monitoring	Document benchmarks informing each assessment cycle's D5 verdicts; flag reduced-confidence assessments when benchmark discrimination degrades
Subsidiary clause PAA-score sensitivity	5.4.1	Monitoring	Reassess whether alternative attribution rules would alter the $\geq 3/\geq 2$ -categories result at each methodology version milestone
Sovereign control principle (governing rule)	5.4.1	Classification	Review territorial vs ownership-chain test assignments if new dimension types are introduced or existing dimensions are restructured
D4 territorial compute requirement	5.4.1	Classification	Monitor for edge cases where territorial compute presence and corporate ownership diverge; review against D2 territorial test for consistency
D4 assessment-date-relative threshold	5.4.1	Calibration	Document prevailing frontier training scale at each assessment cycle; review for consistency across actors assessed at different dates
D6 dual-pathway (export controls + AI-specific legislation)	5.4.1	Clarification	Monitor whether the dual-pathway produces inconsistent assessments across actors
D7 national proportionality standard	5.4.1	Calibration	Review proportionality assessments for cross-actor consistency; monitor for threshold drift
D7 general-infrastructure-fund exclusion	5.4.1	Classification	Apply consistently; monitor for borderline cases where general funds have significant AI-adjacent components
D7 announced-vs-executed investment	5.4.1	Calibration	Verify disbursement data where available; flag actors with execution rates below 25%
D5 single-domain exclusion	5.4.1	Classification	Apply consistently; review if frontier model definition expands to include domain-specific excellence
D2 marketing-node exclusion	5.4.1	Classification	Maintain node-name-to-lithography mapping table for Intel and any other manufacturer using rebranded node names
D3 subsidiary clause	5.4.1	Classification	Monitor for cases where D3 and D2

application			territorial rules produce divergent results for the same actor; ensure consistency
Synthesis majority override rule	5.4.1	Methodology	Document each override with establishing case; review for overuse
Actor Designation Framework temporal disclaimer	5.4.2	Operational	Maintain platform-facing disclaimer; review if retroactive scoring is implemented
Three-gate PAA designation (capability + severance + platform)	5.4.2	Classification	Continue applying to remaining actors; review gate structure against cumulative results
Severance test 24–36 month sustainability window	5.4.2	Calibration	Review at first severance test application; monitor against actual supply-chain disruption events for calibration validation
Three-category spanning requirement (hardware/platform/governance)	5.4.2	Threshold	Review if any actor scores ≥ 3 from a single capability category
Ecosystem inputs definition ((a)/(b) formulation)	5.4.3	Classification	Review if new extraterritorial instruments emerge beyond FDPR/EAR; monitor ecosystem/bloc boundary
Per-PAA severance testing (Option B)	5.4.4	Classification	Monitor for third PAA emergence; review if any designation depends on per-PAA vs simultaneous approach
Deterrence principle with three-part documentation	5.4.5	Classification	Monitor PAA leverage vectors; review supranational entity clause if EU institutional authority evolves
Multi-vector leverage assessment	5.4.5	Classification	Monitor for changes in US leverage vector count for Netherlands and other actors
Supranational entity deterrence clause	5.4.5	Classification	Review if EU develops direct corporate sovereignty instruments applicable to ASML
Cascade analysis boundary (single-level standard)	5.4.6	Classification	Monitor whether sub-tier constraint findings change any designation; review boundary if FDPR enforcement expands
Dimension sustainability taxonomy (5 classifications)	5.4.7	Classification	Monitor whether five levels remain sufficient; review counting rule at boundary cases
AI Infrastructure Keystone classification	5.4.8	Classification	Monitor analytical utility; assess whether AIK-ACS distinction remains stable and informative across assessment cycles
AIK theoretical grounding (Rey and Tirole primary anchor)	5.4.8	Adapted	Review against emerging IO/IR literature; monitor for frameworks that challenge the

			vertical/horizontal foreclosure mapping
Advanced Capability State classification	5.4.9	Adapted	Track whether PAA attribution remains stable; monitor for ACS actors developing multi-ecosystem dependencies
Participant classification (single sub- ≥ 3 tier)	5.4.10	Classification	Monitor whether per-dimension data alone provides sufficient analytical detail for sub- ≥ 3 actors, or whether subdivision is warranted
Evidence standard 1+1 base with escalation	5.4.12	Adapted	Monitor escalation trigger frequency; if >50% of MET dimensions trigger escalation, review base standard
Source authority hierarchy (Grade 1/2/3)	5.4.12	Classification	Review grade assignments against emerging source types
Mandatory source descriptors	5.4.12	Operational	Monitor implementation consistency across assessors
Four escalation triggers (opaque/contested/unaudited/no canonical)	5.4.12	Calibration	Track trigger frequency and outcomes; adjust trigger definitions if producing anomalous results
ENT-2 four-criteria linguistic threshold	8.6	Calibration	Track ENT-2 population at 200 events; test discriminating power against new governance architectures
ENT-5 medium scope boundary	8.9	Classification	If ENT-5 count remains below the \$8.5 provisionality threshold (10) beyond 2028, reassess whether the medium scope boundary is excluding qualifying governance instruments (at V1.5 publication: 5)
ENT-2/ENT-5 governance maturity boundary	8.10	Classification	At 200 events, assess whether all ENT-5 events also carry ENT-2
CC Taxonomy (7 types)	9.1	Novel IP	Protected via Zenodo disclosure
Authority Hierarchy 6-tier structure	9.2	Synthesis	N/A (synthesised from established precedents)
Response Bands (Type 5)	9.4.1	Empirically Grounded	Response CC calibration empirically validated (§9.4.1 Response Band Calibration)
Domain-specific temporal windows (Types 1, 3, 4, 7)	9.4.1	Provisional	100-event milestone fulfilled (corpus audit, June 2026); review at 200-event milestone
Chinese 根据 temporal accommodation (48-month bound)	9.4.1	Adapted	Monitor for 根据 scholarship; reassess if content-dependency evidence emerges

Parallel Policy 3-of-4 threshold with Criterion 1 mandatory gate	9.5	Calibration	100-event milestone fulfilled (corpus audit, June 2026); review at database doubling (~300 CCs)
Criterion 1 mechanism specificity sub-test	9.5.2	Calibration (WEO Synthesis)	Review specificity distribution at database doubling; monitor extension mechanism consistency
Criterion 2 industry-tag binary test	9.5.3	Adapted	Review tag-test adequacy at database doubling; monitor for false negatives from tagging idiosyncrasies
CC-A confidence ceiling for Type 7 (floor = ceiling)	9.3	Category Creator	Monitor routing rule invocations; reassess if coordination evidence consistently fails all Type 1–6 definitions despite being event-pair-specific
Scope Comparator Hierarchy (two-level metric selection)	9.5.6	Synthesis	Review at database doubling milestones
3× scope similarity threshold	9.5.6	Calibration (Adapted)	Review if natural discontinuity shifts
Inapplicable/indeterminate scale assessment distinction	9.5.6	Synthesis	Review indeterminate population at 200-event milestone
Infrastructure Threads (four-type material-link taxonomy)	10	Synthesis	Review link-type coverage at database doubling; monitor for asset types requiring additional predicates

11.3 Validation Framework

All category creator components are subject to:

1. **Continuous monitoring** during database growth
 2. **Systematic review** at designated milestones
 3. **Transparent documentation** of any adjustments
 4. **Academic peer review** as methodology matures
-

Appendix A: Glossary of Key Terms

Term	Definition
ACS (Advanced Capability State)	Nation-state actor meeting ≥ 3 SCP dimensions but failing the per-PAA severance test (§5.3.1)
AIK (AI Infrastructure Keystone)	Nation-state actor meeting ≥ 3 SCP dimensions and passing the severance test but lacking a sustainable platform dimension (§5.3.1)
Category Creator	Methodology component for which WEO exercises novel calibration prerogative
CC (Coordination Connection)	Typed relationship between two events in the WEO database
CC-A	Coordination Connection - Analytical (analyst assessment with evidence)
CC-E	Coordination Connection - Established (authoritative statement documents connection)
CC-V	Coordination Connection - Verified (documentary proof)
Conflict-Era Source Currency Adjustment	Adjusted currency windows (§3.4.12) that apply to source documentation produced within or about conflict-affected territories during formally defined conflict-era conditions, modifying the peacetime ceilings in §3.4.8
Convergent Implementation Evidence Accommodation	Source accommodation (§4.11) accepting multi-stream convergent evidence anchored by an accountability-bearing disclosure as primary evidence for Policy events whose instrument is unpublished; available only after all other verification and qualification pathways are exhausted
Corporate Strategic Disclosure (Option E)	Stage 2 AI Connection Filter Option E — establishes AI connection for Corporate events through regulated financial disclosure explicitly linking a transaction to AI infrastructure, where Options A, B, and C fail
CSTF (Cross-Sector Transmission Frequency)	Proportion of Coordination Connections linking events with different primary industry tags
Defence Appropriation Evidence	Stage 2 AI Connection Filter Option F — establishes AI connection for Policy events through binding defence appropriation documents with explicit AI designation, where Options A, B, and C fail

(Option F)	
ENT (Environmental Nexus Tag)	Binary tag identifying documented environmental intersections across five types: Resource Demand (ENT-1), Environmental Enablement (ENT-2), Environmental Consequence (ENT-3), Environmental Dependency (ENT-4), Environmental Coordination (ENT-5)
Event	Verified happening (coordination or fragmentation)
Impact- Demonstrated (T2)	T2 Restriction classification basis supported by demonstrated cross-bloc fragmentation impact through ≥ 2 other-bloc government policy responses, rather than explicit other-bloc targeting in primary documentation
Industry Similarity (Criterion 2)	Parallel Policy Criterion 2 — events address the same industry, assessed through the industry tagging system (§9.4.3, Chapter 7)
Industry Tagging Principle	Governing standard: events are tagged by the industry whose operational landscape their documented provisions most directly address
Infrastructure Threads	Supplementary factual-linking layer documenting material relationships between assets and events through four link types (Chapter 10)
J-Tier	Jurisdictional significance tier (A/B/C/D/E)
Mechanism Statement	One-sentence statement identifying the document-verifiable pathway from an event's provisions to a secondary industry's operations
OEP (Operational Evidence Protocol)	Stage 2 AI Connection Filter Option D — establishes AI connection for Infrastructure events through operational physical evidence where official documentation is absent, generic, or application-agnostic
Option G (Supply-Chain Response Evidence)	Stage 2 AI Connection Filter Option G — establishes AI connection for Policy events where the instrument controls, secures, or subsidises a documented AI infrastructure input material, and Options A, B, and C fail because the instrument text addresses the material rather than AI (§3.5)
PAA (Principal AI Actor)	Nation-state actor meeting ≥ 3 SCP dimensions, passing the per-PAA severance test, and holding ≥ 1 sustainable platform dimension (§5.3.1)
Participant	Nation-state actor scoring below ≥ 3 on SCP dimensions; the residual designation for actors not qualifying as PAA, AIK, or ACS (§5.3.1)
PIET (Post- Implementation Evidence Track)	Stage 2 qualification pathway for Policy restriction events on physical AI-critical materials or physical technology products where post-implementation empirical evidence establishes the AI connection that documentary sources cannot
PISA-5	Policy and Institutional Significance Assessment (5-factor framework)

Primary Tag	The single industry tag representing the industry most directly addressed by an event's documented provisions
Reduced Verification Lag	60-day reduced verification lag for corporate product release events satisfying the four §3.9.2 conditions, with mandatory maturity re-review (§3.15.13)
Response Bands	Three-band temporal calibration for CC Type 5 Response connections — Band A ≤ 90 days, Band B 91 days–18 months, Band C 18–36 months (§9.4.1)
Scale Assessment Status	Three-value taxonomy (Quantitative / Inapplicable / Indeterminate) recording whether Dimension 2 of Parallel Policy scope similarity was assessed with quantitative data (§9.4.3)
SCP (Sovereign Capability Profiles)	Seven-dimension assessment framework for evaluating sovereign AI infrastructure capability, producing scores used for PAA/AIK/ACS/Participant designation (§5.3)
Secondary Tag	Industry tag (maximum 2 per event) representing an industry whose operational landscape is also identifiably addressed by documented provisions
Transmission Matrix	11×11 matrix recording CC counts between industry category pairs
Warmth Engine Ratio (WER)	Ratio of ENT-1 count to ENT-5 count, measuring the balance between documented environmental resource demand and documented environmental governance

Appendix B: Cross-Reference Verification

All Manual cross-references verified against WEO Methodology Manual:

Rationale Section	Manual Reference	Status
§3.1 Source Quality Hierarchy	§3.1	✓ Verified
§3.2 Claim Verification Tiers	§3.4	✓ Verified
§3.3 Four Event Types	§3.6	✓ Verified
§3.4 Event Type × Tier Matrix	§3.6, Table B	✓ Verified
§3.5 Coordination Event Criteria	§3.7	✓ Verified
§3.5.4 Type 2 Level 5 Operational Evidence	§3.7	✓ Verified
§3.6 Two-Stage AI Filter	§3.5	✓ Verified
§3.6.2 Option E — Corporate Strategic Disclosure	§3.5, Option E	✓ Verified
§3.6.3 Option F — Defence Appropriation Evidence	§3.5, Option F	✓ Verified
§3.6.4 Option G — Supply-Chain Response Evidence	§3.5, Option G	✓ Verified
§3.7 PIET — Post-Implementation Evidence Track	§3.5.1	✓ Verified
§3.8 OEP — Operational Evidence Protocol	§3.5, Option D	✓ Verified
§3.9 6-Month Verification Lag	§3.9	✓ Verified
§3.9.1 Reduced Verification Lag (Verified Product	§3.9.2, §3.15.13	✓

Releases)		Verified
§3.10 Currency Windows	§3.4.8	✓ Verified
§3.10.1 Conflict-Era Source Currency Adjustment	§3.4.12	✓ Verified
§3.11 Non-English Source Verification	§3.3	✓ Verified
§3.12 Bloc Classification	§3.10	✓ Verified
§3.12.1 Universal Membership Categories	§3.10.1	✓ Verified
§3.12.1a Bloc Assignment Gate	§3.10.1	✓ Verified
§3.12.1.1 Evidence Standard for Register Classification Changes	§3.10.1.1	✓ Verified
§3.12.1.2 Register Operations	§3.10.1.2	✓ Verified
§3.12.1.3 Third Bloc Formation	§3.10.1.3	✓ Verified
§3.12.2 Non-Aligned Nation Identification	§3.10.2	✓ Verified
§3.12.3 Event-Level Attribution Scope Limitation	§3.10.3	✓ Verified
§3.12.4 AI Model Events — Critical Technology Determination	§3.10.3 (Priority 2)	✓ Verified
§3.13 Event Evolution Protocol	§3.14	✓ Verified
§3.13.1 Corporate Product Release Diagnostic	§3.14.1 (Supplement)	✓ Verified
§3.14 Post-Qualification Verification	§3.15	✓ Verified
§3.15 Source Accommodation Alignment	§3.1, §3.4.11, §3.5, §3.5.1, §4.11	✓ Verified

§4.1 Basel 5-Factor	§4.3-4.5	✓ Verified
§4.2 \$500M Threshold	§4.4 (Factor 1)	✓ Verified
§4.3 5,000+ GPU Threshold	§4.4	✓ Verified
§4.4 Software Routes 1-4	§4.8	✓ Verified
§4.5 Route 4-CS	§4.8	✓ Verified
§4.6 PISA-5	§4.9	✓ Verified
§4.6.1 Tiebreaker Criteria (TB1-TB5)	§4.9 (Human Review Framework)	✓ Verified
§4.6.2 J-Tier D AI Hub Exception	§4.9	✓ Verified
§4.6.3 F3 Pathway Exemption	§4.9 (Factor 3)	✓ Verified
§4.6.4 PISA-5 F2/F4 Assessment Object	§4.9 (Factor 2, Factor 4 non-regulatory provisions)	✓ Verified
§4.7 T2 Buildout Criteria	§5.5 (T2 Buildout Test)	✓ Verified
§5.1 Three-Tier Architecture	§5.2	✓ Verified
§5.2 Mandatory Standard Test	§5.5 (T2 Mandatory Standard Test)	✓ Verified
§5.3 Bloc Definitions	§3.10 (Bloc Membership Reference)	✓ Verified
§5.4 Principal AI Actors	§5.3	✓ Verified
§5.4 T2 Scope (Principal Actor Asymmetry)	§5.3	✓ Verified

§5.4.1 Dimension Qualification Principles	§5.3.2	✓ Verified
§5.4.2 Three-Gate Designation, Severance Test, and AIK Classification	§5.3.1	✓ Verified
§5.4.3 Ecosystem Inputs Definition	§5.3.1	✓ Verified
§5.4.4 Per-PAA Severance Testing (Option B)	§5.3.1	✓ Verified
§5.4.5 Deterrence Principle	§5.3.1	✓ Verified
§5.4.6 Cascade Analysis Boundary	§5.3.1	✓ Verified
§5.4.7 Dimension Sustainability Taxonomy	§5.3.1	✓ Verified
§5.4.8 AI Infrastructure Keystone (AIK) Designation	§5.3.1	✓ Verified
§5.4.9 Advanced Capability State (ACS) Designation	§5.3.1	✓ Verified
§5.4.10 Participant Designation	§5.3.1	✓ Verified
§5.4.11 Sovereign Capability Profiles Register	§5.3.3	✓ Verified
§5.4.12 Evidence Standard for Dimension Qualification	§5.3.4	✓ Verified
§6.1 Four-Domain Framework	§6.1	✓ Verified
§6.2 Tech vs E/I Distinction	§6.2	✓ Verified
§6.3 Domain Tiebreaker	§6.7	✓ Verified
§7.1 Industry Taxonomy	§7.3	✓ Verified
§7.2 Primary/Secondary Structure	§7.4, §7.7	✓

		Verified
§7.3 Rule-Based Assignment	§7.4	✓ Verified
§7.4 Mechanism Statement	§7.7	✓ Verified
§7.5 CSTF	— (TCS paper; Manual §7.9 nearest context)	✓ Verified
§7.6 Transmission Matrix	— (TCS paper; Manual §7.9 nearest context)	✓ Verified
§7.7 Adversarial Quality Test	§7.9	✓ Verified
§7.8 Workflow Integration	§7.4, §7.9	✓ Verified
§8.1 Five-Type ENT Taxonomy	§8.3	✓ Verified
§8.2 ENT as Enrichment Layer	§8.2	✓ Verified
§8.3 Binary Qualification	§8.2	✓ Verified
§8.4 Architectural Guardrails	§8.2	✓ Verified
§8.5 ENT-1 Resource Demand	§8.3.1	✓ Verified
§8.6 ENT-2 Environmental Enablement	§8.3.2	✓ Verified
§8.7 ENT-3 Environmental Consequence	§8.3.3	✓ Verified
§8.8 ENT-4 Environmental Dependency	§8.3.4	✓ Verified
§8.9 ENT-5 Environmental Coordination	§8.3.5	✓ Verified
§8.10 ENT-2/ENT-5 Boundary	§8.3.2, §8.3.5	✓ Verified

§8.11 Chinese Cascade Model	§8.3.2	✓ Verified
§8.12 Floors and Categorical Tests	§8.3 (cross-cutting)	✓ Verified
§8.13 WER Positioning and Novelty	§8.5	✓ Verified
§8.14 WER Provisionality, n-Disclosure, and Corpus Scope	§8.5	✓ Verified
§9.1 CC Taxonomy Overview	§9.1, §9.2	✓ Verified
§9.1.1 Distinct Actor Requirement	§9.1.6	✓ Verified
§9.2 Seven Connection Types	§9.2.1	✓ Verified
§9.3 CC Confidence Levels	§9.3	✓ Verified
§9.3.1 CC-A Evidence Model for Parallel Policy	§9.3.1, §9.4.3	✓ Verified
§9.3.2 CC-A Confidence Ceiling and Coordination Evidence Routing	§9.3.2, §9.4.3	✓ Verified
§9.3.3 Type 1 Evidence Route Clarification	§9.2.1 (Type 1: Policy Cascade)	✓ Verified
§9.3.4 Independent Adoption Check	§9.1.7	✓ Verified
§9.4 Evidence Requirements	§9.4	✓ Verified
§9.4.1 Chinese Regulatory Temporal Accommodation	§9.4.1	✓ Verified
§9.4.1 Response Band Calibration	§9.4.1 (Response Bands), §9.2.1 (Type 5 temporal window)	✓ Verified
§9.4.2 Funding Attribution Threshold	§9.4.2	✓ Verified

§9.4.3 Linguistic Marker Reference	§9.5.2 (table: Linguistic Markers)	✓ Verified
§9.5 Parallel Policy Criteria	§9.4.3	✓ Verified
§9.5.1 Type 7 — Corporate Product Release Exclusion	§9.1.8	✓ Verified
§9.5.2 Criterion 1 — Mandatory Gate and Specificity Sub-Test	§9.4.3	✓ Verified
§9.5.3 Criterion 2 — Industry Similarity	§9.4.3, Chapter 7	✓ Verified
§9.5.4 Structural Correspondence	§9.4.3	✓ Verified
§9.5.5 Type 7 Analytical Character and Closure	§9.1.2, §9.4.3	✓ Verified
§9.5.6 Criterion 3 — Scope Comparator Hierarchy	§9.4.3	✓ Verified
§10 Infrastructure Threads Rationale	§10.1–10.5	✓ Verified

Appendix C: Note on Source Accuracy

This methodology draws on extensive research across academic literature, regulatory frameworks, and institutional documentation. Whilst every effort has been made to ensure accuracy in citations, dates, and quoted material, minor discrepancies may exist due to:

- Translation variations in non-English source material
- Differing publication dates across editions or jurisdictions
- Paraphrasing or summarisation in secondary sources consulted

Where methodology components are classified as **WEO-SYNTHESIS** or **WEO-ORIGINAL**, the contribution represents novel analytical work informed by, but not directly derived from, external sources. These components are explicitly flagged throughout the document.

Identified errors are corrected in subsequent methodology versions. Users identifying potential inaccuracies are encouraged to report them via the contact information provided on the WEO platform.

End of Methodology Rationale

Warmth Engine Observatory | January 2026 (v1.5, July 2026)